

Robust and Efficient Dynamic Low-Rank Matrix Recovery under Heavy-Tailed Noise

Abstract

Low-rank matrix recovery in high-dimensional settings is often complicated by structural changes and heavy-tailed noise, both of which are prevalent in real-world applications. Existing methods are typically developed under light-tailed assumptions or static model settings without structural changes. To address these challenges, a unified robust framework is developed for simultaneous matrix estimation and change-point detection under a general trace regression model. The proposed approach combines a truncation-based scheme to mitigate the impact of extreme observations with an optimized search strategy that reduces the computational cost of change-point localization. Under mild moment conditions, non-asymptotic error bounds are established for both matrix estimation and change-point detection, covering exact and approximate low-rank structures. The method is further extended to the multiple change-point setting. Numerical studies on synthetic and real datasets demonstrate strong robustness and computational efficiency, particularly in heavy-tailed and large-scale scenarios.

Keywords: Low-rank matrix recovery; change-point detection; heavy-tailed noise; computational efficiency

1. Introduction

High-dimensional data with structural changes are common in many applications, such as finance, genomics, and recommender systems. A substantial body of literature has investigated change-point detection in high-dimensional settings, primarily under sparsity assumptions on the underlying parameters. For example, [Lee et al. \(2016\)](#) studied change-point estimation in high-dimensional linear regression via ℓ_1 -regularization, [Wang and Samworth \(2018\)](#) developed a projection-based approach that leverages sparse changes in the mean vector, [Rinaldo et al. \(2021\)](#) established theoretical guarantees for localizing sparse changes in high-dimensional regression coefficients. These methods typically model the parameter as a sparse vector and exploit entry-wise sparsity or sparse changes to achieve accurate estimation and change-point detection.

While sparsity is convenient for vector-valued parameters, it does not capture the global structure of matrix-valued data. In practice, the underlying signal is more naturally represented by a low-rank or approximately low-rank matrix. This has motivated a growing line of research on dynamic low-rank matrix

recovery. For instance, [Bai et al. \(2020, 2023\)](#) considered vector autoregressive models with evolving low-rank and sparse components, [Enikeeva et al. \(2025\)](#) proposed information-criterion-based procedures for detecting low-rank structural breaks, and [Shi et al. \(2024\)](#) studied a general trace regression framework and developed a two-step method for low-rank matrix recovery with change-points, together with non-asymptotic guarantees.

To model such structured data, we consider the trace regression framework

$$Y = \langle \mathbf{X}, \boldsymbol{\Theta} \rangle + \varepsilon,$$

where $\boldsymbol{\Theta} \in \mathbb{R}^{d_1 \times d_2}$ is a low-rank coefficient matrix, $Y \in \mathbb{R}$ is the response, $\mathbf{X} \in \mathbb{R}^{d_1 \times d_2}$ is the covariate, and ε is the noise. This model is highly general and encompasses compressed sensing (CS), multiresponse regression (MR), and vector autoregressive (VAR) as special cases ([Shi et al., 2024](#)). In dynamic settings, the coefficient matrix is allowed to change across segments, leading to a piecewise low-rank structure.

Existing dynamic low-rank methods typically rely on light-tailed noise assumptions. In practice, heavy-tailed data are common, especially in finance and biology. For example, [Fan et al. \(2021\)](#) analyzed that macroeconomic variables ([Stock and Watson, 2002](#)) and gene expression data ([Gupta et al., 2014](#)) often exhibit pronounced heavy-tailed behavior. Our analysis of the market excess returns in [Section 6](#) also reveals clear deviations from Gaussian distribution, including sharp peaks and heavy tails. Such features indicate that standard least-squares-based methods may be sensitive to extreme observations.

To address heavy-tailed noise, a substantial literature has developed robust methods for low-rank matrix recovery. [Fan et al. \(2021\)](#) proposed a truncation-based shrinkage principle, [Yu et al. \(2024\)](#) introduced convex formulations based on adaptive Huber loss, and [Wang and Fan \(2025\)](#) developed a nonconvex factorization approach with optimal statistical rates under weak moment conditions. These methods improve robustness but are designed for static problems and do not directly extend to change-point settings. In particular, truncation may remove useful signals for change-point detection, while robust loss functions typically require solving non-smooth optimization problems for each candidate change-point.

From a computational perspective, change-point detection is expensive due to the need to search over many candidate locations. In dynamic low-rank models, each candidate evaluation requires solving a nuclear-norm-regularized problem. Although first-order methods such as accelerated proximal gradient algorithms ([Beck and Teboulle, 2009](#); [Toh and Yun, 2010](#)) can speed up each optimization, they do not reduce the number of subproblems to be solved. Recent works have also explored coarse search strategies to reduce the number of candidate evaluations ([Kovács et al., 2024](#)), but such approaches are often tailored to simpler changing-mean models and are difficult to extend to trace regression settings.

Taken together, the coexistence of heavy-tailed noise and structural changes poses fundamental challenges for low-rank matrix recovery. Heavy-tailed noise amplifies the influence of extreme observations and weakens the contrast between segments, making change-point localization more difficult. As illustrated in Figure 1, the standard two-step method (Shi et al., 2024) is sensitive to outliers and can lead to inaccurate change-point detection. Moreover, existing robustness strategies may attenuate the signal when structural breaks are present. At the same time, the computational burden of repeated high-dimensional optimization across candidate change-points limits the scalability of existing approaches. These challenges call for a unified framework that simultaneously achieves robustness to heavy-tailed noise, accurate characterization of structural changes, and computational efficiency.

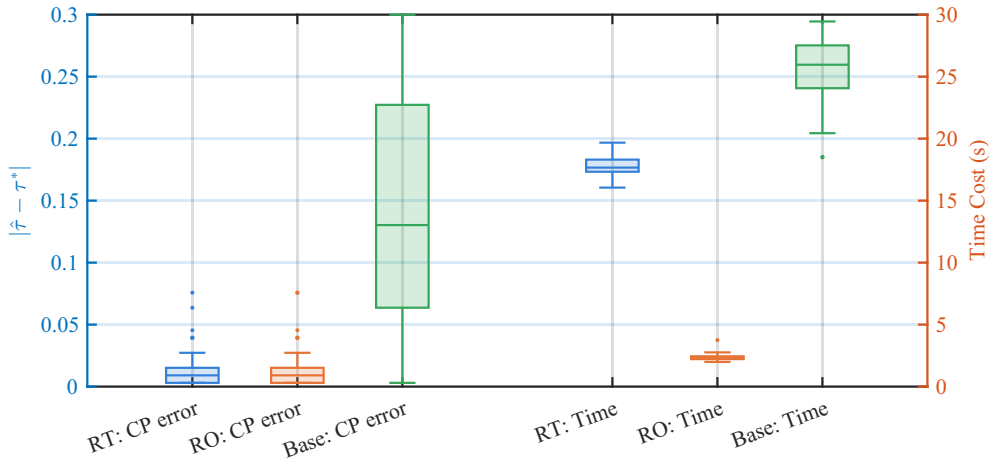


Figure 1: Performance comparison under heavy-tailed noise (Cauchy) in the trace regression model. RT and RO denote the proposed robust two-step method and robust optimized search method, respectively, while Base refers to the standard two-step method in Shi et al. (2024). The box-plots report the estimation error of the change-point location and the computational time across different methods.

To address the challenges outlined above, we develop a unified framework for dynamic low-rank matrix recovery under heavy-tailed noise, integrating statistical robustness with computational efficiency.

From a methodological perspective, we propose a robust two-step method for dynamic low-rank matrix recovery. Starting from the ℓ_2 loss, we first develop a robust loss function that mitigates the impact of heavy-tailed observations. Building on this, we further introduce a candidate-dependent truncation scheme that adapts to each candidate change-point location. This design effectively limits the influence of extreme observations while avoiding the attenuation of change-point signals caused by global truncation. As a result, the proposed method achieves stable estimation under weak moment conditions and preserves the contrast between segments for accurate change-point detection.

From a computational perspective, we develop a robust optimized search algorithm that substantially im-

proves efficiency over exhaustive grid search. By constructing a suitable gain function and employing a coarse-to-fine search strategy, the proposed method significantly reduces the number of nuclear-norm-regularized optimization problems required. As a result, the computational complexity of the robust two-step procedure is reduced from $O((d_1 d_2 N^2 + d_1 d_2 d_{\min} N)/\alpha + N^2)$ to $O\{(d_1 d_2 N \log N + d_1 d_2 d_{\min} \log N)/\alpha + N \log N\}$, where $O(1/\alpha)$ denotes the number of iterations required for coefficient estimation.

From a theoretical perspective, we establish non-asymptotic error bounds for both matrix estimation and change-point localization under the CS and MR cases, highlighting the generality of the proposed methods. The results explicitly characterize how the estimation error depends on the low-rank structure and the tail behavior of the noise distribution. As the moment condition becomes stronger, the rates improve accordingly and recover nearly the optimal rates under sub-Gaussian noise.

In addition, we introduce the notion of a mixed regression matrix to address a key challenge in the theoretical analysis under misspecified change-points. When the candidate change-point deviates from the true location, the observations within each segment are no longer homogeneous, making the estimator behave as if it were fitted to a mixture of different regimes. To capture this effect, we show that the estimator converges to a mixture of segment-specific parameters, which we formalize through the mixed regression matrix. This characterization serves as a key tool in establishing sharp error bounds for both matrix estimation and change-point localization.

1.1. Article Organization and Notation

The rest of the paper is organized as follows. Section 2 introduces the model setup, including the trace regression framework with change-points and the robust loss function for single change-point detection. Section 5 presents the proposed methodology for single change-point detection, including the robust two-step method and the robust optimized search algorithm. Section 4 establishes the theoretical results, where non-asymptotic error bounds are derived for both compressed sensing (CS) and multiresponse regression (MR) settings. Section 5 extends the proposed framework to the multiple change-point scenario. Section 6 reports numerical results on simulated and real datasets, covering both single and multiple change-point detection. Finally, Section 7 concludes the paper.

Notation. We collect the notations used in this paper for ease of reference. For any matrix $\Theta \in \mathbb{R}^{d_1 \times d_2}$, denote $\|\Theta\|_*$, $\|\Theta\|_F$, $\|\Theta\|_{S_q}$, and $\|\Theta\|_{\text{op}}$ as the nuclear norm, Frobenius norm, Schatten- q norm, and operator norm, respectively, and let $\rho_l(\Theta)$ be the l -th singular value. The dimension is denoted by $d = d_1 + d_2$, and $d_{\min} = d_1 \wedge d_2$. For matrices $\mathbf{A}$ and $\mathbf{B}$, define the inner product as $\langle \mathbf{A}, \mathbf{B} \rangle = \text{tr}(\mathbf{A}^\top \mathbf{B})$. For a vector $\mathbf{x}$, $\|\mathbf{x}\|_2$ denotes the Euclidean norm. For a random variable or vector, $\|\cdot\|_{\psi_2}$ and $\|\cdot\|_{\psi_1}$ denote the sub-Gaussian norm and sub-exponential norm, respectively. For a differentiable function $\mathcal{L}(\Theta)$, $\nabla \mathcal{L}(\Theta)$ denotes its gradient with

respect to Θ . For a set $\mathcal{I}$, $|\mathcal{I}|$ denotes its cardinality, and $\mathbf{1}_{\{\cdot\}}$ denotes the indicator function. We write $a \lesssim b$ if $a \leq Cb$ for some constant $C > 0$, and use $a \vee b$ and $a \wedge b$ to denote $\max(a, b)$ and $\min(a, b)$, respectively.

2. Model setup

In this section, we present the trace regression model with change-points studied in this paper, together with the associated loss function for estimating the change-point and coefficient matrices.

2.1. Trace Regression with Change-Points

Suppose that we have N independent samples $\{Y_i, \mathbf{X}_i, t_i\}$, where $Y_i \in \mathbb{R}$ is response variable, $\mathbf{X}_i \in \mathbb{R}^{d_1 \times d_2}$ is design matrix and $0 < t_i \leq 1$ is the threshold variable. Then, the trace regression with change-points can be represented as

$$Y_i = \langle \mathbf{X}_i, \Theta_{(i)}^* \rangle \mathbf{1}_{\{\tau_{s-1}^* < t_i \leq \tau_s^*\}} + \varepsilon_i, \quad (2.1)$$

where $\{\tau_1^*, \tau_2^*, \dots, \tau_{s^*}^*\}$ are locations of true change-points and $\tau_0^*, \tau_{s^*+1}^*$ are denoted as 0, 1. The matrix $\Theta_{(i)}^*$ is the true coefficient matrix to the corresponding segments. For the samples satisfying $\tau_{s-1}^* < t_i \leq \tau_s^*$, the corresponding matrix $\Theta_{(i)}^*$ equals to $\Theta_s^* \in \mathbb{R}^{d_1 \times d_2}$, which is same in the same segment and different for adjacent segments. And $\varepsilon_i \in \mathbb{R}$ is the heavy-tailed error term.

We allow the noise variables ε_i to be heavy-tailed, meaning that their distributions may deviate from the standard sub-Gaussian or sub-exponential regimes. This setting accommodates the presence of extreme observations, which are commonly observed in practice and may substantially affect estimation performance.

At the same time, high-dimensional matrix-valued data often exhibit an intrinsic low-rank structure driven by latent factors. To capture this intrinsic feature, we model the coefficient matrices through a low-rank structure. Rather than imposing an exact low-rank condition, which can be overly restrictive in practice, we adopt a more flexible near low-rank condition that allows for a gradual decay of singular values. Specifically,

$$\Theta_s^* \in \mathcal{B}(R_{s,q}) = \left\{ \Theta \in \mathbb{R}^{d_1 \times d_2} \mid \sum_{l=1}^{d_{\min}} [\rho_l(\Theta)]^q \leq R_{s,q} \right\}, \quad (2.2)$$

where $0 \leq q \leq 1$ is a relaxation parameter and $R_{s,q} > 0$ is the corresponding radius. The set $\mathcal{B}(R_{s,q})$ characterizes matrices whose singular values decay at a controlled rate, allowing Θ_s^* to be well approximated by a low-rank component while retaining small residual singular values. This formulation provides a flexible description of low-rank structure beyond exact low-rankness. In particular, when $q = 0$, it reduces to the standard low-rank condition $\text{rank}(\Theta_s^*) \leq R_{s,0}$.

Meanwhile, the trace regression model itself is sufficiently general to cover a range of important settings, including CS and MR. This generality further broadens the applicability of the proposed robust dynamic low-rank framework under heavy-tailed noise.

(i) *Compressed Sensing.* The proposed framework covers CS problems, where $\mathbf{X}_i$ acts as measurement operators. It is common to assume that $\text{vec}(\mathbf{X}_i)$ are sub-Gaussian or satisfy similar concentration properties. In contrast, we allow for heavy-tailed noise and structural changes in the coefficient matrices.

(ii) *Multiresponse regression.* The MR model can also be embedded into the proposed framework. Consider

$$\mathbf{y}_i = \boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i + \boldsymbol{\epsilon}_i, \quad (2.3)$$

where $\mathbf{y}_i, \boldsymbol{\epsilon}_i \in \mathbb{R}^{d_2}$ and $\mathbf{x}_i \in \mathbb{R}^{d_1}$. By applying a linear aggregation to the response, we define $Y_i = \mathbf{1}_{d_2}^\top \mathbf{y}_i$, which yields $Y_i = \langle \boldsymbol{\Theta}, \mathbf{x}_i \mathbf{1}_{d_2}^\top \rangle + \mathbf{1}_{d_2}^\top \boldsymbol{\epsilon}_i$. Therefore, the MR model can be reformulated as a trace regression model with design matrix $\mathbf{X}_i = \mathbf{x}_i \mathbf{1}_{d_2}^\top$.

2.2. Loss Function for Robust Single Change-Point Detection

To handle heavy-tailed noise and accommodate the dynamic low-rank structure, we replace the standard ℓ_2 loss with a robust alternative. We first introduce the population criterion that motivates our empirical loss. Under the single change-point setting, the population ℓ_2 risk takes the form

$$\begin{aligned} \mathcal{R}(\tau, \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2) &= \mathbb{E} \left[(Y_i - \langle \mathbf{X}_i, \boldsymbol{\Theta}_1 \rangle)^2 \mathbf{1}_{\{t_i \leq \tau\}} + (Y_i - \langle \mathbf{X}_i, \boldsymbol{\Theta}_2 \rangle)^2 \mathbf{1}_{\{t_i > \tau\}} \right] \\ &= \mathbb{E} Y_i^2 + \mathbb{E} \left[-2 \langle Y_i \mathbf{X}_i, \boldsymbol{\Theta}_1 \rangle \mathbf{1}_{\{t_i \leq \tau\}} + \langle \mathbf{X}_i, \boldsymbol{\Theta}_1 \rangle^2 \mathbf{1}_{\{t_i \leq \tau\}} \right] \\ &\quad + \mathbb{E} \left[-2 \langle Y_i \mathbf{X}_i, \boldsymbol{\Theta}_2 \rangle \mathbf{1}_{\{t_i > \tau\}} + \langle \mathbf{X}_i, \boldsymbol{\Theta}_2 \rangle^2 \mathbf{1}_{\{t_i > \tau\}} \right]. \end{aligned}$$

Since the term $\mathbb{E} Y_i^2$ is independent of $(\tau, \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2)$, removing it does not affect the population minimizer. Therefore, we consider the equivalent objective

$$\begin{aligned} \arg \min \mathcal{R}(\tau, \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2) &= \arg \min \mathcal{L}(\tau, \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2) \\ &= \arg \min \left\{ \mathbb{E} \left[-2 \langle Y_i \mathbf{X}_i, \boldsymbol{\Theta}_1 \rangle \mathbf{1}_{\{t_i \leq \tau\}} + \langle \mathbf{X}_i, \boldsymbol{\Theta}_1 \rangle^2 \mathbf{1}_{\{t_i \leq \tau\}} \right] \right. \\ &\quad \left. + \mathbb{E} \left[-2 \langle Y_i \mathbf{X}_i, \boldsymbol{\Theta}_2 \rangle \mathbf{1}_{\{t_i > \tau\}} + \langle \mathbf{X}_i, \boldsymbol{\Theta}_2 \rangle^2 \mathbf{1}_{\{t_i > \tau\}} \right] \right\}. \end{aligned}$$

We next characterize the population minimizers of $\mathcal{L}(\tau, \Theta_1, \Theta_2)$ under a possibly misspecified change-point. To this end, we first fix τ and minimize $\mathcal{L}(\tau, \Theta_1, \Theta_2)$ with respect to Θ_1 and Θ_2 . For ease of analysis, we assume that t_i follows a uniform distribution on $(0, 1)$. When the candidate change-point τ differs from the true change-point τ^* , the samples assigned to one candidate segment may in fact come from two different regimes. As a result, the corresponding population minimizer is generally a weighted average of the true segment-specific coefficient matrices. We introduce the mixed regression matrices $\bar{\Theta}_{1,\tau}$ and $\bar{\Theta}_{2,\tau}$ to describe these effective population targets.

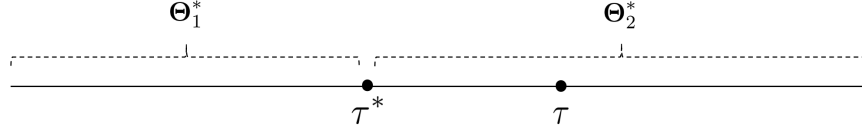


Figure 2: The relationship between the true regression matrices and the mixed regression matrix when $\tau > \tau^*$.

Figure 2 illustrates the relationship between the true coefficient matrices and the mixed matrices when $\tau > \tau^*$. In this case, the first candidate segment contains observations from both true regimes, whereas the second candidate segment only contains observations from the second regime. We define

$$\bar{\Theta}_{1,\tau} = \frac{\tau^*}{\tau} \Theta_1^* + \frac{\tau - \tau^*}{\tau} \Theta_2^*, \quad \text{and} \quad \bar{\Theta}_{2,\tau} = \Theta_2^*.$$

Similarly, when $\tau \leq \tau^*$, the first candidate segment only contains observations from the first regime, whereas the second candidate segment contains observations from both true regimes. We define

$$\bar{\Theta}_{1,\tau} = \Theta_1^*, \quad \text{and} \quad \bar{\Theta}_{2,\tau} = \frac{\tau^* - \tau}{1 - \tau} \Theta_1^* + \frac{1 - \tau^*}{1 - \tau} \Theta_2^*.$$

For any fixed τ , the minimizers of $\mathcal{L}(\tau, \Theta_1, \Theta_2)$ with respect to Θ_1 and Θ_2 satisfy $\Theta_s = \bar{\Theta}_{s,\tau}$, $s = 1, 2$. Substituting these optimal values into $\mathcal{L}(\tau, \Theta_1, \Theta_2)$, we further obtain that the objective is minimized at $\tau = \tau^*$. At this point, the mixed matrices reduce to the true coefficient matrices, namely $\bar{\Theta}_{1,\tau^*} = \Theta_1^*$ and $\bar{\Theta}_{2,\tau^*} = \Theta_2^*$. Therefore, minimizing $\mathcal{L}(\tau, \Theta_1, \Theta_2)$ leads to the same population target as minimizing the original least-squares risk, and hence remains a valid criterion for simultaneous change-point localization and coefficient matrix estimation.

Motivated by this population equivalence, we consider the empirical counterpart by replacing the expec-

tations with sample averages, which leads to the loss function

$$L(\tau, \Theta_1, \Theta_2) = \sum_{s=1}^2 \sum_{i \in \mathcal{I}_{s,\tau}} [-2\langle Y_i \mathbf{X}_i, \Theta_s \rangle + \langle \mathbf{X}_i, \Theta_s \rangle^2],$$

where $\mathcal{I}_{1,\tau} = \{i | t_i \leq \tau\}$ and $\mathcal{I}_{2,\tau} = \{i | t_i > \tau\}$ denote the two segments induced by τ . At the empirical level, using the loss without the response-square term is also desirable for change-point detection under heavy-tailed responses. Although $\sum_i Y_i^2$ is a common constant in the ordinary least-squares criterion, the response-square component can be highly unstable when the responses have heavy tails. In Appendix [Appendix A](#), we provide an additional numerical study based on the proposed robust procedures in Section [3](#), comparing the variants with and without the response-square term. The results show that removing this term does not degrade performance in the CS setting and can lead to more stable estimation in the MR setting. Nevertheless, removing the response-square term alone is not sufficient to address heavy-tailed noise, since the cross term involving $Y_i \mathbf{X}_i$ may still be affected by extreme responses. This motivates the robust procedures developed in the next section.

3. Robust Methodology for Single Change Point Detection

In this section, we study the single change-point setting and develop robust methods for dynamic low-rank recovery. We begin with a robust two-step procedure for change-point regression in Section [3.1](#), which ensures reliable estimation under heavy-tailed noise. To further enhance computational efficiency, we then propose a robust optimized search method in Section [3.2](#) that reduces the search complexity while preserving statistical accuracy.

3.1. Robust Two-step Method

Recalling the existing methods for dynamic low-rank estimation, the two-step procedure remains the predominant classical approach (see, e.g., [Shi et al. \(2024\)](#)). Specifically, it first fixes a candidate change-point, estimates the corresponding regression matrices, and then selects the optimal change-point by comparing the total loss across candidates. However, in the presence of heavy-tailed noise, additional treatment is required within this framework. Broadly speaking, there are two main strategies to address heavy-tailed data. One approach is to adopt robust loss functions to mitigate the influence of outliers ([Sun et al., 2020](#); [Yu et al., 2024](#)). The other is to directly truncate the observations or loss function to control extreme values ([Brownlees et al., 2015](#); [Fan et al., 2021](#)). While robust loss functions are effective, they typically increase the computational complexity and introduce additional tuning parameters. Motivated by this trade-off, we

instead integrate a truncation scheme into the two-step procedure, leading to a robust and computationally efficient method for dynamic low-rank estimation.

To mitigate the influence of heavy-tailed observations, we truncate the response variable as $\tilde{Y}_i(\tau) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{s,\tau})$, where $\tilde{Y}_i(\tau)$ denotes the truncated response. Specially, we truncate the response vector as $\tilde{\mathbf{y}} = \min(\gamma_{s,\tau}, \|\mathbf{y}_i\|_2) \mathbf{y}_i / \|\mathbf{y}_i\|_2$ for MR model. Importantly, the truncation parameter $\gamma_{s,\tau}$ is allowed to depend on both the segment index s and the candidate change-point τ , since each τ induces a different segmentation of the data and hence leads to segment sizes $N_{s,\tau}$ that vary with τ .

Next, we define the grid search domain $\mathcal{T}$. For the grid search step, we restrict attention to a set of candidate change-points $\mathcal{T}$. To avoid extremely small segments that may lead to unstable estimation, we further restrict $\mathcal{T}$ so that the segment sizes induced by each $\tau \in \mathcal{T}$ remain sufficiently large. Specifically, when $\tau < \tau^*$, we require $\tau^* - \tau < c_\tau \min(1 - \tau, \tau^*)$, and when $\tau \geq \tau^*$, we require $\tau - \tau^* < c_\tau \min(1 - \tau^*, \tau)$, for some constant $c_\tau \in (0, 1)$.

Based on the truncated samples, the robust two-step procedure proceeds as follows. For a fixed candidate change-point τ , we first construct the truncated responses $\{\tilde{Y}_i(\tau)\}_{i=1}^N$. Then, for each segment $s = 1, 2$, we estimate the corresponding regression matrix by solving a nuclear norm penalized optimization problem based on the truncated quadratic loss

$$\hat{\Theta}_{s,\tau} = \arg \min_{\Theta} \underbrace{\sum_{i \in \mathcal{I}_{s,\tau}} \left(-2 \langle \tilde{Y}_i(\tau) \mathbf{X}_i, \Theta \rangle + \langle \mathbf{X}_i, \Theta \rangle^2 \right)}_{\tilde{L}_{s,\tau}(\Theta)} + \lambda_{s,\tau} \|\Theta\|_*. \quad (3.1)$$

After obtaining the segment-wise estimators, we select the optimal change-point by minimizing the total penalized loss over the candidate set $\mathcal{T}$ as

$$\hat{\tau} = \arg \min_{\tau \in \mathcal{T}} \tilde{L}_{1,\tau}(\hat{\Theta}_{1,\tau}) + \tilde{L}_{2,\tau}(\hat{\Theta}_{2,\tau}). \quad (3.2)$$

To solve problem (3.1), we employ proximal gradient descent (PGD). For each segment s and candidate change-point τ , the update takes the form

$$\Theta_{s,\tau}^{(t+1)} = \text{Prox}_{\gamma\lambda_{s,\tau}} \left(\Theta_{s,\tau}^{(t)} - \gamma \nabla \tilde{L}_{s,\tau}(\Theta_{s,\tau}^{(t)}) \right),$$

where $\Theta_{s,\tau}^{(t)}$ denotes the estimate of the coefficient matrix for segment s under candidate change-point τ in t -th iteration, and $\Theta_{s,\tau}^{(t+1)}$ is the updated iterate. The proximal mapping $\text{Prox}_{\gamma\lambda_{s,\tau}}(\cdot)$ associated with the nuclear norm admits a closed-form solution via singular value thresholding, namely, $\text{Prox}_{\gamma\lambda_{s,\tau}}(\mathbf{A}) = \mathbf{U} \text{diag}((\sigma_i -$

$\gamma\lambda_{s,\tau})_+ \mathbf{V}^\top$, where $\mathbf{A} = \mathbf{U}\text{diag}(\sigma_i)\mathbf{V}^\top$ is the singular value decomposition and $(x)_+ = \max(x, 0)$. The step size $\gamma > 0$ controls the gradient descent step.

Compared with the standard two-step method, the robust version introduces an additional truncation step with cost $O(N)$. For the truncated sub-problem, each PGD iteration involves computing the gradient and applying the proximal operator. The gradient $\nabla \tilde{L}_{s,\tau}(\boldsymbol{\Theta}_{s,\tau}^{(t)})$ requires $O(d_1 d_2 N)$ operations, while the proximal step based on singular value thresholding costs $O(d_1 d_2 d_{\min})$. Therefore, each iteration has complexity $O(d_1 d_2 N + d_1 d_2 d_{\min})$, and $O(1/\alpha)$ iterations are needed to achieve an α -optimal solution (Toh and Yun, 2010). Hence, for a fixed τ , the total cost is $O((d_1 d_2 N + d_1 d_2 d_{\min})/\alpha + N)$. Searching over all $\tau \in \mathcal{T}$ with $|\mathcal{T}| = O(N)$, the overall computational complexity is $O((d_1 d_2 N^2 + d_1 d_2 d_{\min} N)/\alpha + N^2)$.

3.2. Robust Optimized Search Method

Recalling the robust two-step method, it requires solving $O(N)$ sub-problems over all candidate change-points, which leads to substantial computational cost due to repeatedly solving high-dimensional nuclear norm regularized optimization problems. However, such exhaustive evaluation is often unnecessary, since candidate change-points far from the true location τ^* typically yield much larger losses, while nearby candidates tend to produce similar estimates, resulting in considerable redundant computations. In addition, evaluating a large number of candidate change-points requires performing the truncation step repeatedly, which further adds redundant computations in practice.

Algorithm 1 Robust Optimized Search Method with Single Change-Point

Require: $\tau^{(0)} \in [0, 1]$, candidate set $\mathcal{T}$, minimal length m

- 1: Truncate $\tilde{Y}_i(\tau^{(0)})$ and estimate $\boldsymbol{\Theta}_s^{(0)} \leftarrow \arg \min_{\boldsymbol{\Theta}} \tilde{L}_{s,\tau^{(0)}}(\boldsymbol{\Theta}) + \lambda_{s,\tau^{(0)}} \|\boldsymbol{\Theta}\|_*$, $s = 1, 2$
- 2: $f_{\tau^{(0)}}(\boldsymbol{\Theta}_1^{(0)}, \boldsymbol{\Theta}_2^{(0)}) \leftarrow \tilde{L}_{1,\tau}(\boldsymbol{\Theta}_1^{(0)}) + \tilde{L}_{2,\tau}(\boldsymbol{\Theta}_2^{(0)})$, $t \leftarrow 0$
- 3: **while** $|\mathcal{T}| > m$ **do**
- 4: Choose probe point u as midpoint of the longer segment of $\mathcal{T}$ split by $\tau^{(t)}$
- 5: Truncate $\tilde{Y}_i(u)$ and estimate $\boldsymbol{\Theta}_{s,u} \leftarrow \arg \min_{\boldsymbol{\Theta}} \tilde{L}_{s,u}(\boldsymbol{\Theta}) + \lambda_{s,u} \|\boldsymbol{\Theta}\|_*$, $s = 1, 2$
- 6: $f_u(\boldsymbol{\Theta}_{1,u}, \boldsymbol{\Theta}_{2,u}) \leftarrow \tilde{L}_{1,u}(\boldsymbol{\Theta}_{1,u}) + \tilde{L}_{2,u}(\boldsymbol{\Theta}_{2,u})$
- 7: **if** $f_u(\boldsymbol{\Theta}_{1,u}, \boldsymbol{\Theta}_{2,u}) < f_{\tau^{(t)}}(\boldsymbol{\Theta}_1^{(t)}, \boldsymbol{\Theta}_2^{(t)})$ **then**
- 8: $\tau^{(t+1)} \leftarrow u$, $\boldsymbol{\Theta}_s^{(t+1)} \leftarrow \boldsymbol{\Theta}_{s,u}$, $f_{\tau^{(t+1)}}(\boldsymbol{\Theta}_1^{(t+1)}, \boldsymbol{\Theta}_2^{(t+1)}) \leftarrow f_u(\boldsymbol{\Theta}_{1,u}, \boldsymbol{\Theta}_{2,u})$
- 9: $\mathcal{T} \leftarrow \mathcal{T} \cap [\tau^{(t)}, 1]$ if $u > \tau^{(t)}$, else $\mathcal{T} \leftarrow \mathcal{T} \cap [0, \tau^{(t)}]$
- 10: **else**
- 11: $\tau^{(t+1)} \leftarrow \tau^{(t)}$, $\boldsymbol{\Theta}_s^{(t+1)} \leftarrow \boldsymbol{\Theta}_s^{(t)}$, $f_{\tau^{(t+1)}}(\boldsymbol{\Theta}_1^{(t+1)}, \boldsymbol{\Theta}_2^{(t+1)}) \leftarrow f_{\tau^{(t)}}(\boldsymbol{\Theta}_1^{(t)}, \boldsymbol{\Theta}_2^{(t)})$
- 12: $\mathcal{T} \leftarrow \mathcal{T} \cap (0, u]$ if $u > \tau^{(t)}$, else $\mathcal{T} \leftarrow \mathcal{T} \cap [u, 1]$
- 13: **end if**
- 14: $t \leftarrow t + 1$
- 15: **end while**
- 16: $(\hat{\tau}, \hat{\boldsymbol{\Theta}}_1, \hat{\boldsymbol{\Theta}}_2)$ are obtained by applying the robust two-step method in Section 3.1 to the refined candidate set $\mathcal{T}$.

Ensure: $\hat{\tau}, \hat{\boldsymbol{\Theta}}_1, \hat{\boldsymbol{\Theta}}_2$

To address this issue, we propose a robust optimized search method inspired by divide-and-conquer strategies. Instead of scanning all candidates, the algorithm iteratively compares only a small number of candidate change-points and uses the comparison outcome to infer the direction of τ^* , thereby progressively shrinking the candidate set $\mathcal{T}$. The detailed procedure is given in Algorithm 1.

Remark 3.1. The core idea of Algorithm 1 lies in lines 7–13, which implement an iterative shrinking procedure for the candidate set $\mathcal{T}$. The key intuition is that, among the selected probe points, the one with a smaller loss is more likely to be closer to the true change-point. Based on this comparison, the algorithm discards the less promising side of $\mathcal{T}$ and progressively refines the search region.

Next, we analyze the computational complexity of the robust optimized search method. At each iteration, the candidate set is reduced by at least a constant proportion. In particular, the size of the candidate set decreases to at most $(3/4)^t N$ after t iterations. The algorithm terminates once the number of candidates falls below m , which implies that the total number of iterations is at most $O(\log N)$. In each iteration, the truncation step costs $O(N)$, and solving the corresponding sub-problem requires $O\{(d_1 d_2 N + d_1 d_2 d_{\min})/\alpha + N\}$ as discussed in Section 3.1. Therefore, the overall computational complexity of the robust optimized search method is $O\{(d_1 d_2 N \log N + d_1 d_2 d_{\min} \log N)/\alpha + N \log N\}$.

4. Theoretical results

In this section, we establish theoretical guarantees for the proposed methods in Section 3, with two special cases including CS and MR. A key challenge arises when the candidate change-point $\tau \neq \tau^*$, under which the corresponding estimators are no longer centered around Θ_1^* and Θ_2^* due to sample misallocation across segments. As introduced in Section 2, we address this issue by considering the mixed regression matrices $\bar{\Theta}_{1,\tau}$ and $\bar{\Theta}_{2,\tau}$, which serve as the effective targets under a misspecified change-point and will be used throughout the analysis.

4.1. Compressed Sensing

We now present the theoretical guarantees for the proposed methods. We begin with the CS setting under sub-Gaussian designs and introduce the assumptions needed for the analysis.

Assumption 1. The covariates $\{\mathbf{X}_i\}_{i=1}^N$ satisfy that $\{\text{vec}(\mathbf{X}_i)\}_{i=1}^N$ are i.i.d. sub-Gaussian random vectors with $\|\text{vec}(\mathbf{X}_i)\|_{\psi_2} \leq K_0 < \infty$ and $\mathbb{E}\mathbf{X}_i = \mathbf{0}$. Moreover, their covariance matrix satisfies $0 < \underline{\lambda} \leq \lambda_{\min}(\mathbb{E} \text{vec}(\mathbf{X}_i) \text{vec}(\mathbf{X}_i)^\top) \leq \lambda_{\max}(\mathbb{E} \text{vec}(\mathbf{X}_i) \text{vec}(\mathbf{X}_i)^\top) \leq \bar{\lambda} < \infty$. The threshold variables $\{t_i\}_{i=1}^N$ are i.i.d. random variables following the uniform distribution on $[0, 1]$. The noise variables $\{\varepsilon_i\}_{i=1}^N$ satisfy $\mathbb{E}(\varepsilon_i | \mathbf{X}_i) = 0$.

Assumption 2. For all $i = 1, \dots, N$, the response variables satisfy $\mathbb{E}(|Y_i|^{2k}) \leq M < \infty$, for some $k > 2$.

Assumption 3 (Signal-to-noise condition). The signal strength satisfies

$$\xi^* N \|\Delta^*\|_F^2 \geq c_N R_{\max} (d_1 + d_2)^{1-\frac{a}{2}} (\xi^* N)^{\frac{a}{2}},$$

where $\xi^* = \min(\tau^*, 1 - \tau^*)$ denotes the minimal segment proportion, $\Delta^* = \Theta_2^* - \Theta_1^*$ is the jump size, $R_{\max} = \max(R_{1,q}, R_{2,q})$ characterizes the maximal complexity level, $a = 2/k + q$, and $c_N > 0$ is a sufficiently large constant.

These assumptions play distinct roles in the analysis, which we briefly explain below. Assumption 1 imposes a standard sub-Gaussian design condition on the covariates, which ensures suitable concentration properties and is essential for deriving high-probability error bounds. It also assumes that the noise variables have zero conditional mean, as is standard in regression models. To simplify the analysis of segment proportions, the threshold variables are taken to be uniformly distributed on $[0, 1]$. Assumption 2 allows the response variable to be heavy-tailed by only requiring the existence of a finite $2k$ -th moment. This condition is weaker than sub-Gaussian or sub-exponential assumptions and permits the presence of large outliers. We impose the moment condition on Y_i , which facilitates the analysis of the truncation procedure applied to the response variable and allows direct control of the tail behavior of the truncated samples.

Assumption 3 specifies a signal-to-noise condition to ensure identifiability of the change-point. The left-hand side represents the effective signal strength, determined by the jump size $\|\Delta^*\|_F$ and the sample size within the smaller segment. The right-hand side captures the intrinsic difficulty of the problem, reflecting the model complexity and the heavy-tailedness of the data through α . This condition requires the signal to be sufficiently strong relative to this complexity level, and is standard in change-point analysis and high-dimensional estimation.

Based on the above assumptions, we first study the relationship between the estimator $\hat{\Theta}_{s,\tau}$ and the corresponding mixed matrix $\bar{\Theta}_{s,\tau}$. To this end, we define the estimation error with respect to the mixed matrix as $\bar{\Delta}_{s,\tau} = \hat{\Theta}_{s,\tau} - \bar{\Theta}_{s,\tau}$. The following theorem provides high-probability bounds for $\bar{\Delta}_{s,\tau}$ under both the Frobenius norm and the nuclear norm.

Theorem 4.1. Suppose Assumptions 1–3 hold and the true coefficient matrix Θ_s^* belongs to $\mathcal{B}_q(R_{s,q})$. Choose the penalty parameter $\lambda_{s,\tau}$ as $c_\lambda \sqrt{(d_1 + d_2)N_{s,\tau}}$ and choose the truncation parameter $\gamma_{s,\tau} = c_\gamma [MN_{s,\tau}/(d_1 + d_2)]^{1/(2k)}$, where c_λ and c_γ are constants and $N_{s,\tau}$ represents the sample size to the corresponding segments with candidate change-point τ . Then there exists constants C_1 and C_2 , with probability

at least $1 - C_1 \exp\{-C_2(d_1 + d_2)\}$, we have

$$\|\bar{\Delta}_{s,\tau}\|_F^2 \lesssim \frac{R_{\max}(d_1 + d_2)^{1-q/2}}{N_{s,\tau}^{1-q/2}}, \quad \text{and} \quad \|\bar{\Delta}_{s,\tau}\|_*^2 \lesssim \frac{R_{\max}^2(d_1 + d_2)^{1-q}}{N_{s,\tau}^{1-q}}. \quad (4.1)$$

Theorem 4.1 establishes non-asymptotic error bounds for the estimator around the mixed matrix $\bar{\Theta}_{s,\tau}$ with fixed τ . The result shows that, even under heavy-tailed responses, the estimator achieves a convergence rate comparable to that in standard low-rank estimation problems with the effective sample size $N_{s,\tau}$ (see, e.g., Corollary 5 in [Negahban and Wainwright \(2011\)](#)). The error bounds depend on the complexity parameter $R_{\max}$ and the dimensions $(d_1 + d_2)$, which capture the overall scale of the problem, while the parameter q governs the deviation from exact low-rankness. As q increases, the recovery problem becomes more challenging and the rate correspondingly deteriorates. In the special case $q = 0$, the condition reduces to exact low-rankness, and the bounds recover the corresponding rates for exact low-rank matrix estimation.

Importantly, the theorem justifies the use of the mixed matrix as the target under a misspecified change-point. It shows that $\hat{\Theta}_{s,\tau}$ concentrates around $\bar{\Theta}_{s,\tau}$ rather than the true coefficient matrix, providing a key step for analyzing the change-point estimation error in subsequent results. Building upon the error bounds with respect to the mixed matrices in Theorem 4.1, we now establish estimation error bounds for the true coefficient matrices and the change-point.

Theorem 4.2. Under the same conditions as in Theorem 4.1, there exist positive constants C_1 and C_2 such that the estimators $\hat{\Theta}_s$ and $\hat{\tau}$ obtained from the robust two-step method satisfy, with probability at least $1 - C_1 \exp -C_2(d_1 + d_2)$,

$$\left\| \hat{\Theta}_s - \Theta_s^* \right\|_F^2 \lesssim R_{\max} \left(\frac{d_1 + d_2}{\xi^* N} \right)^{1 - \frac{q}{2} \vee \frac{1}{k}}, \quad \left\| \hat{\Theta}_s - \Theta_s^* \right\|_*^2 \lesssim R_{\max}^2 \left(\frac{d_1 + d_2}{\xi^* N} \right)^{1 - q \vee \frac{2}{k}},$$

and

$$|\hat{\tau} - \tau^*| \lesssim \frac{R_{\max}}{\|\Delta^*\|_F^2} \left(\frac{d_1 + d_2}{N} \right)^{1 - \frac{q}{2} \vee \frac{1}{k}}.$$

Theorem 4.2 provides non-asymptotic error bounds for both the coefficient matrices and the change-point estimator. For the coefficient matrices, the estimation error is governed by the effective sample size $\xi^* N$, where $\xi^* = \min(\tau^*, 1 - \tau^*)$ denotes the minimal segment proportion. This quantity reflects the number of observations correctly assigned under the true change-point and determines the scale of the estimation error.

The convergence rate further depends on both the near low-rank structure and the heavy-tailedness of the response. Specifically, the exponent $1 - \frac{q}{2} \vee \frac{1}{k}$ captures the joint effect of the low-rank complexity parameter q and the moment condition parameter k . As q increases, the underlying structure deviates further from exact

low-rankness and the rate deteriorates, while as k increases, the tail condition becomes lighter and the rate improves. In the special case $q = 0$, the model reduces to exact low-rankness, and the bound recovers the corresponding rate up to the effect of heavy-tailedness; as $k \rightarrow \infty$, the rate matches that under sub-Gaussian conditions with no change-point (see, e.g., Corollary 5 in [Negahban and Wainwright \(2011\)](#)).

For the change-point estimator, the error bound is inversely proportional to the jump size $\|\Delta^*\|_F^2$, indicating that larger separation between the two segments leads to more accurate localization. Moreover, the rate inherits the same dependence on q and k , showing that reliable change-point detection requires both sufficient signal strength and accurate estimation of the segment-wise coefficients.

We now turn to the theoretical properties of the robust optimized search method. The following result establishes the estimation error bound for the change-point under the optimized search scheme.

Theorem 4.3. Under the same conditions as in Theorem 4.1, there exist positive constants C_1 – C_3 such that the estimator $\hat{\tau}$ obtained from the robust optimized search method satisfy, with probability at least $1 - C_1 \exp\{-C_2(d_1 + d_2)\} - C_3(d_1 + d_2)^{-k}$,

$$|\hat{\tau} - \tau^*| \lesssim \frac{R_{\max}}{\|\Delta^*\|_F^2} \left(\frac{d_1 + d_2}{N} \right)^{1 - \frac{q}{2} \vee \frac{1}{k}}.$$

Theorem 4.3 shows that the proposed robust optimized search method achieves the same convergence rate for the change-point estimator as the robust two-step method. Compared with Theorem 4.2, the main difference lies in the form of the probability bound. In addition to the exponential term, the bound now involves an extra polynomial term of order $(d_1 + d_2)^{-k}$, which arises from the truncation nature of the data and the iterative search procedure.

Remark 4.4. The error bounds for the coefficient matrices under the robust optimized search method can be obtained by following the same argument as in Theorem 4.2, once the change-point estimation error is properly controlled. We omit the details here, as they directly follow from Step 2 of the proof for Theorem 4.2 in Appendix [Appendix D.2](#).

4.2. Multiresponse Regression

We next turn to the theoretical analysis under the MR setting. We state the assumptions below, which ensure the identifiability of the model as well as the concentration behavior required for establishing theoretical guarantees.

Assumption 4. The covariates $\{\mathbf{x}_i\}_{i=1}^n$ are i.i.d. sub-Gaussian random vectors satisfying $\|\mathbf{x}_i\|_{\psi_2} \leq K_0 < \infty$, $\mathbb{E}\mathbf{x}_i = \mathbf{0}$, and $0 < \underline{\lambda} \leq \lambda_{\min}(\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]) \leq \lambda_{\max}(\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top]) \leq \bar{\lambda} < \infty$. The threshold variables $\{t_i\}_{i=1}^n$ are i.i.d.

random variables following the uniform distribution on $[0, 1]$. The error vectors $\{\varepsilon_i\}_{i=1}^n$ satisfy $\mathbb{E}(\varepsilon_i \mid \mathbf{x}_i) = \mathbf{0}$.

Assumption 5. For all $i = 1, \dots, n$, the response variables satisfy $\lambda_{\max}(\mathbb{E}(\mathbf{y}_i \mathbf{y}_i^\top)) \leq R < \infty$. Moreover, for all $j_1, j_2 = 1, \dots, d_2$ with $j_1 \neq j_2$, $\varepsilon_{ij_1} \perp\!\!\!\perp \varepsilon_{ij_2} \mid \mathbf{x}_i$, and for all $j = 1, \dots, d_2$, $\mathbb{E}(\varepsilon_{ij}^{2k} \mid \mathbf{x}_i) \leq M < \infty$, for some $k > 2$.

Assumption 6 (Signal-to-noise condition). The signal strength is assumed to satisfy

$$\xi^* n \|\Delta^*\|_F^2 \geq c_n R_{\max} (d \log d)^{1-\frac{a}{2}} (\xi^* n)^{\frac{a}{2}},$$

where $d = d_1 + d_2$, $\xi^* = \min(\tau^*, 1 - \tau^*)$ denotes the minimal segment proportion, $\Delta^* = \Theta_2^* - \Theta_1^*$ is the jump size, $R_{\max} = \max(R_{1,q}, R_{2,q})$ characterizes the maximal complexity level, $a = 2/k + q$, and $c_n > 0$ is a sufficiently large constant.

Assumptions 4–6 largely parallel those in the CS setting and are stated here for completeness, with appropriate modifications to accommodate the multiresponse structure. Based on these conditions, we study the bias $\bar{\Delta}_{s,\tau} = \hat{\Theta}_{s,\tau} - \bar{\Theta}_{s,\tau}$ under MR setting first.

Theorem 4.5. Suppose Assumptions 4–6 hold, and the true coefficient matrix Θ_s^* belongs to $\mathcal{B}_q(R_{s,q})$. Let the penalty parameter be chosen as $\lambda_{s,\tau} = c_\lambda \sqrt{(d \log d) n_{s,\tau}}$, and the truncation parameter as $\gamma_{s,\tau} = c_\gamma \sqrt{d_2} [n_{s,\tau} / (d \log d)]^{1/(2k)}$, where c_λ and c_γ are positive constants, and $n_{s,\tau}$ denotes the sample size of the corresponding segment under the candidate change-point τ . Then there exist constants C_1 – C_4 such that, with probability at least $1 - C_1 d^{-C_2} - C_3 \exp\{-C_4 d\}$, we have

$$\|\bar{\Delta}_{s,\tau}\|_F^2 \lesssim \frac{R_{\max} (d \log d)^{1-q/2}}{n_{s,\tau}^{1-q/2}}, \quad \text{and} \quad \|\bar{\Delta}_{s,\tau}\|_*^2 \lesssim \frac{R_{\max}^2 (d \log d)^{1-q}}{n_{s,\tau}^{1-q}}. \quad (4.2)$$

The choice of the truncation parameter $\gamma_{s,\tau}$ differs from that in the CS setting by an additional factor $\sqrt{d_2}$. This is because, in the multiresponse setting, truncation is applied to $\|\mathbf{y}_i\|_2$, whose typical magnitude is of order $\sqrt{d_2}$ for $\mathbf{y}_i \in \mathbb{R}^{d_2}$.

Theorem 4.5 shows that, for a fixed candidate change-point τ , the estimator $\hat{\Theta}_{s,\tau}$ is centered around the mixed matrix $\bar{\Theta}_{s,\tau}$ rather than the true coefficient Θ_s^* . Moreover, the theorem provides upper bounds for this deviation, which match the estimation rates in the no-change-point setting (see, e.g., Theorem 5 in Fan et al. (2021)). Building on this result, we further analyze the bias of the change-point estimator $\hat{\tau}$ and the corresponding coefficient estimators $\hat{\Theta}_s$, leading to Theorem 4.6.

Theorem 4.6. Under the same conditions as in Theorem 4.5, there exist positive constants C_1 – C_4 such that the estimators $\hat{\Theta}_s$ and $\hat{\tau}$ obtained from the robust two-step method satisfy, with probability at least

$$1 - C_1 d^{-C_2} - C_3 \exp\{-C_4 d\},$$

$$\left\| \widehat{\Theta}_s - \Theta_s^* \right\|_F^2 \lesssim R_{\max} \left(\frac{d \log d}{\xi^* n} \right)^{1 - \frac{q}{2} \vee \frac{1}{k}}, \quad \left\| \widehat{\Theta}_s - \Theta_s^* \right\|_*^2 \lesssim R_{\max}^2 \left(\frac{d \log d}{\xi^* n} \right)^{1 - q \vee \frac{2}{k}},$$

and

$$|\widehat{\tau} - \tau^*| \lesssim \frac{R_{\max}}{\|\Delta^*\|_F^2} \left(\frac{d \log d}{n} \right)^{1 - \frac{q}{2} \vee \frac{1}{k}}.$$

The rates in Theorem 4.6 are comparable to those in existing results for low-rank change-point problems (see, e.g., Theorem 11 in Shi et al. (2024)), up to an additional $\log d$ factor. This logarithmic term arises from handling heavy-tailed responses and reflects the price paid for achieving robustness under weaker moment conditions. For completeness, we also present the theoretical results for the robust optimized search method under MR setting, as stated in Theorem 4.7.

Theorem 4.7. Under the same conditions as in Theorem 4.5, there exist positive constants C_1 – C_5 such that the estimator $\widehat{\tau}$ obtained from the robust optimized search method satisfy, with probability at least $1 - C_1 d^{-C_2} - C_3 (d \log d)^{-(k-1)} - C_4 \exp\{-C_5 d\}$,

$$|\widehat{\tau} - \tau^*| \lesssim \frac{R_{\max}}{\|\Delta^*\|_F^2} \left(\frac{d \log d}{n} \right)^{1 - \frac{q}{2} \vee \frac{1}{k}}.$$

Theorem 4.7 shows that the robust optimized search method achieves the same localization rate as the robust two-step estimator in Theorem 4.6. In particular, the error bound inherits the same dependence on the model complexity $R_{\max}$, the jump size $\|\Delta^*\|_F^2$, and the parameters (q, k) that capture the near low-rank structure and heavy-tailedness. Compared with the two-step procedure, the optimized search method avoids exhaustive evaluation over all candidate change-points, leading to a substantial reduction in computational cost while achieving the same statistical accuracy.

5. Robust Methodology for Multiple Change Points Detection

So far, we have focused on the single change-point scenario under heavy-tailed settings, and established both computational properties and statistical guarantees of the proposed robust methods. More generally, multiple change-points may be present. A natural extension of our framework is to embed the proposed single change-point procedures into a recursive segmentation scheme. In particular, we adopt the Wild Binary Segmentation (WBS) framework (Fryzlewicz, 2014), which is widely used for multiple change-point detection.

The key idea is to treat our robust single change-point methods as a single change-point solver within WBS. By applying the single change-point algorithm on randomly sampled sub-intervals and recursively partitioning the data, we obtain a computationally efficient approach for detecting multiple change-points under heavy-tailed observations. The detailed procedure is summarized in Algorithm 2.

Algorithm 2 WBS-based Multiple Change-Point Detection

Require: Data $\{(Y_i, \mathbf{X}_i, t_i)\}_{i=1}^N$, single change-point solver SINGLECP, number of intervals M , threshold ζ , minimal length δ

Ensure: Estimated change-points $\{\hat{\tau}\}$

```

1: Draw  $M$  random intervals  $\{(L_m, R_m]\}_{m=1}^M \subset [0, 1]$  such that  $R_m - L_m \geq \delta$ 
2: function SEG( $L, R$ )
3:   if  $R - L < \delta$  then
4:     return
5:   end if
6:   for  $m = 1, \dots, M$  with  $(L_m, R_m] \subset (L, R]$  do
7:      $\mathcal{I}_m = \{i : L_m < t_i \leq R_m\}$ 
8:      $(\hat{\tau}_m, \hat{\Theta}_{1,m}, \hat{\Theta}_{2,m}) \leftarrow \text{SINGLECP}(\mathcal{I}_m)$ 
9:     Compute gain function  $G_m$ ; set score  $T_m = G_m/|\mathcal{I}_m|$ 
10:  end for
11:   $\hat{m} = \arg \max_m T_m$ ,  $\hat{\tau} = \hat{\tau}_{\hat{m}}$ 
12:  if  $T_{\hat{m}} \leq \zeta$  then
13:    return
14:  end if
15:  Record  $\hat{\tau}$ 
16:  SEG( $L, \hat{\tau}$ ); SEG( $\hat{\tau}, R$ )
17: end function
18: SEG( $0, 1$ )
19: return sorted  $\{\hat{\tau}\}$ 

```

In Algorithm 2, the gain function G_m is defined as

$$G_m = \min \left\{ f_{\hat{\tau}_m}(\hat{\Theta}_{1,m}, \hat{\Theta}_{1,m}), f_{\hat{\tau}_m}(\hat{\Theta}_{2,m}, \hat{\Theta}_{2,m}) \right\} - f_{\hat{\tau}_m}(\hat{\Theta}_{1,m}, \hat{\Theta}_{2,m}),$$

where the terms involved are computed over the sub-interval $(L_m, R_m]$ using the truncated responses $\tilde{Y}_i(\hat{\tau}_m)$. Define the index sets $\mathcal{I}_m = \{i \mid L_m < t_i \leq R_m\}$, $\mathcal{I}_{1,\tau}(m) = \{i \mid L_m < t_i \leq \tau\}$, and $\mathcal{I}_{2,\tau}(m) = \{i \mid \tau < t_i \leq R_m\}$. For $s \in \{1, 2\}$, we have

$$f_{\hat{\tau}_m}(\hat{\Theta}_{s,m}, \hat{\Theta}_{s,m}) = \sum_{i \in \mathcal{I}_m} \left[-2 \langle \mathbf{X}_i, \hat{\Theta}_{s,m} \rangle \tilde{Y}_i(\hat{\tau}_m) + \langle \mathbf{X}_i, \hat{\Theta}_{s,m} \rangle^2 \right],$$

and

$$f_{\hat{\tau}_m}(\hat{\Theta}_{1,m}, \hat{\Theta}_{2,m}) = \sum_{s=1}^2 \sum_{i \in \mathcal{I}_{s,\hat{\tau}_m}(m)} \left[-2 \langle \mathbf{X}_i, \hat{\Theta}_{s,m} \rangle \tilde{Y}_i(\hat{\tau}_m) + \langle \mathbf{X}_i, \hat{\Theta}_{s,m} \rangle^2 \right].$$

Intuitively, a standard gain function compares the loss under a no-change model, obtained by fitting a single coefficient matrix over the entire interval, with that under a one-change model. However, recomputing the no-change estimator for each candidate interval would incur a substantial computational cost.

To alleviate this issue, we approximate the no-change loss by $\min\{f_{\hat{\tau}_m}(\hat{\Theta}_{1,m}, \hat{\Theta}_{1,m}), f_{\hat{\tau}_m}(\hat{\Theta}_{2,m}, \hat{\Theta}_{2,m})\}$, which reuses the segment-wise estimators and avoids recomputing a global estimator for each interval. When no change-point is present, $\hat{\Theta}_{1,m}$ and $\hat{\Theta}_{2,m}$ are expected to be close, so that this approximation provides a reliable surrogate for the true no-change loss. When a change-point is present, the two estimators capture different regimes, and the above minimum effectively selects the one that is closer to the overall structure of the data on $(L_m, R_m]$, thereby still serving as a reasonable proxy for the no-change loss while maintaining computational efficiency.

Thus, G_m measures the reduction in loss achieved by introducing a change-point within the sub-interval $(L_m, R_m]$. Since the lengths of sub-intervals may vary, we further normalize the gain by the number of samples in the sub-interval and define the score $T_m = G_m/|\mathcal{I}_m|$, which allows for a fair comparison across different intervals.

6. Simulation and Real Data Application

In this section, we evaluate the performance of the proposed methods through both simulation studies and real data applications. The experiments are designed to evaluate the performance of change-point localization and coefficient matrix estimation across different noise distributions, including heavy-tailed settings.

6.1. Single Change-Point

We consider a single change-point setting under the CS and MR setting, where the regression model follows a piecewise low-rank structure indexed by a threshold variable $t_i \in (0, 1]$. The observations are generated according to a two-segment model with a single change-point at $\tau^* = 0.5$. Specifically, the samples are divided into two segments $(0 < t_i \leq \tau)$ and $(\tau < t_i \leq 1)$. The threshold variable is taken as a deterministic grid, i.e., $t_i = i/N$ for $i = 1, \dots, N$, ensuring evenly spaced observations across the unit interval.

We set $d_1 = d_2 = d$ and consider rank r^* . The coefficient matrices are generated via a low-rank SVD form $\Theta_s^* = \mathbf{U}_s \mathbf{V}_s^\top / \sqrt{r^*}$, $s = 1, 2$, where $\mathbf{U}_s, \mathbf{V}_s \in \mathbb{R}^{d \times r^*}$ have orthonormal columns obtained from the left and right singular vectors of Gaussian random matrices. To control the change magnitude, we consider two settings. In the *large jump* case, $(\mathbf{U}_1, \mathbf{V}_1)$ and $(\mathbf{U}_2, \mathbf{V}_2)$ are generated independently, leading to a pronounced structural change. In the *small jump* case, we construct $(\mathbf{U}_2, \mathbf{V}_2)$ by keeping the first $k = 2$ singular directions

the same as $(\mathbf{U}_1, \mathbf{V}_1)$ and regenerating the remaining $r^* - k$ columns independently, so that only part of the singular structure is changed.

For the CS design matrix, each covariate matrix $X_i \in \mathbb{R}^{d_1 \times d_2}$ is generated independently with i.i.d. standard Gaussian entries. The noise ε_i is generated under three different settings to reflect both light-tailed and heavy-tailed behaviors:

- **Gaussian:** $\varepsilon_i \sim \mathcal{N}(0, 0.04)$, serving as a light-tailed benchmark;
- **Lognormal:** $\varepsilon_i = (\exp(Z_i) - \exp(6))/800$, where $Z_i \sim \mathcal{N}(0, 12)$, leading to a right-skewed and heavy-tailed distribution;
- **Cauchy:** $\varepsilon_i = Z_i/7$, where Z_i follows standard Cauchy distribution, representing an extremely heavy-tailed case.

Compared with the Gaussian noise, the latter two settings exhibit significantly heavier tails and are used to assess the robustness of the proposed method. For the hyper-parameters, we set $c_\lambda = 1.5$ and $c_\gamma = 2.3$, corresponding to the scaling constants in the regularization parameter and the truncation level, respectively.

Under Cauchy noise, we first consider a low-dimensional setting with $d = 6$, $r^* = 2$ and $N = 500$ to visualize the estimation performance. We compare the true matrix Θ_1^* , the estimator $\hat{\Theta}_1$ from the proposed robust optimized search method, and a direct estimator $\hat{\Theta}$ based on truncated observations, similar to (Fan et al., 2021), defined as

$$\hat{\Theta} = \arg \min_{\Theta} \sum_{i=1}^N \left(-2 \langle \tilde{Y}_i \mathbf{X}_i, \Theta \rangle + \langle \mathbf{X}_i, \Theta \rangle^2 \right) + \lambda_N \|\Theta\|_*,$$

where $\tilde{Y}_i$ is truncated by γ_N , $\gamma_N = c_\gamma (N/d)^{1/(2k)}$ and $\lambda_N = c_\lambda \sqrt{Nd}$. Figure 3 shows that the proposed estimator better recovers the low-rank structure under heavy-tailed noise. The estimated matrix $\hat{\Theta}_1$ largely matches the true matrix Θ_1^* in both sign pattern and magnitude, whereas the direct estimator $\hat{\Theta}$ exhibits noticeable distortion.

We then consider a higher-dimensional setting with $d = 20$, $r^* = 5$ and $N = 500$. Table 1 reports the performance under different noise distributions and signal strengths, where RT, RO, and Base denote the robust two-step method, the robust optimized search method, and the standard two-step method (Shi et al., 2024), respectively.

Under Gaussian noise, all methods achieve comparable performance in terms of coefficient estimation and change-point localization. In particular, the Frobenius norm error, nuclear norm error, and change-point estimation error are all well controlled. Meanwhile, RO significantly reduces the computational cost,

Table 1: Method comparison under the CS setting across different noise distributions and signal strengths.

Method	Signal	$ \hat{\tau} - \tau^* $	$\ \hat{\Theta}_1 - \Theta_1^*\ _F$	$\ \hat{\Theta}_1 - \Theta_1^*\ _*$	$\text{rank}(\hat{\Theta}_1)$	$\ \hat{\Theta}_2 - \Theta_2^*\ _F$	$\ \hat{\Theta}_2 - \Theta_2^*\ _*$	$\text{rank}(\hat{\Theta}_2)$	Time Cost (s)
Gaussian									
RT	large	0.004 (0.002)	0.619 (0.039)	1.892 (0.057)	5.020 (0.247)	0.605 (0.060)	1.873 (0.092)	5.040 (0.198)	15.493
	small	0.006 (0.005)	0.625 (0.045)	1.900 (0.070)	5.000 (0.286)	0.615 (0.054)	1.887 (0.074)	5.020 (0.247)	15.005
RO	large	0.004 (0.002)	0.619 (0.039)	1.892 (0.057)	5.020 (0.247)	0.605 (0.060)	1.873 (0.092)	5.040 (0.198)	2.017
	small	0.006 (0.006)	0.626 (0.046)	1.900 (0.071)	5.000 (0.286)	0.615 (0.054)	1.887 (0.074)	5.020 (0.247)	1.862
Base	large	0.004 (0.002)	0.619 (0.039)	1.892 (0.057)	5.020 (0.247)	0.605 (0.060)	1.873 (0.092)	5.040 (0.198)	15.485
	small	0.006 (0.005)	0.625 (0.046)	1.899 (0.071)	5.000 (0.286)	0.615 (0.054)	1.887 (0.075)	5.020 (0.247)	15.002
Lognormal									
RT	large	0.007 (0.009)	0.650 (0.042)	2.054 (0.085)	5.740 (0.600)	0.642 (0.049)	2.046 (0.091)	5.740 (0.633)	14.751
	small	0.019 (0.028)	0.660 (0.059)	2.086 (0.113)	6.000 (0.639)	0.652 (0.052)	2.063 (0.088)	5.740 (0.527)	14.351
RO	large	0.007 (0.006)	0.650 (0.042)	2.054 (0.084)	5.740 (0.600)	0.641 (0.048)	2.046 (0.089)	5.760 (0.625)	1.935
	small	0.019 (0.022)	0.661 (0.061)	2.087 (0.116)	6.000 (0.639)	0.653 (0.054)	2.065 (0.092)	5.740 (0.527)	1.966
Base	large	0.106 (0.097)	153.576 (747.216)	15.556 (43.195)	9.820 (3.652)	40.013 (197.082)	9.039 (21.227)	8.940 (3.542)	18.580
	small	0.142 (0.100)	108.109 (375.253)	18.286 (33.845)	11.000 (4.295)	62.087 (317.596)	12.263 (26.087)	10.500 (3.934)	19.627
Cauchy									
RT	large	0.011 (0.015)	0.673 (0.057)	2.162 (0.124)	6.400 (0.700)	0.687 (0.060)	2.196 (0.106)	6.480 (0.677)	15.221
	small	0.030 (0.037)	0.685 (0.055)	2.187 (0.118)	6.160 (0.766)	0.693 (0.065)	2.204 (0.138)	6.320 (0.713)	15.126
RO	large	0.011 (0.014)	0.672 (0.057)	2.163 (0.124)	6.400 (0.700)	0.687 (0.060)	2.193 (0.107)	6.420 (0.731)	2.697
	small	0.029 (0.036)	0.686 (0.056)	2.188 (0.118)	6.160 (0.738)	0.692 (0.063)	2.202 (0.131)	6.340 (0.717)	2.652
Base	large	0.119 (0.091)	149.184 (538.411)	20.874 (39.578)	11.840 (3.678)	293.711 (895.136)	27.956 (57.495)	11.720 (3.817)	20.883
	small	0.142 (0.109)	787.091 (3017.334)	38.551 (98.176)	11.560 (3.845)	41.048 (130.109)	13.004 (19.193)	11.780 (3.183)	20.529

Results are based on 50 replications. Each entry reports the mean, with standard deviation given in parentheses.

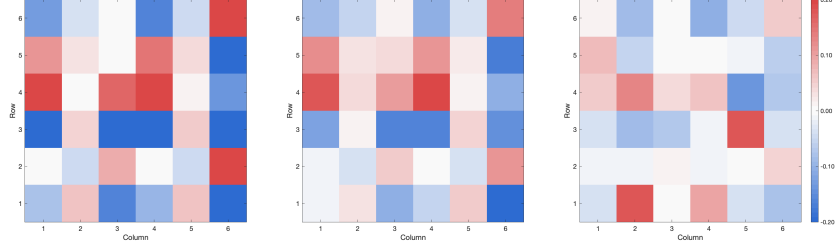


Figure 3: Heatmap comparison of coefficient matrices under Cauchy noise. Left: true coefficient matrix Θ_1^* . Middle: estimator $\hat{\Theta}_1$ obtained by the proposed robust optimized search method. Right: direct estimator $\hat{\Theta}$ based on truncated observations without accounting for the change-point.

being nearly an order of magnitude faster than both Base and RT. Under heavy-tailed noise (Lognormal and Cauchy), the advantage of the robust methods becomes evident. The Base method suffers from severe performance degradation, with substantially inflated coefficient estimation errors and noticeably less accurate change-point estimation, especially under Cauchy noise due to its extreme tail behavior. In contrast, both RT and RO remain stable across all noise settings, yielding reliable coefficient recovery and accurate change-point localization. Moreover, RO preserves a clear computational advantage over RT with substantially reduced running time.

For the MR setting, the covariates $\{\mathbf{x}_i\}_{i=1}^N$ are generated independently from $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. The noise vectors $\varepsilon_i \in \mathbb{R}^d$ are generated under three different settings to reflect both light-tailed and heavy-tailed behaviors:

- **Gaussian:** $\varepsilon_i \sim \mathcal{N}(\mathbf{0}, 0.04 \mathbf{I}_d)$;
- **Lognormal:** $\varepsilon_i = (\exp(\mathbf{z}_i) - \exp(9/2)\mathbf{1})/500$, where $\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, 9 \mathbf{I}_d)$ and the exponential is applied entry-wise;
- **Cauchy:** $\varepsilon_i = \mathbf{z}_i/22$, where $\mathbf{z}_i$ has i.i.d. standard Cauchy entries.

For the hyper-parameters, we set $c_\lambda = 1.5$ and $c_\gamma = 2.5$, corresponding to the scaling constants in the regularization parameter and the truncation level, respectively. Table 2 reports the estimation performance in the MR setting. Under Gaussian noise, all methods achieve comparable estimation accuracy, while RO consistently exhibits a clear computational advantage over both Base and RT.

When the noise becomes heavy-tailed, the differences across methods become more pronounced. The standard two-step method (Base) is highly sensitive to heavy-tailed disturbances, leading to substantially inflated estimation errors, especially under Cauchy noise where extreme observations are more frequent in the multivariate setting. In contrast, both RT and RO maintain stable performance across all noise distributions, indicating that the truncation mechanism effectively mitigates the impact of heavy-tailed

noise in high dimensions. Moreover, RO achieves comparable estimation accuracy to RT while requiring significantly less computational time, further demonstrating its efficiency advantage.

6.2. Multiple Change-Points Detection

We further evaluate the proposed method in a multiple change-point setting. To address this scenario, we employ the robust two-step method or robust optimized search method as the single change-point solver within the segmentation procedure. Specifically, we consider three true change-points located at $\{0.25, 0.5, 0.75\}$. The coefficient matrices across segments are constructed to be low-rank with rank $r^* = 5$, while the differences between adjacent segments are designed to have relatively large jump sizes, following the same construction as in the single change-point setting to ensure clear separation between regimes.

We conduct simulations under both CS and MR settings, with the same design generation mechanism as described in the single change-point experiments. For the CS setting, the design matrices $\{\mathbf{X}_i\}_{i=1}^N$ are generated independently with entries drawn from a standard Gaussian distribution. We set the matrix dimensions to $d_1 = d_2 = 20$ and the sample size to $N = 1000$. To assess robustness, we consider both light-tailed and heavy-tailed noise distributions. Specifically, the noise ε_i is generated as

- **Gaussian:** $\varepsilon_i \sim \mathcal{N}(0, 0.01)$;
- **Lognormal:** $\varepsilon_i = (\exp(Z_i) - \exp(7/2))/500$, where $Z_i \sim \mathcal{N}(0, 7)$;
- **Cauchy:** $\varepsilon_i = Z_i/50$, where Z_i follows the standard Cauchy distribution.

For the MR setting, the covariates $\{\mathbf{x}_i\}_{i=1}^N$ are generated independently from a multivariate normal distribution, and the responses are constructed according to the corresponding matrix regression model. We set $d_1 = d_2 = 30$ and the sample size to $N = 2000$. The noise vectors $\varepsilon_i \in \mathbb{R}^{d_2}$ are generated with i.i.d. entries following the same distributions as in the CS setting.

Table 3 reports the estimation performance under different noise distributions for both CS and MR settings. The proposed method accurately recovers the number of change-points across all scenarios, with $\hat{s}$ consistently close to the true value. Here $d_H(\cdot, \cdot)$ denotes the Hausdorff distance. The localization error $d_H(\hat{\tau}, \tau^*)$ remains small across all noise settings, indicating the robustness of the proposed method. A similar trend is observed for the coefficient estimation error $d_H(\hat{\Theta}, \Theta^*)$, while the overall performance remains stable. In terms of rank recovery, we report the maximum and minimum estimated ranks across segments, denoted by $\hat{r}_{\max}$ and $\hat{r}_{\min}$, respectively. The estimate $\hat{r}_{\min}$ is close to the true rank across all settings, whereas $\hat{r}_{\max}$ tends to be slightly inflated under heavy-tailed noise. The results show that the proposed method performs well in both CS and MR settings, achieving accurate estimation across different noise conditions.

Table 2: Method comparison under the MR setting across different noise distributions and signal strengths.

Method	Signal	$ \hat{\tau} - \tau^* $	$\ \hat{\Theta}_1 - \Theta_1^*\ _F$	$\ \hat{\Theta}_1 - \Theta_1^*\ _*$	$\text{rank}(\hat{\Theta}_1)$	$\ \hat{\Theta}_2 - \Theta_2^*\ _F$	$\ \hat{\Theta}_2 - \Theta_2^*\ _*$	$\text{rank}(\hat{\Theta}_2)$	Time Cost (s)
Gaussian									
RT	large	0.003 (0.000)	0.465 (0.029)	1.537 (0.045)	5.000 (0.000)	0.462 (0.034)	1.531 (0.054)	5.000 (0.000)	7.492
	small	0.003 (0.000)	0.458 (0.033)	1.526 (0.052)	5.000 (0.000)	0.462 (0.030)	1.532 (0.048)	5.000 (0.000)	7.395
RO	large	0.003 (0.000)	0.465 (0.029)	1.537 (0.045)	5.000 (0.000)	0.462 (0.034)	1.531 (0.054)	5.000 (0.000)	1.361
	small	0.003 (0.000)	0.458 (0.033)	1.526 (0.052)	5.000 (0.000)	0.462 (0.030)	1.532 (0.048)	5.000 (0.000)	1.336
Base	large	0.003 (0.000)	0.465 (0.029)	1.537 (0.045)	5.000 (0.000)	0.462 (0.034)	1.531 (0.054)	5.000 (0.000)	7.510
	small	0.003 (0.000)	0.458 (0.033)	1.526 (0.052)	5.000 (0.000)	0.462 (0.030)	1.532 (0.048)	5.000 (0.000)	7.398
Lognormal									
RT	large	0.004 (0.003)	0.675 (0.063)	2.017 (0.098)	5.420 (0.538)	0.675 (0.072)	2.021 (0.127)	5.560 (0.577)	7.900
	small	0.013 (0.020)	0.687 (0.063)	2.052 (0.097)	5.720 (0.573)	0.655 (0.057)	1.990 (0.086)	5.560 (0.644)	8.120
RO	large	0.005 (0.006)	0.675 (0.062)	2.018 (0.098)	5.400 (0.495)	0.676 (0.074)	2.022 (0.128)	5.580 (0.609)	1.406
	small	0.016 (0.037)	0.689 (0.063)	2.054 (0.097)	5.720 (0.573)	0.657 (0.058)	1.995 (0.096)	5.580 (0.673)	1.435
Base	large	0.215 (0.091)	148.982 (428.341)	13.445 (13.896)	10.260 (2.709)	4851.788 (31838.089)	24.729 (70.144)	10.600 (2.231)	9.437
	small	0.242 (0.080)	321.664 (1184.844)	16.292 (17.912)	10.440 (1.929)	102.428 (226.959)	12.216 (10.366)	10.920 (2.389)	9.758
Cauchy									
RT	large	0.008 (0.009)	0.745 (0.067)	2.161 (0.106)	5.920 (0.601)	0.733 (0.071)	2.149 (0.110)	6.160 (0.650)	8.154
	small	0.018 (0.022)	0.729 (0.062)	2.131 (0.104)	5.960 (0.669)	0.734 (0.062)	2.155 (0.095)	6.060 (0.740)	7.969
RO	large	0.008 (0.009)	0.745 (0.067)	2.161 (0.106)	5.920 (0.601)	0.733 (0.071)	2.149 (0.110)	6.160 (0.650)	1.446
	small	0.018 (0.023)	0.727 (0.064)	2.127 (0.104)	5.940 (0.652)	0.734 (0.061)	2.156 (0.095)	6.060 (0.740)	1.400
Base	large	0.238 (0.086)	9688.381 (36101.567)	62.292 (101.704)	12.200 (2.814)	10232.682 (40468.425)	57.216 (93.711)	12.260 (2.834)	9.989
	small	0.217 (0.096)	3764.597 (13571.461)	45.657 (69.461)	12.040 (2.249)	16324.572 (76071.820)	59.129 (122.232)	12.340 (2.715)	9.795

Results are based on 50 replications. Each entry reports the mean, with standard deviation given in parentheses.

Table 3: Performance under multiple change-points with different noise types.

Model	Noise	Method	$\hat{s}$	$d_H(\hat{\tau}, \tau^*)$	$d_H(\hat{\Theta}, \Theta^*)$	$\hat{r}_{\max}$	$\hat{r}_{\min}$	Time Cost (s)
CS	Gaussian	WBS-RT	3.000 (0.000)	0.007 (0.006)	0.821 (0.023)	7.600	5.000	1939.849
		WBS-RO	3.020 (0.141)	0.009 (0.012)	0.825 (0.031)	8.460	5.000	325.019
	Lognormal	WBS-RT	3.020 (0.141)	0.009 (0.018)	0.826 (0.029)	7.360	5.000	1904.696
		WBS-RO	3.020 (0.141)	0.008 (0.011)	0.826 (0.029)	7.080	5.020	320.805
	Cauchy	WBS-RT	3.060 (0.240)	0.014 (0.028)	0.833 (0.039)	8.280	5.000	1681.604
		WBS-RO	3.080 (0.274)	0.019 (0.036)	0.836 (0.043)	8.240	5.000	288.893
MR	Gaussian	WBS-RT	3.000 (0.000)	0.001 (0.000)	0.324 (0.007)	5.000	5.000	1304.539
		WBS-RO	3.000 (0.000)	0.001 (0.000)	0.628 (0.012)	5.000	5.000	205.070
	Lognormal	WBS-RT	3.000 (0.000)	0.002 (0.003)	0.517 (0.033)	10.740	8.320	1298.571
		WBS-RO	3.040 (0.198)	0.009 (0.036)	0.699 (0.059)	6.560	5.100	213.605
	Cauchy	WBS-RT	3.180 (0.388)	0.026 (0.041)	0.698 (0.077)	13.340	11.040	1401.701
		WBS-RO	3.640 (0.631)	0.088 (0.077)	0.884 (0.102)	10.860	6.220	245.847

Results are based on 50 replications. Each entry reports the mean, with standard deviation given in parentheses when applicable. WBS-RT and WBS-RO denote the WBS procedure using the robust two-step method and the robust optimized search method as the single change-point solver, respectively.

6.3. Real Data Application

We apply the proposed method (Algorithm 2) to the Fama–French 49 industry portfolio data over the period from 2005 to 2015, obtained from the `frenchdata` R package. For each trading day t , the response variable Y_t is taken as the market excess return, which is further standardized to have zero mean and unit variance after alignment. The matrix-valued predictor $\mathbf{X}_t \in \mathbb{R}^{49 \times 49}$ is constructed as the rolling correlation matrix of the 49 industry portfolio returns, computed over a 60-day window prior to time t , capturing the evolving dependence structure among industries.

As illustrated in Figure 4, the empirical distribution of Y_t exhibits a more pronounced peak than the Gaussian benchmark, while the Q–Q plot shows noticeable deviations from the reference line in the tails. Such features, commonly observed in financial return series, are indicative of a peaked and heavy-tailed distribution. The resulting sequence of matrix–response pairs $\{(Y_t, \mathbf{X}_t)\}$ therefore exhibits both time-varying dependence structures and heavy-tailed noise, posing additional challenges for statistical modeling and inference in dynamic low-rank settings.

We apply the proposed method to the Fama–French 49 industry portfolio data, using the robust optimized search procedure as the single change-point solver within the multiple change-point algorithm. The method detects three change-points on May 30, 2008, March 9, 2009, and October 4, 2011, which divide the sample into four segments. These dates coincide with periods of pronounced market instability around the global financial crisis and its aftermath, suggesting that the detected structural changes capture meaningful shifts in the relationship between market excess returns and the industry-level dependence structure. Figure 5 displays the standardized response series together with the detected change-points. The segmentation reveals clear

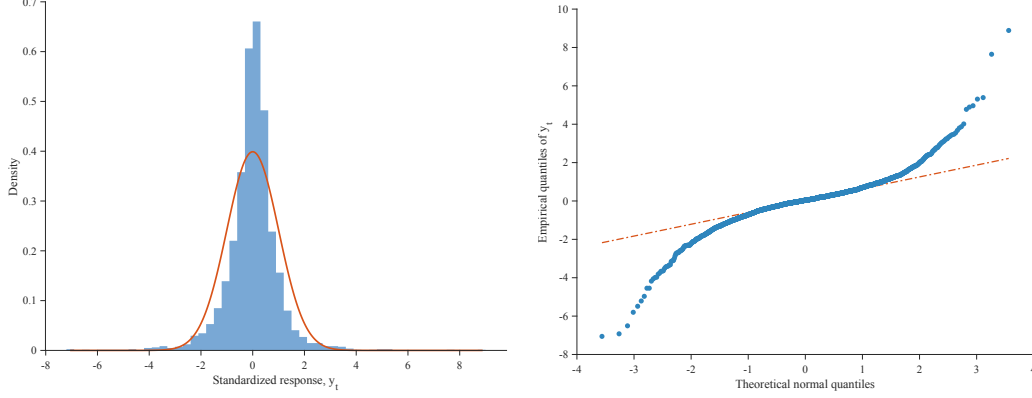


Figure 4: Empirical distribution of the response variable in the Fama–French 49 data. Left panel: Empirical density of the standardized response Y_t and the corresponding Gaussian density with matched mean and standard deviation. Right panel: Normal Q–Q plot of the standardized response Y_t .

differences in fluctuation patterns across regimes, supporting the presence of multiple structural changes.

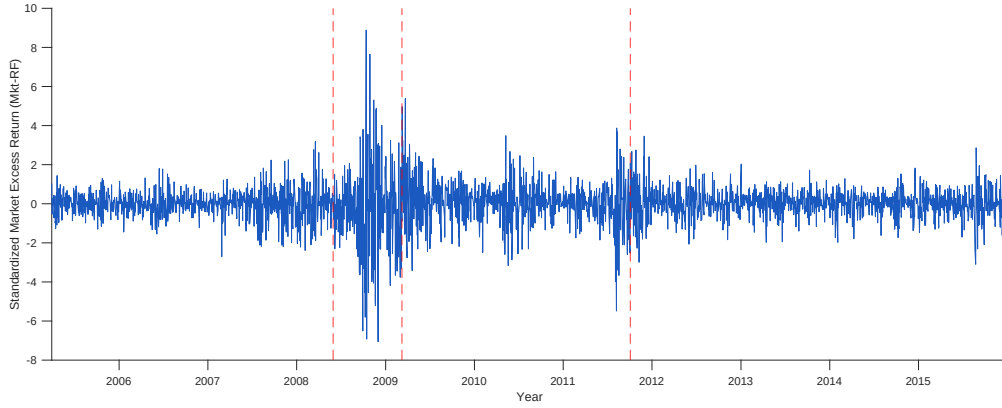


Figure 5: Standardized market excess return series with detected change-points for the Fama–French 49 industry portfolio data. The vertical dashed lines mark the detected change-points on May 30, 2008, March 9, 2009, and October 4, 2011.

To further investigate the nature of these structural changes, we examine the estimated coefficient matrices within each segment. Recall that the matrix-valued predictor $\mathbf{X}_t$ represents the rolling correlation matrix of industry returns, so the corresponding coefficient matrix reflects how cross-industry dependence contributes to the market excess return in each regime.

Figure 6 presents the heatmaps of the estimated coefficient matrices across the four segments. In the first segment, the coefficients are weak and close to zero, indicating a limited role of cross-industry dependence. In the second segment, most coefficients become negative, while a few industries, notably Gold and Agric, exhibit positive values, forming distinct row and column patterns that deviate from the dominant structure. In the third segment, this pattern is largely reversed, with predominantly positive coefficients, whereas

several industries, including Guns, Gold, Smoke, and PerSv, display negative values. In the fourth segment, the coefficients become more homogeneous again. These pronounced changes highlight the need to account for structural breaks, as such regime-dependent structures would be obscured under static methods.

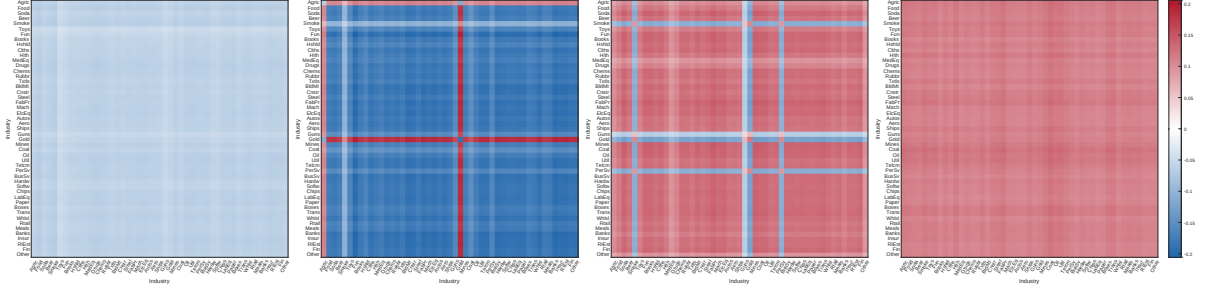


Figure 6: Heatmaps of the estimated coefficient matrices for each segment of the Fama–French 49 industry portfolio data. From left to right, the panels correspond to the four segments determined by the detected change-points.

7. Conclusion

In this paper, we study dynamic low-rank matrix recovery under heavy-tailed noise and propose a unified framework that combines statistical robustness with computational efficiency. We develop a truncation-based two-step method to mitigate the impact of extreme observations, together with an optimized search procedure that significantly reduces the computational cost of change-point detection. We establish non-asymptotic error bounds for both matrix estimation and change-point localization. Numerical experiments on simulated and real datasets demonstrate that the proposed methods achieve strong statistical accuracy while maintaining high computational efficiency. Several problems remain worthy of further investigation. The current framework is primarily designed for offline change-point analysis, where the full sample is available before estimation. Extending the proposed robust methodology to online or streaming settings would be important for applications in which observations arrive sequentially and structural changes need to be detected in real time. Another interesting direction is to move beyond purely low-rank models and consider more general matrix structures, such as combinations of low-rank and sparse components, which may better capture heterogeneous patterns in real applications.

References

- Bai, P., Safikhani, A., Michailidis, G., 2020. Multiple change points detection in low rank and sparse high dimensional vector autoregressive models. *IEEE Transactions on Signal Processing* 68, 3074–3089.
- Bai, P., Safikhani, A., Michailidis, G., 2023. Multiple change point detection in reduced rank high dimensional vector autoregressive models. *Journal of the American Statistical Association* 118, 2776–2792.
- Beck, A., Teboulle, M., 2009. A fast iterative shrinkage-thresholding algorithm for linear inverse problems. *SIAM journal on imaging sciences* 2, 183–202.
- Boucheron, S., Lugosi, G., Massart, P., 2013. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press.
- Brownlees, C., Joly, E., Lugosi, G., 2015. Empirical risk minimization for heavy-tailed losses. *The Annals of Statistics* 43, 2507–2536.
- Dirksen, S., 2015. Tail bounds via generic chaining. *Electron. J. Probab* 20, 1–29.
- Enikeeva, F., Klopp, O., Rousselot, M., 2025. Change-point detection in low-rank var processes. *Bernoulli* 31, 1058–1083.
- Fan, J., Wang, W., Zhu, Z., 2021. A shrinkage principle for heavy-tailed data: High-dimensional robust low-rank matrix recovery. *Annals of statistics* 49, 1239.
- Fryzlewicz, P., 2014. Wild binary segmentation for multiple change-point detection .
- Gupta, S., Ellis, S.E., Ashar, F.N., Moes, A., Bader, J.S., Zhan, J., West, A.B., Arking, D.E., 2014. Transcriptome analysis reveals dysregulation of innate immune response genes and neuronal activity-dependent genes in autism. *Nature communications* 5, 5748.
- Ibragimov, R., Sharakhmetov, S., 1998. On an exact constant for the rosenthal inequality. *Theory of Probability & Its Applications* 42, 294–302.
- Kovács, S., Li, H., Haubner, L., Munk, A., Bühlmann, P., 2024. Optimistic search: Change point estimation for large-scale data via adaptive logarithmic queries. *Journal of Machine Learning Research* 25, 1–64.
- Lee, S., Seo, M.H., Shin, Y., 2016. The lasso for high dimensional regression with a possible change point. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 78, 193–210.

- Negahban, S., Wainwright, M.J., 2011. Estimation of (near) low-rank matrices with noise and high-dimensional scaling .
- Negahban, S.N., Ravikumar, P., Wainwright, M.J., Yu, B., 2012. A unified framework for high-dimensional analysis of m-estimators with decomposable regularizers .
- Rinaldo, A., Wang, D., Wen, Q., Willett, R., Yu, Y., 2021. Localizing changes in high-dimensional regression models, in: International Conference on Artificial Intelligence and Statistics, PMLR. pp. 2089–2097.
- Shi, L., Wang, G., Zou, C., 2024. Low-rank matrix estimation in the presence of change-points. *Journal of Machine Learning Research* 25, 1–71.
- Stock, J.H., Watson, M.W., 2002. Macroeconomic forecasting using diffusion indexes. *Journal of business & economic statistics* 20, 147–162.
- Sun, Q., Zhou, W.X., Fan, J., 2020. Adaptive huber regression. *Journal of the American Statistical Association* 115, 254–265.
- Toh, K.C., Yun, S., 2010. An accelerated proximal gradient algorithm for nuclear norm regularized linear least squares problems. *Pacific Journal of optimization* 6, 15.
- Tsybakov, A., Rohde, A., 2011. Estimation of high-dimensional low-rank matrices. *Annals of Statistics* 39, 887–930.
- Vershynin, R., 2018. High-dimensional probability: An introduction with applications in data science. volume 47. Cambridge university press.
- Wainwright, M.J., 2019. High-dimensional statistics: A non-asymptotic viewpoint. volume 48. Cambridge university press.
- Wang, B., Fan, J., 2025. Robust matrix completion with heavy-tailed noise. *Journal of the American Statistical Association* 120, 922–934.
- Wang, T., Samworth, R.J., 2018. High dimensional change point estimation via sparse projection. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 80, 57–83.
- Yu, M., Sun, Q., Zhou, W.x., 2024. Low-rank matrix recovery under heavy-tailed errors. *Bernoulli* 30, 2326–2345.

Appendix A. Additional Numerical Study

In this section, we provide an additional numerical study to examine the empirical effect of retaining or removing the response-square term in the robust objective. The simulation setup is the same as the single change-point experiment under Cauchy noise in Section 6.1. We compare two variants of the proposed robust procedures. The first variant retains the response-square term and uses the empirical loss

$$L_{Y^2}(\tau, \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2) = \sum_{s=1}^2 \sum_{i \in \mathcal{I}_{s,\tau}} \left[\tilde{Y}_i(\tau) - \langle \mathbf{X}_i, \boldsymbol{\Theta}_s \rangle \right]^2.$$

The second variant removes the response-square term and uses

$$L_{\text{no-}Y^2}(\tau, \boldsymbol{\Theta}_1, \boldsymbol{\Theta}_2) = \sum_{s=1}^2 \sum_{i \in \mathcal{I}_{s,\tau}} \left[-2\langle \tilde{Y}_i(\tau) \mathbf{X}_i, \boldsymbol{\Theta}_s \rangle + \langle \mathbf{X}_i, \boldsymbol{\Theta}_s \rangle^2 \right].$$

This no- Y^2 criterion is exactly the robust loss adopted in the proposed methods in the main text; see (3.1). All other implementation details, including the truncation rule, tuning parameters, and search procedures, are kept the same. The results are summarized in Table A.4.

For the CS setting, the two variants yield nearly identical localization and estimation accuracy, indicating that removing the response-square term does not lead to performance loss. For the MR setting, however, retaining this term results in substantially larger change-point localization errors, while the variant without this term remains stable. These results support the use of the response-square-free loss adopted in the main text.

Table A.4: Comparison of robust procedures with and without the response-square term under Cauchy noise.

Model	Loss	Method	$ \hat{\tau} - \tau^* $	$\ \hat{\boldsymbol{\Theta}}_1 - \boldsymbol{\Theta}_1^*\ _F$	$\ \hat{\boldsymbol{\Theta}}_1 - \boldsymbol{\Theta}_1^*\ _*$	$\text{rank}(\hat{\boldsymbol{\Theta}}_1)$	$\ \hat{\boldsymbol{\Theta}}_2 - \boldsymbol{\Theta}_2^*\ _F$	$\ \hat{\boldsymbol{\Theta}}_2 - \boldsymbol{\Theta}_2^*\ _*$	$\text{rank}(\hat{\boldsymbol{\Theta}}_2)$
CS	L_{Y^2}	RT	0.009 (0.014)	0.671 (0.057)	2.161 (0.124)	6.380 (0.667)	0.685 (0.060)	2.190 (0.107)	6.380 (0.725)
		RO	0.010 (0.014)	0.671 (0.056)	2.161 (0.123)	6.380 (0.667)	0.686 (0.059)	2.191 (0.106)	6.400 (0.728)
	$L_{\text{no-}Y^2}$	RT	0.011 (0.015)	0.673 (0.057)	2.162 (0.124)	6.400 (0.700)	0.687 (0.060)	2.196 (0.106)	6.480 (0.677)
		RO	0.011 (0.014)	0.672 (0.057)	2.163 (0.124)	6.400 (0.700)	0.687 (0.060)	2.193 (0.107)	6.420 (0.731)
MR	L_{Y^2}	RT	0.296 (0.013)	0.888 (0.045)	2.370 (0.133)	5.600 (0.926)	0.873 (0.062)	2.365 (0.141)	5.740 (0.899)
		RO	0.237 (0.103)	0.873 (0.066)	2.261 (0.086)	5.300 (0.789)	0.834 (0.084)	2.388 (0.206)	6.080 (0.829)
	$L_{\text{no-}Y^2}$	RT	0.008 (0.009)	0.745 (0.067)	2.161 (0.106)	5.920 (0.601)	0.733 (0.071)	2.149 (0.110)	6.160 (0.650)
		RO	0.008 (0.009)	0.745 (0.067)	2.161 (0.106)	5.920 (0.601)	0.733 (0.071)	2.149 (0.110)	6.160 (0.650)

Results are based on 50 replications. Each entry reports the mean, with standard deviation given in parentheses when applicable. RT and RO denote the robust two-step method and the robust optimized search method, respectively.

Appendix B. Useful Lemmas

Lemma Appendix B.1. Let $\mathbf{S} \in \mathbb{R}^{d_1 \times d_2}$. Then

$$\|\mathbf{S}\|_{\text{op}} \leq \frac{16}{7} \max_{\mathbf{u} \in \mathcal{N}_1} \max_{\mathbf{v} \in \mathcal{N}_2} \langle \mathbf{u}, \mathbf{S}\mathbf{v} \rangle,$$

where $\mathcal{N}_1, \mathcal{N}_2$ are $1/4$ -nets of the unit spheres.

This is a standard ε -net argument; see, e.g., [Fan et al. \(2021\)](#) and [Vershynin \(2018\)](#).

Lemma Appendix B.2. Let $\Theta \in \mathbb{R}^{d_1 \times d_2}$ and $0 \leq q \leq 1$. Then

$$\|\Theta\|_* \leq \|\Theta\|_{S_q}^{\frac{q}{2-q}} \|\Theta\|_F^{\frac{2(1-q)}{2-q}}.$$

Proof. Let $\{\rho_l(\Theta)\}_{l=1}^{d_{\min}}$ be the singular values of Θ . Then

$$\|\Theta\|_* = \sum_{l=1}^{d_{\min}} \rho_l(\Theta), \quad \|\Theta\|_{S_q}^q = \sum_{l=1}^{d_{\min}} \rho_l(\Theta)^q, \quad \|\Theta\|_F^2 = \sum_{l=1}^{d_{\min}} \rho_l(\Theta)^2.$$

We write $\rho_l = \rho_l^{\frac{q}{2-q}} \cdot \rho_l^{\frac{2(1-q)}{2-q}}$. Applying Hölder's inequality with exponents $p = \frac{2-q}{q}$, $p' = \frac{2-q}{2(1-q)}$, which satisfy $\frac{1}{p} + \frac{1}{p'} = 1$, we obtain

$$\sum_{l=1}^{d_{\min}} \rho_l \leq \left(\sum_{l=1}^{d_{\min}} \rho_l^q \right)^{\frac{1}{p}} \left(\sum_{l=1}^{d_{\min}} \rho_l^2 \right)^{\frac{1}{p'}}.$$

Substituting p, p' gives

$$\sum_{l=1}^{d_{\min}} \rho_l \leq \left(\sum_{l=1}^{d_{\min}} \rho_l^q \right)^{\frac{q}{2-q}} \left(\sum_{l=1}^{d_{\min}} \rho_l^2 \right)^{\frac{2(1-q)}{2-q}}.$$

□

Appendix C. Lemmas for Section [Appendix D](#)

Lemma Appendix C.1. Suppose Assumption [1](#) and [3](#) hold. There exist constant C_1 and then with the probability at least $1 - 2N \exp\{-C_1(d_1 + d_2)\}$, we have

$$\frac{1}{2} N_{s,\tau} \lambda \|\Delta\|_F^2 - c'(d_1 + d_2) \|\Delta\|_*^2 \leq \sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \Delta \rangle^2 \leq \frac{3}{2} N_{s,\tau} \bar{\lambda} \|\Delta\|_F^2 + c'(d_1 + d_2) \|\Delta\|_*^2,$$

for all $\tau \in \mathcal{T}$ and $s = 1, 2$, where $\Delta \in \mathbb{R}^{d_1 \times d_2}$ is any matrix and c' is some positive constant.

Lemma Appendix C.2. Suppose that Assumption 1-3 hold. There exists some constant C_2 and with probability at least $1 - C_1 N \exp\{-C_2(d_1 + d_2)\}$, the inequality

$$\|\nabla \tilde{L}_{s,\tau}(\bar{\Theta}_{s,\tau})\|_{\text{op}} \leq \frac{1}{2} \lambda_{s,\tau},$$

holds uniformly for all $\tau \in \mathcal{T}$ and $s = 1, 2$, where $\lambda_{s,\tau} = c_\lambda \sqrt{(d_1 + d_2)N_{s,\tau}}$ and $\gamma_{s,\tau} = c_\gamma \left(\frac{N_{s,\tau}M}{d_1 + d_2}\right)^{1/2k}$ with constants c_λ and c_γ .

Lemma Appendix C.3. Suppose that we have samples $(Y_i, \mathbf{X}_i), i \in \mathcal{I}$ and $Y_i = \langle \mathbf{X}_i, \Theta^* \rangle + \varepsilon_i$. Define the truncated response variable as $\tilde{Y}_i = Y_i \mathbf{1}_{\{|Y_i| \leq \gamma_{s,\tau}\}} + \text{sgn}(Y_i) \gamma_{s,\tau} \mathbf{1}_{\{|Y_i| > \gamma_{s,\tau}\}}$ and $\gamma_{s,\tau} = c_\gamma \left(\frac{N_{s,\tau}M}{d_1 + d_2}\right)^{1/2k}$. Define the random variable Z_i as $\langle \mathbf{X}_i, \mathbf{A} \rangle$, where $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$ is a matrix with $\|\mathbf{A}\|_F = 1$. If $N_I = |\mathcal{I}| \gtrsim d_1 + d_2$, then there exists some positive constants C_1, C_2 and with probability at least $1 - C_1 \exp\{-C_2(d_1 + d_2)\}$, we have

$$\sum_{i \in \mathcal{I}} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta^* \rangle) Z_i \lesssim \gamma_{s,\tau} \sqrt{N_I(d_1 + d_2)}.$$

Lemma Appendix C.4. Suppose that the candidate change-points u and v satisfying $0 \leq u < v \leq 1$. For $i \in \mathcal{I}_{1,v} = \{i | 0 \leq t_i < u\}$, the upper bound for the difference can be written as

$$|\tilde{Y}_i(v) - \tilde{Y}_i(u)| \leq \frac{c_\gamma M^{\frac{1}{2k}}}{2k} (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0 \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}}.$$

For $i \in \mathcal{I}_0 = \{i | u \leq t_i < v\}$, we define $\gamma_{\min} = \min(\gamma_{1,v}, \gamma_{2,u})$ and the upper bound for the difference can be written as

$$|\tilde{Y}_i(v) - \tilde{Y}_i(u)| \leq c_\gamma M^{\frac{1}{2k}} (d_1 + d_2)^{-\frac{1}{2k}} |N_{1,v} - N_{2,u}|^{\frac{1}{2k}} \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}}.$$

For $i \in \mathcal{I}_{2,u} = \{i | v \leq t_i \leq 1\}$, the upper bound for the difference can be written as

$$|\tilde{Y}_i(v) - \tilde{Y}_i(u)| \leq \frac{c_\gamma M^{\frac{1}{2k}}}{2k} (d_1 + d_2)^{-\frac{1}{2k}} N_{2,v}^{\frac{1}{2k}-1} N_0 \mathbf{1}_{\{|Y_i| > \gamma_{2,v}\}}.$$

Appendix D. Proof of main results for CS case

Appendix D.1. Proof of Theorem 4.1

The proof of this Theorem is an expansion of the theoretical framework established in [Negahban et al. \(2012\)](#). Thus, we mainly focus on the difference and present the whole proof here. For simplicity, we write

$\hat{\Theta}_{s,\tau} - \bar{\Theta}_{s,\tau}$ as $\bar{\Delta}_{s,\tau}$ and write $\tilde{Y}_i(\tau)$ as $\tilde{Y}_i$. By the definition of $\hat{\Theta}_{s,\tau}$, we have

$$\begin{aligned} L_{s,\tau}(\hat{\Theta}_{s,\tau}) + \lambda_{s,\tau} \|\hat{\Theta}_{s,\tau}\|_* &\leq L_{s,\tau}(\bar{\Theta}_{s,\tau}) + \lambda_{s,\tau} \|\bar{\Theta}_{s,\tau}\|_*, \\ \left\langle \nabla \tilde{L}_{s,\tau}(\bar{\Theta}_{s,\tau}), \bar{\Delta}_{s,\tau} \right\rangle + \sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \bar{\Delta}_{s,\tau} \rangle^2 &\leq \lambda_{s,\tau} \left[\|\bar{\Theta}_{s,\tau}\|_* - \|\hat{\Theta}_{s,\tau}\|_* \right]. \end{aligned}$$

Thus, we can derive that

$$\begin{aligned} \sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \bar{\Delta}_{s,\tau} \rangle^2 &\leq_{(1)} \left\| \nabla \tilde{L}_{s,\tau}(\bar{\Theta}_{s,\tau}) \right\|_{op} \|\bar{\Delta}_{s,\tau}\|_* + \lambda_{s,\tau} \left[\|\bar{\Theta}_{s,\tau}\|_* - \|\hat{\Theta}_{s,\tau}\|_* \right] \\ &\leq_{(2)} \frac{1}{2} \lambda_{s,\tau} \|\bar{\Delta}_{s,\tau}\|_* + \lambda_{s,\tau} \left[\|\bar{\Theta}_{s,\tau}\|_* - \|\hat{\Theta}_{s,\tau}\|_* \right] \end{aligned} \quad (\text{D.1})$$

$$\leq_{(3)} \frac{3}{2} \lambda_{s,\tau} \|\bar{\Delta}_{s,\tau}\|_*, \quad (\text{D.2})$$

where the inequality (1) is derived by $\langle \mathbf{A}, \mathbf{B} \rangle \leq \|\mathbf{A}\|_{op} \|\mathbf{B}\|_*$ for any matrix $\mathbf{A}$ and $\mathbf{B}$; the inequality (2) is derived by Lemma [Appendix C.2](#); the inequality (3) is derived by triangle inequality. To bound the RHS of the above inequality, we follow Lemma 1 of [Negahban and Wainwright \(2011\)](#). First, we decompose $\bar{\Theta}_{s,\tau}$ as $\mathbf{U}_{s,\tau} \Sigma_{s,\tau} \mathbf{V}_{s,\tau}^\top$ and $\mathbf{U}_{s,\tau}$, $\mathbf{V}_{s,\tau}$ are matrices consisted by singular vectors, $\Sigma_{s,\tau}$ is the diagonal matrix consisted by singular values. We define the first $r_{s,\tau}$ columns of $\mathbf{U}_{s,\tau}$ and $\mathbf{V}_{s,\tau}$ as $\mathbf{U}_{s,\tau}^{r_{s,\tau}}$ and $\mathbf{V}_{s,\tau}^{r_{s,\tau}}$. Define the subspaces as

$$\begin{aligned} \mathcal{M}_{s,\tau} &:= \left\{ \Theta \in \mathbb{R}^{d_1 \times d_2} \mid \text{row}(\Theta) \subseteq \text{col}(\mathbf{V}_{s,\tau}^{r_{s,\tau}}), \text{col}(\Theta) \subseteq \text{col}(\mathbf{U}_{s,\tau}^{r_{s,\tau}}) \right\}, \\ \bar{\mathcal{M}}_{s,\tau}^\perp &:= \left\{ \Theta \in \mathbb{R}^{d_1 \times d_2} \mid \text{row}(\Theta) \perp \text{col}(\mathbf{V}_{s,\tau}^{r_{s,\tau}}), \text{col}(\Theta) \perp \text{col}(\mathbf{U}_{s,\tau}^{r_{s,\tau}}) \right\}. \end{aligned}$$

For simplicity, we define the orthogonal projection matrix of the subspaces as $\Pi_{\mathcal{M}}$ and $\Pi_{\bar{\mathcal{M}}^\perp}$. Then, by the proof for Lemma 1 of [Negahban and Wainwright \(2011\)](#), we can obtain the following decomposition that

$$\bar{\Theta}_{s,\tau} = \Pi_{\mathcal{M}}(\bar{\Theta}_{s,\tau}) + \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau}), \quad (\text{D.3})$$

and

$$\begin{aligned} \|\hat{\Theta}_{s,\tau}\|_* &= \|\bar{\Delta}_{s,\tau} + \bar{\Theta}_{s,\tau}\|_* \\ &=_{(1)} \left\| \Pi_{\mathcal{M}}(\bar{\Theta}_{s,\tau}) + \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau}) + \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Delta}_{s,\tau}) + \Pi_{\mathcal{M}}(\bar{\Delta}_{s,\tau}) \right\|_* \\ &\geq_{(2)} \left\| \Pi_{\mathcal{M}}(\bar{\Theta}_{s,\tau}) + \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Delta}_{s,\tau}) \right\|_* - \left\| \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau}) + \Pi_{\mathcal{M}}(\bar{\Delta}_{s,\tau}) \right\|_* \\ &\geq_{(3)} \left\| \Pi_{\mathcal{M}}(\bar{\Theta}_{s,\tau}) \right\|_* + \left\| \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Delta}_{s,\tau}) \right\|_* - \left\| \Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau}) \right\|_* - \left\| \Pi_{\mathcal{M}}(\bar{\Delta}_{s,\tau}) \right\|_*, \end{aligned} \quad (\text{D.4})$$

where the equation (1) is derived by (D.3) and the property of orthogonal projection matrix; the inequality (2) is derived by triangle inequality; the inequality (3) is derived by Negahban et al. (2012) that nuclear norm is decomposable. Recalling (D.2), we can derive

$$\begin{aligned}
0 &\leq \sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \bar{\Delta}_{s,\tau} \rangle^2 \\
&\leq_{(1)} \frac{1}{2} \lambda_{s,\tau} [\|\bar{\Delta}_{s,\tau}\|_* + 4\|\Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau})\|_* - 2\|\Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Delta}_{s,\tau})\|_* + 2\|\Pi_{\bar{\mathcal{M}}}(\bar{\Delta}_{s,\tau})\|_*] \\
&\leq_{(2)} \frac{1}{2} \lambda_{s,\tau} [4\|\Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau})\|_* - \|\Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Delta}_{s,\tau})\|_* + 3\|\Pi_{\bar{\mathcal{M}}}(\bar{\Delta}_{s,\tau})\|_*],
\end{aligned}$$

where the inequality (1) is derived by replacing the RHS of (D.2) by (D.3) and (D.4); the inequality (2) is derived by triangle inequality. Thus, it is naturally to obtain

$$\|\bar{\Delta}_{s,\tau}\|_* \leq 4\|\Pi_{\bar{\mathcal{M}}}(\bar{\Delta}_{s,\tau})\|_* + 4\|\Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau})\|_*. \quad (\text{D.5})$$

Following the proof of Corollary 2 in Negahban and Wainwright (2011), we choose the effective rank $r_{s,\tau}$ with threshold $t_{s,\tau} > 0$ as

$$r_{s,\tau} = \text{card} \{l \in [1, 2, \dots, d] \mid \rho_l(\bar{\Theta}_{s,\tau}) \geq t_{s,\tau}\}.$$

Then, by the definition of orthogonal projection matrix, we write $\|\Pi_{\bar{\mathcal{M}}^\perp}(\bar{\Theta}_{s,\tau})\|_*$ as $\sum_{l > r_{s,\tau}} \rho_l(\bar{\Theta}_{s,\tau})$ and control it by

$$\begin{aligned}
\sum_{l > r_{s,\tau}} \rho_l(\bar{\Theta}_{s,\tau}) &\leq t_{s,\tau} \sum_{l > r_{s,\tau}} \frac{\rho_l(\bar{\Theta}_{s,\tau})}{t_{s,\tau}} \leq t_{s,\tau} \sum_{l > r_{s,\tau}} \left[\frac{\rho_l(\bar{\Theta}_{s,\tau})}{t_{s,\tau}} \right]^q \leq t_{s,\tau}^{1-q} \sum_{l > r_{s,\tau}} [\rho_l(\bar{\Theta}_{s,\tau})]^q \\
&\leq_{(1)} t_{s,\tau}^{1-q} \left\{ \sum_{l > 0} [\rho_l(\Theta_1^*)]^q + \sum_{l > 0} [\rho_l(\Theta_2^*)]^q \right\} \\
&\leq_{(2)} t_{s,\tau}^{1-q} (R_{1,q} + R_{2,q}),
\end{aligned}$$

where the inequality (1) is derived by (2.1) of Tsybakov and Rohde (2011); the inequality (2) is derived by near low-rank condition of Θ_s^* . Moreover, we also have $R_{1,q} + R_{2,q} \geq \sum_{l > r_{s,\tau}} [\rho_l(\bar{\Theta}_{s,\tau})]^q \geq r_{s,\tau} t_{s,\tau}^q$, which implies $r_{s,\tau} \leq (R_{1,q} + R_{2,q}) t_{s,\tau}^{-q}$. Taking the threshold $t_{s,\tau} = \lambda_{s,\tau} / \underline{\lambda}$, we can obtain that

$$\sum_{l > r_{s,\tau}} \rho_l(\bar{\Theta}_{s,\tau}) \leq \frac{c_\lambda^{1-q} (d_1 + d_2)^{\frac{1}{2} - \frac{q}{2}}}{\underline{\lambda}^{1-q} N_{s,\tau}^{\frac{1}{2} - \frac{q}{2}}} (R_{1,q} + R_{2,q}), \quad (\text{D.6})$$

and

$$r_{s,\tau} \leq \frac{c_\lambda^{-q}(d_1 + d_2)^{-\frac{q}{2}}}{\underline{\lambda}^{-q}N_{s,\tau}^{-\frac{q}{2}}}(R_{1,q} + R_{2,q}). \quad (\text{D.7})$$

Then, we turn to derive the upper bound for $\|\bar{\Delta}_{s,\tau}\|_F$ and $\|\bar{\Delta}_{s,\tau}\|_*$. By Lemma [Appendix C.1](#), (D.2) can be transformed into

$$\frac{1}{2}\underline{\lambda}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 - c'(d_1 + d_2)\|\bar{\Delta}_{s,\tau}\|_*^2 \leq \sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \bar{\Delta}_{s,\tau} \rangle^2 \leq \frac{3}{2}\lambda_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_*$$

Combining with (D.5), we can derive

$$\begin{aligned} \frac{1}{2}\underline{\lambda}N_{j,\tau}\|\bar{\Delta}_j\|_F^2 &\leq_{(1)} 6c_\lambda\sqrt{(d_1 + d_2)N_{s,\tau}} \left[\sqrt{2r_{s,\tau}}\|\bar{\Delta}_{s,\tau}\|_F + \sum_{l>r_{s,\tau}} \rho_l(\bar{\Theta}_{s,\tau}) \right] \\ &\quad + 32c'(d_1 + d_2) \left\{ 2r_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + \left[\sum_{l>r_{s,\tau}} \rho_l(\bar{\Theta}_{s,\tau}) \right]^2 \right\} \\ &\leq_{(2)} \frac{1}{4}\underline{\lambda}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + \frac{1152c_\lambda^2r_{s,\tau}(d_1 + d_2)}{\underline{\lambda}} + 6c_\lambda\sqrt{(d_1 + d_2)N_{j,\tau}} \sum_{l=r_j+1}^d \rho_l(\bar{\Theta}_{j,\tau}) \\ &\quad + 64c'r_j(d_1 + d_2)\|\bar{\Delta}_j\|_F^2 + 32c'(d_1 + d_2) \left[\sum_{l=r_j+1}^d \rho_l(\bar{\Theta}_{j,\tau}) \right]^2 \\ &\leq_{(3)} \frac{1}{4}\underline{\lambda}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + \frac{1152c_\lambda^{2-q}(d_1 + d_2)^{1-\frac{q}{2}}}{\underline{\lambda}^{1-q}N_{s,\tau}^{-\frac{q}{2}}}(R_{1,q} + R_{2,q}) \\ &\quad + \frac{6c_\lambda^{2-q}(d_1 + d_2)^{1-\frac{q}{2}}}{\underline{\lambda}^{1-q}N_{s,\tau}^{-\frac{q}{2}}}(R_{1,q} + R_{2,q}) + \frac{64c'c_\lambda^{-q}(d_1 + d_2)^{1-\frac{q}{2}}}{\underline{\lambda}^{-q}N_{s,\tau}^{-\frac{q}{2}}}(R_{1,q} + R_{2,q})\|\bar{\Delta}_{s,\tau}\|_F^2 \\ &\quad + \frac{32c'c_\lambda^{2-2q}(d_1 + d_2)^{2-q}}{\underline{\lambda}^{2-2q}N_{s,\tau}^{1-q}}(R_{1,q} + R_{2,q}) \\ &\leq_{(4)} \frac{1}{4}\underline{\lambda}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + \frac{1158c_\lambda^{2-q}(d_1 + d_2)^{1-\frac{q}{2}}}{\underline{\lambda}^{1-q}N_{s,\tau}^{-\frac{q}{2}}}(R_{1,q} + R_{2,q}) \\ &\quad + \frac{512c'(1 - c_\tau)^{\frac{q}{2}-1}c_\lambda^{-q}M^{\frac{1}{k}}}{c_N\underline{\lambda}^{1-q}}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 \\ &\quad + \frac{256c'(1 - c_\tau)^{\frac{q}{2}-1}c_\lambda^{2-2q}M^{\frac{1}{k}}(d_1 + d_2)^{1-\frac{q}{2}}}{c_N\underline{\lambda}^{3-2q}N_{s,\tau}^{-\frac{q}{2}}}(R_{1,q} + R_{2,q}) \\ &\leq_{(5)} \frac{1}{4}\underline{\lambda}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + \frac{2316c_\lambda^{2-q}(d_1 + d_2)^{1-\frac{q}{2}}R_{\max}}{\underline{\lambda}^{1-q}N_{s,\tau}^{-\frac{q}{2}}} \\ &\quad + \frac{1}{8}\underline{\lambda}N_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + \frac{c_\lambda^{2-q}(d_1 + d_2)^{1-\frac{q}{2}}R_{\max}}{8\underline{\lambda}^{1-q}N_{s,\tau}^{-\frac{q}{2}}}, \end{aligned}$$

where the inequality (1) is derived by Lemma 1 of [Negahban and Wainwright \(2011\)](#) that $\text{rank}(\mathbf{\Pi}_{\mathcal{M}}(\bar{\Delta}_{s,\tau})) \leq 2r_{s,\tau}$; the inequality (2) is derived by the inequality that $ab \leq 2a^2 + 2b^2$ for any $a, b \in \mathbb{R}$; the inequality (3) is derived by (D.6) and (D.7); the inequality (4) is derived by Assumption 3 and search domain condition, Assumption 1 and Assumption 2 can derive $M^{\frac{1}{k}} \geq \mathbb{E}Y_i^2 \geq \mathbb{E}\langle \mathbf{X}_i, \Theta_s^* \rangle^2 \geq \lambda \|\Theta_s^*\|_F^2$ and $\|\Delta^*\|_F^2 \leq 2\|\Theta_1^*\|_F^2 + 2\|\Theta_2^*\|_F^2 \leq 4M^{\frac{1}{k}}/\lambda$; the inequality (5) is depending on the enough large constant c_N in the signal-to-noise ratio condition, which satisfying $c_N > 2^{12}c'(1 - c_\tau)^{\frac{q}{2}-1}c_\lambda^{-q}M^{\frac{1}{k}}/\lambda^{2-q}$.

Thus, the above inequality can be simplified as

$$\|\bar{\Delta}_{s,\tau}\|_F^2 \lesssim \frac{c_\lambda^{2-q}(d_1 + d_2)^{1-\frac{q}{2}}R_{\max}}{\lambda^{2-q}N_{s,\tau}^{1-\frac{q}{2}}}. \quad (\text{D.8})$$

And (D.5) can be written as

$$\begin{aligned} \|\bar{\Delta}_{s,\tau}\|_*^2 &\leq_{(1)} 64r_{s,\tau}\|\bar{\Delta}_{s,\tau}\|_F^2 + 32 \left[\sum_{l>r_{s,\tau}} \rho_l(\bar{\Theta}_{s,\tau}) \right]^2 \\ &\leq_{(2)} \frac{c_\lambda^{2-2q}(d_1 + d_2)^{1-q}R_{\max}^2}{\lambda^{2-2q}N_{s,\tau}^{1-q}}, \end{aligned}$$

where the inequality (1) is derived by Lemma 1 of [Negahban and Wainwright \(2011\)](#) that $\text{rank}(\mathbf{\Pi}_{\mathcal{M}}(\bar{\Delta}_{s,\tau})) \leq 2r_{s,\tau}$ and the inequality that $(a+b)^2 \leq 2a^2 + 2b^2$ for any $a, b \in \mathbb{R}$; the inequality (2) is derived by (D.6)-(D.8).

Appendix D.2. Proof of Theorem 4.2

To derive the upper bound for estimation bias of change-point and coefficient matrices, we introduce the mixed matrices $\bar{\Theta}_{s,\hat{\tau}}$ as the intermediate term combining $\hat{\Theta}_{s,\hat{\tau}}$ and Θ_s^* . For simplicity, we write $\hat{\tau}$ as τ , $\hat{\Theta}_{s,\hat{\tau}}$ as $\hat{\Theta}_s$, $\bar{\Theta}_{s,\hat{\tau}}$ as $\bar{\Theta}_s$ and $\hat{\Theta}_{s,\hat{\tau}} - \bar{\Theta}_{s,\hat{\tau}}$ as $\bar{\Delta}_s$. We denote the index set of samples satisfying $\{i \mid \min(\tau, \tau^*) \leq t_i < \max(\tau, \tau^*)\}$ as $\mathcal{I}_0$ and the truncated samples $\tilde{Y}_i(\tau)$ as $\tilde{Y}_i$ with $N_0 = |\mathcal{I}_0|$. The proof is mainly consisted by two steps. We derive the upper bound for estimation bias of change-point in the first step and utilize the result of first step to derive the upper bound for estimation bias of coefficient matrices in the second step. Without loss of generality, suppose that $\tau > \tau^*$ and the proof for the corresponding case $\tau \leq \tau^*$ is same.

Step 1. In this step, we employs the idea of proof by contradiction to demonstrate the upper bound for the estimation bias of the change-point. And first assume that

$$|\tau - \tau^*|N > cR_{\max}(d_1 + d_2)^{1-\frac{q}{2}\vee\frac{1}{k}}N_{1,\tau}^{\frac{q}{2}\vee\frac{1}{k}}\|\Delta^*\|_F^{-2}. \quad (\text{D.9})$$

Then, we can decompose the difference of loss function as

$$\begin{aligned}
0 &\geq_{(1)} \tilde{L}_{1,\tau}(\Theta_1) + \tilde{L}_{2,\tau}(\Theta_2) - \tilde{L}_{1,\tau^*}(\Theta_1) - \tilde{L}_{2,\tau^*}(\Theta_2) \\
&= \sum_{s=1}^2 \left[\tilde{L}_{s,\tau}(\Theta_s) - \tilde{L}_{s,\tau}(\bar{\Theta}_s) + \tilde{L}_{s,\tau}(\bar{\Theta}_s) - \tilde{L}_{s,\tau^*}(\bar{\Theta}_s) + \tilde{L}_{s,\tau^*}(\bar{\Theta}_s) - \tilde{L}_{s,\tau^*}(\Theta_s) \right] \\
&= -2 \sum_{i \in \mathcal{I}_{1,\tau}} (\tilde{Y}_i - \langle \mathbf{X}_i, \bar{\Theta}_1 \rangle) \langle \mathbf{X}_i, \bar{\Delta}_1 \rangle + \sum_{i \in \mathcal{I}_{1,\tau}} \langle \mathbf{X}_i, \bar{\Delta}_1 \rangle^2 - 2 \sum_{i \in \mathcal{I}_{2,\tau}} (\tilde{Y}_i - \langle \mathbf{X}_i, \bar{\Theta}_2 \rangle) \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle \\
&\quad + \sum_{i \in \mathcal{I}_{2,\tau}} \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle^2 + \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \bar{\Theta}_1 \rangle)^2 - \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \bar{\Theta}_2 \rangle)^2 - \sum_{i \in \mathcal{I}_{2,\tau^*}} \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle^2 \\
&\quad + 2 \sum_{i \in \mathcal{I}_{1,\tau^*}} (\tilde{Y}_i - \langle \mathbf{X}_i, \bar{\Theta}_1 \rangle) \langle \mathbf{X}_i, \bar{\Delta}_1 \rangle - \sum_{i \in \mathcal{I}_{1,\tau^*}} \langle \mathbf{X}_i, \bar{\Delta}_1 \rangle^2 + 2 \sum_{i \in \mathcal{I}_{2,\tau^*}} (\tilde{Y}_i - \langle \mathbf{X}_i, \bar{\Theta}_2 \rangle) \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle \\
&= 2 \underbrace{\sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle) \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle}_{U_2} - 2 \underbrace{\sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle) \langle \mathbf{X}_i, \bar{\Delta}_1 \rangle}_{U_3} \\
&\quad + 2 \underbrace{\frac{\tau^*}{\tau} \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle) \langle \mathbf{X}_i, \Delta^* \rangle}_{U_4} + 2 \underbrace{\frac{\tau^*}{\tau} \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle \langle \mathbf{X}_i, \bar{\Delta}_1 \rangle}_{U_5} \\
&\quad - \underbrace{\sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle^2}_{U_1} + \underbrace{\left(\frac{\tau^*}{\tau} \right)^2 \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle^2}_{U_6},
\end{aligned}$$

where the inequality (1) is derived by the definition of the estimated change-point τ . Then, the above inequality can be simplified as

$$U_6 \leq U_1 + |U_2| + |U_3| + |U_4| + |U_5|. \quad (\text{D.10})$$

Next, we turn to derive the upper bound for each term of RHS and lower bound for LHS. For the term U_6 in the LHS, we can obtain that $\langle \mathbf{X}_i, \Delta^* \rangle$ is a sub-Gaussian random variable with $\|\langle \mathbf{X}_i, \Delta^* \rangle\|_{\psi_2} \lesssim K_0 \|\Delta^*\|_F$ by Assumption 1. And by the relationship between sub-Gaussian and sub-Exponential (see, eg. Lemma 2.7.7 in Vershynin (2018)), $\langle \mathbf{X}_i, \Delta^* \rangle^2$ is a sub-Exponential random variable with $\|\langle \mathbf{X}_i, \Delta^* \rangle^2\|_{\psi_1} \lesssim K_0^2 \|\Delta^*\|_F^2$. By the moment property of sub-Exponential random variable (see, eg. Proposition 2.7.1 in Vershynin (2018)), the corresponding expectation can be derived as $\mathbb{E} \langle \mathbf{X}_i, \Delta^* \rangle^2 \leq 2K_0^2 \|\Delta^*\|_F^2$. Then, by Bernstein's inequality (see, eg. Theorem 2.8.3 in Vershynin (2018)), we have

$$\mathbb{P} \{ |U_6| \geq c_6 N_0 \|\Delta^*\|_F^2 \} \leq 2 \exp \{ -C_1 (d_1 + d_2) \}.$$

For the first term U_1 in RHS, its upper bound can be derived as

$$\begin{aligned}
\sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle^2 &\leq \sum_{i \in \mathcal{I}_{2,\tau^*}} \langle \mathbf{X}_i, \bar{\Delta}_2 \rangle^2 \\
&\leq_{(1)} \frac{3}{2} \bar{\lambda} N_{2,\tau^*} \|\bar{\Delta}_2\|_F^2 + c'(d_1 + d_2) \|\bar{\Delta}_2\|_*^2 \\
&\leq_{(2)} \frac{3}{2} c_F \bar{\lambda} R_{\max} (d_1 + d_2)^{1-\frac{q}{2}} N_{2,\tau}^{\frac{q}{2}-1} N_{2,\tau^*} + c' c_* R_{\max}^2 (d_1 + d_2)^{2-q} N_{2,\tau}^{q-1} \\
&\leq_{(3)} \frac{3c_F \bar{\lambda}}{2(1-c_\tau)} R_{\max} (d_1 + d_2)^{1-\frac{q}{2}} N_{2,\tau}^{\frac{q}{2}} \\
&\quad + \frac{2c' c_* (1-c_\tau)^{q-1}}{c_N} N_0 \|\Delta^*\|_F^2 (d_1 + d_2)^{1-q+\frac{q}{2}\vee\frac{1}{k}} (\xi^* N)^{q-\frac{q}{2}\vee\frac{1}{k}-1} \\
&\leq_{(4)} \frac{3c_F \bar{\lambda}}{2c(1-c_\tau)} N_0 \|\Delta^*\|_F^2 \\
&\quad + \frac{2c' c_* (1-c_\tau)^{q-1}}{c_N} M^{\frac{1}{k}} \left[\frac{4M^{\frac{1}{k}}}{c_N R_{\max} \underline{\lambda}} \right]^{\frac{\max(2, 4-kq)}{2k-qk-2}} N_0 \|\Delta^*\|_F^2 \\
&\leq_{(5)} \frac{1}{4} c_6 N_0 \|\Delta^*\|_F^2,
\end{aligned} \tag{D.11}$$

where the inequality (1) is derived by the URSM condition in Lemma [Appendix C.1](#); the inequality (2) is derived by Theorem [4.1](#) and the constants c_F , c_* are corresponding the constants for the upper bound of Frobenius norm and nuclear norm; the inequality (3) is derived by Assumption [3](#) and search domain condition; the inequality (4) is derived by [\(D.9\)](#) and the SNR condition can be transformed into $c_N R_{\max} (d_1 + d_2)^{1-\alpha/2} (\xi^* N)^{\alpha/2-1} \leq \|\Delta^*\|_F^2 \leq 4M^{1/k}/\underline{\lambda}$; the inequality (5) is derived by enough large constants c_N of SNR condition and c of [\(D.9\)](#).

For the second term U_2 , we can derive the upper bound as

$$\begin{aligned}
|U_2| &\leq_{(1)} 2 \left\| \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle) \mathbf{X}_i \right\|_{op} \|\bar{\Delta}_2\|_* \\
&\leq_{(2)} \frac{32}{7} \|\bar{\Delta}_2\|_* \max_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle) Z_i,
\end{aligned} \tag{D.12}$$

where the inequality (1) is derived by the inequality $\langle \mathbf{A}, \mathbf{B} \rangle \leq \|\mathbf{A}\|_{op} \|\mathbf{B}\|_*$ for any $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d_1 \times d_2}$; the inequality (2) is derived by Lemma [Appendix B.1](#) and $\mathbf{u} \mathbf{X}_i \mathbf{v}^\top$ is denoted as Z_i . By Lemma [Appendix C.3](#), we can obtain that

$$\mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle) Z_i \right| \geq c'_2 \gamma_{1,\tau} \sqrt{N_0 (d_1 + d_2)} \right\} \leq C_1 \exp\{-C_2 (d_1 + d_2)\}.$$

Based on above inequality, a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$ yields that

$$\mathbb{P} \left\{ \left\| \sum_{i \in \mathbb{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \boldsymbol{\Theta}_2^* \rangle) \mathbf{X}_i \right\|_{op} \geq c'_2 \gamma_{1,\tau} \sqrt{N_0(d_1 + d_2)} \right\} \leq C_1 \exp\{-(C_2 - \log 8)(d_1 + d_2)\}.$$

Recalling (D.12), we can derive the upper bound as

$$\begin{aligned} |U_2| &\leq \frac{32}{7} c'_2 \gamma_{1,\tau} \sqrt{N_0(d_1 + d_2)} \|\bar{\Delta}_2\|_* \\ &\leq_{(1)} \frac{32}{7} c'_2 c_\gamma M^{\frac{1}{2k}} (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,\tau}^{\frac{1}{2k}} N_0^{\frac{1}{2}} c_* R_{\max} (d_1 + d_2)^{\frac{1}{2} - \frac{q}{2}} N_{2,\tau}^{\frac{q}{2} - \frac{1}{2}} \\ &\leq_{(2)} c_2 R_{\max} (d_1 + d_2)^{1 - \frac{q}{2} - \frac{1}{2k}} N_{1,\tau}^{\frac{1}{2k}} N_{2,\tau}^{\frac{q}{2} - \frac{1}{2}} N_0^{\frac{1}{2}}, \end{aligned}$$

where the inequality (1) is derived by Theorem 4.1, the definition of truncation parameter $\gamma_{1,\tau}$ and the constant c_* is the corresponding constant for the upper bound of nuclear norm; the constant c_2 of the inequality (2) absorbs the constants of the former inequality.

For the third term U_3 , we can derive the upper bound as same as the third term U_3 and obtain

$$|U_3| \leq c_3 R_{\max} (d_1 + d_2)^{1 - \frac{q}{2} - \frac{1}{2k}} N_{1,\tau}^{\frac{q}{2} + \frac{1}{2k} - \frac{1}{2}} N_0^{\frac{1}{2}}.$$

For the forth term U_4 , we can write the upper bound of $|U_4|$ as

$$\begin{aligned} |U_4| &\leq 2 \frac{\tau^*}{\tau} \left| \sum_{i \in \mathcal{I}_0} (\tilde{Y}_i - \langle \mathbf{X}_i, \boldsymbol{\Theta}_2^* \rangle) \left\langle \mathbf{X}_i, \frac{\Delta^*}{\|\Delta^*\|_F} \right\rangle \right| \|\Delta^*\|_F \\ &\leq_{(1)} c'_4 \gamma_{1,\tau} \sqrt{N_0(d_1 + d_2)} \|\Delta^*\|_F \\ &\leq_{(2)} c_4 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,\tau}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \|\Delta^*\|_F, \end{aligned}$$

where the inequality (1) holds with probability at least $1 - C_1 \exp\{-C_2(d_1 + d_2)\}$, and it is derived by search domain condition and Lemma Appendix C.3; the inequality (2) is derived by the definition of the truncation parameter $\gamma_{1,\tau}$.

For the fifth term U_5 , we can write its upper bound as

$$\begin{aligned} |U_5| &\leq_{(1)} 2 \frac{\tau^*}{\tau} \left\| \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle \mathbf{X}_i \right\|_{op} \|\bar{\Delta}_1\|_* \\ &\leq_{(2)} \frac{32}{7} \|\bar{\Delta}_1\|_* \max_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle Z_i, \end{aligned} \tag{D.13}$$

where the inequality (1) is derived by the inequality $\langle \mathbf{A}, \mathbf{B} \rangle \leq \|\mathbf{A}\|_{op} \|\mathbf{B}\|_*$ for any $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d_1 \times d_2}$; the inequality (2) is derived by Lemma [Appendix B.1](#) and search domain condition. By Assumption [1](#) and the definition of sub-Gaussian random variable, it is easy to obtain that $\langle \mathbf{X}_i, \mathbf{\Delta}^* \rangle$ and Z_i are both sub-Gaussian random variables with $\|\langle \mathbf{X}_i, \mathbf{\Delta}^* \rangle\|_{\psi_2} \lesssim K_0 \|\mathbf{\Delta}^*\|_F$ and $\|Z_i\|_{\psi_2} \lesssim K_0$. Then, $\langle \mathbf{X}_i, \mathbf{\Delta}^* \rangle Z_i$ is a sub-Exponential random variable with $\|\langle \mathbf{X}_i, \mathbf{\Delta}^* \rangle Z_i\|_{\psi_1} \lesssim K_0^2 \|\mathbf{\Delta}^*\|_F$ by the relationship between sub-Gaussians and sub-Exponentials (see, eg. Lemma 2.7.7 in [Vershynin \(2018\)](#)). By Bernstein's inequality (see, eg. Theorem 2.8.1 in [Vershynin \(2018\)](#)), we can derive that

$$\mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \mathbf{\Delta}^* \rangle Z_i \right| \geq c'_5 \sqrt{N_0(d_1 + d_2)} \|\mathbf{\Delta}^*\|_F \right\} \leq 2 \exp\{-C_1(d_1 + d_2)\}.$$

Taking a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$, the above inequality can be written as

$$\mathbb{P} \left\{ \left\| \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \mathbf{\Delta}^* \rangle \mathbf{X}_i \right\|_{op} \geq c'_5 \sqrt{N_0(d_1 + d_2)} \|\mathbf{\Delta}^*\|_F \right\} \leq 2 \exp\{-(C_1 - \log 8)(d_1 + d_2)\}.$$

Recalling [\(D.13\)](#), we can derive the upper bound as

$$\begin{aligned} |U_5| &\leq \frac{32}{7} c'_5 \sqrt{N_0(d_1 + d_2)} \|\mathbf{\Delta}^*\|_F \|\bar{\mathbf{\Delta}}_1\|_* \\ &\leq_{(1)} \frac{32}{7} c'_5 c_* N_0^{\frac{1}{2}} (d_1 + d_2)^{\frac{1}{2}} \|\mathbf{\Delta}^*\|_F R_{\max}(d_1 + d_2)^{\frac{1}{2} - \frac{q}{2}} N_{1,\tau}^{\frac{q}{2} - \frac{1}{2}} \\ &\leq_{(2)} c_5 R_{\max}(d_1 + d_2)^{1 - \frac{q}{2}} N_{1,\tau}^{\frac{q}{2} - \frac{1}{2}} N_0^{\frac{1}{2}} \|\mathbf{\Delta}^*\|_F, \end{aligned}$$

where the inequality (1) is derived by Theorem [4.1](#); the constant c_5 of the inequality (2) absorbs the constants of the former inequality.

Recalling (D.10), we have

$$\begin{aligned}
c_6 N_0 \|\Delta^*\|_F^2 &\leq \frac{1}{4} c_6 N_0 \|\Delta^*\|_F^2 + c_2 R_{\max}(d_1 + d_2)^{1-\frac{q}{2}-\frac{1}{2k}} N_{1,\tau}^{\frac{1}{2k}} N_{2,\tau}^{\frac{q}{2}-\frac{1}{2}} N_0^{\frac{1}{2}} \\
&\quad + c_3 R_{\max}(d_1 + d_2)^{1-\frac{q}{2}-\frac{1}{2k}} N_{1,\tau}^{\frac{q}{2}+\frac{1}{2k}-\frac{1}{2}} N_0^{\frac{1}{2}} + c_4 (d_1 + d_2)^{\frac{1}{2}-\frac{1}{2k}} N_{1,\tau}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \|\Delta^*\|_F \\
&\quad + c_5 R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} N_{1,\tau}^{\frac{q}{2}-\frac{1}{2}} N_0^{\frac{1}{2}} \|\Delta^*\|_F \\
&\leq_{(1)} \frac{1}{2} c_6 N_0 \|\Delta^*\|_F^2 + \frac{16c_2^2}{c_6} R_{\max}^2(d_1 + d_2)^{2-q-\frac{1}{k}} N_{2,\tau}^{q-1} N_{1,\tau}^{\frac{1}{k}} \|\Delta^*\|_F^{-2} \\
&\quad + \frac{16c_3^2}{c_6} R_{\max}^2(d_1 + d_2)^{2-q-\frac{1}{k}} N_{1,\tau}^{q+\frac{1}{k}-1} \|\Delta^*\|_F^{-2} + \frac{16c_4^2}{c_6} (d_1 + d_2)^{1-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}} \\
&\quad + \frac{16c_5^2}{c_6} R_{\max}^2(d_1 + d_2)^{2-q} N_{1,\tau}^{q-1} \\
&\leq_{(2)} \frac{1}{2} c_6 N_0 \|\Delta^*\|_F^2 + \underbrace{\frac{16(c_2^2 + c_3^2)}{c_6 c_N (1 - c_\tau)^{1-q}} R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} (\xi^* N)^{\frac{q}{2}-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}}}_I \\
&\quad + \underbrace{\frac{16c_4^2}{c_6} (d_1 + d_2)^{1-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}}}_{II} + \underbrace{\frac{32c_5^2 M^{\frac{1}{k}}}{c_6 c_N (1 - c_\tau)^{1-q}} R_{\max}(d_1 + d_2)^{1-\frac{q}{2}+\frac{1}{k}} (\xi^* N)^{\frac{q}{2}-\frac{1}{k}}}_{III}, \quad (D.14)
\end{aligned}$$

where the inequality (1) is derived by mean inequality $2ab \leq a^2 + b^2$, for any $a, b \in \mathbb{R}$; the inequality (2) is derived by Assumption 3 and search domain condition. For the last three terms in RHS of the above inequality, we consider different cases depending on the relative magnitude of $1/k$ and $q/2$ and combine the results accordingly.

If $1/k > q/2$, we have

$$\begin{aligned}
I + II + III &\leq_{(1)} c_I R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} (d_1 + d_2)^{\frac{q}{2}-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}} + c_{II} R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} (\xi^* N)^{\frac{q}{2}} \\
&\quad + c_{III} R_{\max}(d_1 + d_2)^{1-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}} \\
&\leq_{(2)} c'_0 R_{\max}(d_1 + d_2)^{1-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}},
\end{aligned}$$

where the inequality (1) is derived by Assumption 3 and c_I, c_{II}, c_{III} are some suitable constants; the inequality (2) is derived by $1/k > q/2$ and the constant c'_0 absorbs the constants of the former inequality.

If $1/k \leq q/2$, we similarly derive the upper bound as

$$\begin{aligned}
I + II + III &\leq_{(1)} c_I R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} (\xi^* N)^{\frac{q}{2}} + c_{II} R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} N_{1,\tau}^{\frac{q}{2}} \\
&\quad + c_{III} R_{\max}(d_1 + d_2)^{1-\frac{1}{k}} N_{1,\tau}^{\frac{1}{k}} \\
&\leq_{(2)} c'_0 R_{\max}(d_1 + d_2)^{1-\frac{q}{2}} N_{1,\tau}^{\frac{q}{2}}.
\end{aligned}$$

Then, the inequality (D.14) can be written as

$$\frac{1}{2}c_6N_0\|\Delta^*\|_F^2 \leq c'_0R_{\max}(d_1+d_2)^{1-\frac{q}{2}\vee\frac{1}{k}}N_{1,\tau}^{\frac{q}{2}\vee\frac{1}{k}},$$

which is contrary to the assumption of N_0 with enough large c . Thus, the assumption is wrong and we can obtain

$$\begin{aligned} |\tau - \tau^*| &\leq cR_{\max}(d_1+d_2)^{1-\frac{q}{2}\vee\frac{1}{k}}N_{1,\tau}^{\frac{q}{2}\vee\frac{1}{k}}N^{-1}\|\Delta^*\|_F^{-2} \\ &\leq_{(1)} cR_{\max}(d_1+d_2)^{1-\frac{q}{2}\vee\frac{1}{k}}N^{\frac{q}{2}\vee\frac{1}{k}-1}\|\Delta^*\|_F^{-2}, \end{aligned} \quad (\text{D.15})$$

where the inequality (1) is derived by $N > N_{1,\tau}$.

Step 2. In this step, we find the upper bound for Frobenius and nuclear norm for coefficient estimation bias. We first derive the upper bound for Frobenius norm as

$$\begin{aligned} \|\widehat{\Theta}_1 - \Theta_1^*\|_F^2 &\leq_{(1)} 2\|\widehat{\Theta}_1 - \bar{\Theta}_1\|_F^2 + 2\|\bar{\Theta}_1 - \Theta_1^*\|_F^2 \\ &\leq_{(2)} 2\|\widehat{\Theta}_1 - \bar{\Theta}_1\|_F^2 + 2\left(\frac{\tau - \tau^*}{\tau}\right)^2 \|\Delta^*\|_F^2 \\ &\lesssim_{(3)} R_{\max}(d_1+d_2)^{1-\frac{q}{2}}N_{1,\tau}^{\frac{q}{2}-1} + \frac{2N_0^2}{N_{1,\tau}^2} \|\Delta^*\|_F^2 \\ &\lesssim_{(4)} R_{\max}(d_1+d_2)^{1-\frac{q}{2}}(\xi^*N)^{\frac{q}{2}-1} + R_{\max}^2(d_1+d_2)^{2-q\vee\frac{1}{k}}(\xi^*N)^{q\vee\frac{1}{k}-2} \|\Delta^*\|_F^{-2} \\ &\lesssim_{(5)} R_{\max}(d_1+d_2)^{1-\frac{q}{2}\vee\frac{1}{k}}(\xi^*N)^{\frac{q}{2}\vee\frac{1}{k}-1}, \end{aligned}$$

where the inequality (1) is derived by the inequality $(a+b)^2 \leq 2a^2 + 2b^2$ for any $a, b \in \mathbb{R}$; the inequality (2) is derived by the definition of mixed matrix $\bar{\Theta}_1$; the inequality (3) is derived by Theorem 4.1; the inequality (4) is derived by (D.15) and search domain condition; the inequality (5) is derived by Assumption 3.

Similarly, we can derive the upper bound for nuclear norm as

$$\begin{aligned}
\|\hat{\Theta}_1 - \Theta_1^*\|_*^2 &\leq_{(1)} 2\|\hat{\Theta}_1 - \bar{\Theta}_1\|_*^2 + 2\|\bar{\Theta}_1 - \Theta_1^*\|_*^2 \\
&\leq_{(2)} 2\|\hat{\Theta}_1 - \bar{\Theta}_1\|_*^2 + 2\left(\frac{\tau - \tau^*}{\tau}\right)^2 \|\Delta^*\|_*^2 \\
&\lesssim_{(3)} R_{\max}^2 (d_1 + d_2)^{1-q} N_{1,\tau}^{q-1} + \frac{2N_0^2}{N_{1,\tau}^2} \|\Delta^*\|_{S_q}^{\frac{2q}{2-q}} \|\Delta^*\|_F^{\frac{4(1-q)}{2-q}} \\
&\lesssim_{(4)} R_{\max}^2 (d_1 + d_2)^{1-q} (\xi^* N)^{q-1} + R_{\max}^2 (d_1 + d_2)^{2-q-\frac{2}{k}} (\xi^* N)^{q+\frac{2}{k}-2} \|\Delta^*\|_{S_q}^{\frac{2q}{2-q}} \|\Delta^*\|_F^{\frac{4(1-q)}{2-q}} \\
&\lesssim_{(5)} R_{\max}^2 (d_1 + d_2)^{1-q} (\xi^* N)^{q-1} + R_{\max}^2 (d_1 + d_2)^{2-q\vee\frac{2}{k}} (\xi^* N)^{q\vee\frac{2}{k}-2} \|\Delta^*\|_F^{\frac{-4}{2-q}} \left[\|\Theta_1^*\|_{S_q}^q + \|\Theta^*\|_{S_q}^q \right]^{\frac{2}{2-q}} \\
&\lesssim_{(6)} R_{\max}^2 (d_1 + d_2)^{1-q} (\xi^* N)^{q-1} + R_{\max}^2 (d_1 + d_2)^{2-q\vee\frac{2}{k}} (\xi^* N)^{q\vee\frac{2}{k}-2} \|\Delta^*\|_F^{\frac{-4}{2-q}} R_{\max}^{\frac{2}{2-q}} \\
&\lesssim_{(7)} R_{\max}^2 (d_1 + d_2)^{1-q\vee\frac{2}{k}} (\xi^* N)^{q\vee\frac{2}{k}-1},
\end{aligned}$$

where the inequality (1) is derived by the inequality $(a+b)^2 \leq 2a^2 + 2b^2$ for any $a, b \in \mathbb{R}$; the inequality (2) is derived by the definition of $\bar{\Theta}_1$; the inequality (3) is derived by Theorem 4.1 and Lemma Appendix B.2; the inequality (4) is derived by the result of Step 1 and search domain condition; the inequality (5) is derived by (2.1) of Tsybakov and Rohde (2011); the inequality (6) is derived by the near low-rank setting of Θ_s^* ; the inequality (7) is derived by Assumption 3.

Appendix D.3. Proof of Theorem OS

The proof is consist of two steps.

Step 1. In this step, we analyze the case where the true change-point τ^* lies within the interval before the probe point first crosses to the left-hand side of τ^* . For simplicity and clarity in interval partitioning, the three intervals divided by the current probe points have relative lengths either $1 : 1 : 1$, $1 : 1 : 2$, or $2 : 1 : 1$. Denote by I_k the interval corresponding to the k -th iteration in the k -th optimized search, and write the estimated coefficient matrix $\hat{\Theta}_{s,v}$ and $\hat{\Theta}_{s,u}$ as $\Theta_{s,v}$ and $\Theta_{s,u}$, which is corresponding to the probe change-point v and u . We define k^* as

$$k^* = \max\{k : u \wedge v \leq \tau^* \text{ with } (u,v) \text{ are the probe points of } I_k\},$$

which is the index of the inner loop that the probe point first crosses to the left-hand side of τ^* . Define the set of probe points before k^* as

$$\mathcal{D}_1 = \{(u, v) : (u, v) \text{ are the probe points of } I_k \text{ with } u < v \text{ and } k < k^*\}.$$

According to the optimized search, we need to compare the loss function values $f_u(\Theta_{1,u}, \Theta_{2,u})$ and $f_v(\Theta_{1,v}, \Theta_{2,v})$ to determine which side of the current interval will be retained for the next iteration. Let $\mathcal{I}_0 = \{i \mid \min(u, v) \leq t_i \leq \max(u, v)\}$ and denote $N_0 = |\mathcal{I}_0|$. We claim that $N_0 \geq \frac{1}{2}\tau^*N$. Suppose, for contradiction, that $N_0 < \frac{1}{2}\tau^*N$ and consider two possible scenarios. If the second probe point is in the left-hand of the first probe point, we have $v \geq 1 - v$ and $uN = N_0 < \frac{1}{2}\tau^*N$, which is contrary to $(u, v) \in \mathcal{D}_1$. If the second probe point is in the right-hand of the first probe point, we have $1 - u \geq u$ and $(1 - u)N < \tau^*N$, which is contrary to $(1 - u)N \geq uN$. Hence, we conclude that $N_0 \geq \frac{1}{2}\tau^*N$ must hold.

Comparing the loss function, we have

$$\begin{aligned}
& f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u}) \\
&= \sum_{s=1}^2 \left[\tilde{L}_{s,v}(\Theta_{1,v}, \Theta_{2,v}) - \sum_{i \in \mathcal{I}_{s,v}} [\tilde{Y}_i(v)]^2 - \tilde{L}_{s,u}(\Theta_{1,u}, \Theta_{2,u}) + \sum_{i \in \mathcal{I}_{s,u}} [\tilde{Y}_i(u)]^2 \right] \\
&\geq_{(1)} \sum_{s=1}^2 \left[\tilde{L}_{s,v}(\Theta_{1,v}, \Theta_{2,v}) - \sum_{i \in \mathcal{I}_{s,v}} [\tilde{Y}_i(v)]^2 - \tilde{L}_{s,u}(\Theta_{1,v}, \Theta_{2,v}) + \sum_{i \in \mathcal{I}_{s,u}} [\tilde{Y}_i(u)]^2 \right] \\
&=_{(2)} \underbrace{\sum_{s=1}^2 \left[\sum_{i \in \mathcal{I}_{s,v}} \left(\tilde{Y}_i(v) - \langle \mathbf{X}_i, \Theta_{s,v} \rangle \right)^2 - \sum_{i \in \mathcal{I}_{s,u}} \left(\tilde{Y}_i(v) - \langle \mathbf{X}_i, \Theta_{s,v} \rangle \right)^2 \right]}_I \\
&\quad + \underbrace{\sum_{s=1}^2 \left\{ \sum_{i \in \mathcal{I}_{s,u}} \left[\left(\tilde{Y}_i(v) - \langle \mathbf{X}_i, \Theta_{s,v} \rangle \right)^2 - \left(\tilde{Y}_i(v) \right)^2 \right] - \sum_{i \in \mathcal{I}_{s,u}} \left[\left(\tilde{Y}_i(u) - \langle \mathbf{X}_i, \Theta_{s,v} \rangle \right)^2 - \left(\tilde{Y}_i(u) \right)^2 \right] \right\}}_{II}.
\end{aligned} \tag{D.16}$$

To derive the lower bound for term I in RHS of above inequality, we can follow the process of the proof of Theorem 4.2 in Section Appendix D.2 and define $\bar{\Delta}_{s,v} = \Theta_{s,v} - \bar{\Theta}_{s,v}$. Then, term I can be decomposed as

$$\begin{aligned}
I &= \left(\frac{\tau^*}{v} \right)^2 \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle^2 - \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \bar{\Delta}_{2,v} \rangle^2 + 2 \sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \langle \mathbf{X}_i, \Theta_2^* \rangle \right) \langle \mathbf{X}_i, \bar{\Delta}_{2,v} \rangle \\
&\quad - 2 \sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i - \langle \mathbf{X}_i, \Theta_2^* \rangle \right) \langle \mathbf{X}_i, \bar{\Delta}_{1,v} \rangle + \frac{2\tau^*}{v} \sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \langle \mathbf{X}_i, \Theta_2^* \rangle \right) \langle \mathbf{X}_i, \Delta^* \rangle \\
&\quad + \frac{2\tau^*}{v} \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle \langle \mathbf{X}_i, \bar{\Delta}_{1,v} \rangle \\
&\geq_{(1)} c_1^* N_0 \|\Delta^*\|_F^2 - c_2^* R_{\max}(d_1 + d_2)^{1 - \frac{q}{2} \vee \frac{1}{k}} N_{1,v}^{\frac{q}{2} \vee \frac{1}{k}},
\end{aligned} \tag{D.17}$$

where the inequality (1) is derived by Step 1 of the proof of Theorem 4.2, because of the same decomposition

terms before. Thus, we only need to focus on term II and it can be decompose as

$$\begin{aligned}
II &= -2 \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \boldsymbol{\Theta}_{1,v} \rangle - 2 \sum_{i \in \mathcal{I}_{2,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \boldsymbol{\Theta}_{2,v} \rangle \\
&= -2 \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Delta}}_{1,v} \rangle - 2 \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{1,v} \rangle \\
&\quad - 2 \sum_{i \in \mathcal{I}_{2,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Delta}}_{2,v} \rangle - 2 \sum_{i \in \mathcal{I}_{2,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle \\
&= -2 \underbrace{\sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Delta}}_{1,v} \rangle}_{U_4} - 2 \underbrace{\sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{1,v} \rangle}_{U_3} \\
&\quad - 2 \underbrace{\sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Delta}}_{2,v} \rangle}_{U_2} - 2 \underbrace{\sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle}_{U_1} \\
&\quad - 2 \underbrace{\sum_{i \in \mathcal{I}_{2,v}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Delta}}_{2,v} \rangle}_{U_6} - 2 \underbrace{\sum_{i \in \mathcal{I}_{2,v}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle}_{U_5}
\end{aligned}$$

The above inequality can be written as

$$II \geq -|U_1| - |U_2| - |U_3| - |U_4| - |U_5| - |U_6|, \quad (\text{D.18})$$

so it suffices to bound each U_j in absolute value. We first focus on the term $|U_1|$ and define $\gamma_{\min} = \min(\gamma_{1,v}, \gamma_{2,u})$, $N_{\min} = \min(N_{1,v}, N_{2,u})$. And its upper bound can be derived as

$$\begin{aligned}
|U_1| &\leq 2 \sum_{i \in \mathcal{I}_0} \left| \tilde{Y}_i(v) - \tilde{Y}_i(u) \right| \left| \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle \right| \\
&\leq_{(1)} \frac{c_\gamma M^{\frac{1}{2k}}}{k} (d_1 + d_2)^{-\frac{1}{2k}} |N_{1,v} - N_{2,u}|^{\frac{1}{2k}} \sum_{i \in \mathcal{I}_0} \underbrace{\mathbf{1}_{\{|Y_i| > \gamma_{\min}\}} \left| \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle \right|}_{W_{1i}}. \quad (\text{D.19})
\end{aligned}$$

Based on the definition of W_{1i} , we define $V_{1i} = W_{1i} - \mathbb{E}W_{1i}$. Recalling Markov's inequality, for any $t > 0$, we have

$$\mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} [W_{1i} - \mathbb{E}W_{1i}] \right| > t \right\} \leq \frac{\mathbb{E} \left(\sum_{i \in \mathcal{I}_0} V_{1i} \right)^{2k}}{t^{2k}}. \quad (\text{D.20})$$

Then, the remaining thing is derive the upper bounds for the moments. And we decompose the expectation

$\mathbb{E}W_{1i}$ as

$$\begin{aligned}
\mathbb{E}W_{1i} &= \mathbb{E} \left[|\langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle| \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}} \right] \\
&\leq_{(1)} \left[\mathbb{E} \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle^{2k} \right]^{\frac{1}{2k}} [\mathbb{P}(|Y_i| > \gamma_{\min})]^{1 - \frac{1}{2k}} \\
&\leq_{(2)} c_1'' K_0 \left[\frac{\mathbb{E}|Y_i|^{2k}}{\gamma_{\min}^{2k}} \right]^{\frac{2k-1}{2k}} \\
&\leq_{(3)} c_1' (d_1 + d_2)^{1 - \frac{1}{2k}} N_{\min}^{\frac{1}{2k} - 1},
\end{aligned} \tag{D.21}$$

where the inequality (1) is derived by Hölder's inequality; the inequality (2) is derived by Assumption 2, the moment property of sub-Gaussian random variable and Markov's inequality; the inequality (3) is derived by Assumption 2; the constant c_1'' is some suitable constant and the constant c_1' of inequality (3) absorbs the constants before.

To derive the upper bound for $\mathbb{E} \left(\sum_{i \in \mathcal{I}_0} V_{1i} \right)^{2k}$, we need to control $\mathbb{E}W_{1i}^2$ and $\mathbb{E}W_{1i}^{2k}$. By the mean value inequality, we have

$$W_{1i}^2 \leq 2 \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle^2 \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}},$$

and upper bound for its expectation can be derive as

$$\begin{aligned}
\mathbb{E}W_{1i}^2 &\leq 2 \mathbb{E} \left[\langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle^2 \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}} \right] \\
&\leq_{(1)} c_2' (d_1 + d_2)^{1 - \frac{1}{k}} N_{\min}^{\frac{1}{k} - 1},
\end{aligned} \tag{D.22}$$

where the inequality (1) is derived as same as (D.21) and the constant c_2' absorbs the constants before. For W_{1i}^{2k} , we can derive as

$$W_{1i}^{2k} = |\langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle|^{2k} \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}} \leq |\langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle|^{2k}.$$

Thus, the upper bound for its expectation can be derive as

$$\mathbb{E}W_{1i}^{2k} \leq \mathbb{E} \langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{2,v} \rangle^{2k} \leq_{(1)} c_3', \tag{D.23}$$

where the inequality (1) is derived by Assumption 1, 2 and the moments property for sub-Gaussian random variable; the constant c_3' absorbs the constants contained. Then, we derive the upper bound for $\mathbb{E}V_{1i}^2$ and

$\mathbb{E}V_{1i}^{2k}$ and have

$$\mathbb{E}V_{1i}^2 = \mathbb{E}(W_{1i} - \mathbb{E}W_{1i})^2 \leq 2\mathbb{E}W_{1i}^2 = 2c'_2(d_1 + d_2)^{1-\frac{1}{k}} N_{\min}^{\frac{1}{k}-1},$$

where the last inequality is derived by (D.22). For $\mathbb{E}V_{1i}^{2k}$, it is more complex and we derive as

$$\mathbb{E}V_{1i}^{2k} \leq_{(1)} 2^{2k-1} [\mathbb{E}|W_{1i}|^{2k} + |\mathbb{E}W_{1i}|^{2k}] \leq_{(2)} 2^{2k} \mathbb{E}|W_{1i}|^{2k} \leq_{(3)} c'_3 2^{2k},$$

where the inequality (1) is derived by the convexity inequality; the inequality (2) is derived by Jensen's inequality; the inequality (3) is derived by (D.23). Thus, by Rosenthal's inequality (see, eg. [Ibragimov and Sharakhmetov \(1998\)](#)), we have

$$\begin{aligned} \mathbb{E} \left| \sum_{i \in \mathcal{I}_0} V_{1i} \right|^{2k} &\leq 2^{2k} \left[\left(\sum_{i \in \mathcal{I}_0} \mathbb{E}V_{1i}^2 \right)^k + \sum_{i \in \mathcal{I}_0} \mathbb{E}V_{1i}^{2k} \right] \\ &\leq 2^{2k} \left\{ \left[c'_2(d_1 + d_2)^{1-\frac{1}{k}} N_{\min}^{\frac{1}{k}} \right]^k + c'_3 2^{2k} N_0 \right\}. \end{aligned}$$

Recalling (D.20) and taking $t = c'_4(d_1 + d_2)^{1-\frac{1}{2k}} N_{\min}^{\frac{1}{2k}}$, the inequality can be written as

$$\begin{aligned} \mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} V_{1i} \right| > c'_4(d_1 + d_2)^{1-\frac{1}{2k}} N_{\min}^{\frac{1}{2k}} \right\} &\leq \frac{2^{2k} \left\{ \left[c'_2(d_1 + d_2)^{1-\frac{1}{k}} N_{\min}^{\frac{1}{k}} \right]^k + c'_3 2^{2k} N_0 \right\}}{c'_4{}^{2k} (d_1 + d_2)^{2k-1} N_{\min}} \\ &\leq \left(\frac{C_1}{d_1 + d_2} \right)^k. \end{aligned}$$

Recalling (D.19) and (D.21), we have

$$|U_1| \leq c_1(d_1 + d_2)^{1-\frac{1}{k}} |N_{1,v} - N_{2,u}|^{\frac{1}{2k}} N_{\min}^{\frac{1}{2k}} \leq c_1(d_1 + d_2)^{1-\frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}},$$

with probability at least $1 - \left(\frac{C_1}{d_1 + d_2} \right)^k$, where the constants C_1 and c_1 absorb the constants before.

For the second term $|U_2|$, we derive its upper bound as

$$\begin{aligned} |U_2| &\leq_{(1)} \sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right)^2 + \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \bar{\Delta}_{2,v} \rangle^2 \\ &\leq_{(2)} \sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right)^2 + \frac{1}{4} c_1^* N_0 \|\Delta^*\|_F^2, \end{aligned} \tag{D.24}$$

where the inequality (1) is derived by the mean value inequality; the inequality (2) is derived as same as

(D.11). Next, we only need to focus on the first term in RHS of above inequality and we can derive that

$$\sum_{i \in \mathcal{I}_0} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right)^2 \leq_{(1)} c'_1 (d_1 + d_2)^{-\frac{1}{k}} |N_{1,v} - N_{1,u}|^{\frac{1}{k}} \sum_{i \in \mathcal{I}_0} \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}}, \quad (\text{D.25})$$

where the inequality (1) is derived by Lemma [Appendix C.4](#) and c'_1 is the corresponding constant. Let $W_{2i} = \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}}$ and $V_{2i} = W_{2i} - \mathbb{E}W_{2i}$. The upper bound for the moments is easy to derive that

$$\mathbb{E}W_{2i} = \mathbb{E}W_{2i}^2 = \mathbb{E}W_{2i}^{2k} \leq c'_2 (d_1 + d_2) N_{\min}^{-1},$$

where the last inequality is derived by Markov's inequality and c'_2 is some suitable constant. Similarly, we can derive the upper bound for $\mathbb{E}V_{2i}^2$ and $\mathbb{E}V_{2i}^{2k}$ as

$$\mathbb{E}V_{2i}^2 \leq 2c'_2 (d_1 + d_2) N_{\min}^{-1}, \mathbb{E}V_{2i}^{2k} \leq 2^{2k} c'_2 (d_1 + d_2) N_{\min}^{-1}.$$

By Rosenthal's inequality, we can obtain

$$\begin{aligned} \mathbb{E} \left| \sum_{i \in \mathcal{I}_0} V_{2i} \right|^{2k} &\leq 2^{2k} \left[\left(\sum_{i \in \mathcal{I}_0} \mathbb{E}V_{2i}^2 \right)^k + \sum_{i \in \mathcal{I}_0} \mathbb{E}V_{2i}^{2k} \right] \\ &\leq 2^{4k} \left[N_0^k (d_1 + d_2)^k N_{\min}^{-k} + N_0 (d_1 + d_2) N_{\min}^{-1} \right]. \end{aligned}$$

Then, by Markov's inequality, we can obtain

$$\begin{aligned} \mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} V_{2i} \right| > c'_3 (d_1 + d_2) \right\} &\leq \frac{\mathbb{E} \left| \sum_{i \in \mathcal{I}_0} V_{2i} \right|^{2k}}{c'_3{}^{2k} (d_1 + d_2)^{2k}} \\ &\leq \frac{2^{4k} \left[N_0^k (d_1 + d_2)^k N_{\min}^{-k} + N_0 (d_1 + d_2) N_{\min}^{-1} \right]}{c'_3{}^{2k} (d_1 + d_2)^{2k}} \\ &\leq \left(\frac{C_2}{d_1 + d_2} \right)^k. \end{aligned}$$

Recalling (D.24) and (D.25), we can obtain that

$$|U_2| \leq c_2 (d_1 + d_2)^{1-\frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}} + \frac{1}{4} c_1^* N_0 \|\Delta^*\|_F^2.$$

For the third term $|U_3|$, we derive its upper bound following the process of the term $|U_1|$ as

$$\begin{aligned} |U_3| &\leq 2 \sum_{i \in \mathcal{I}_{1,u}} \left| \tilde{Y}_i(v) - \tilde{Y}_i(u) \right| \left| \langle \mathbf{X}_i, \boldsymbol{\Theta}_{(i)}^* \rangle \right| \\ &\leq \frac{c_\gamma M^{\frac{1}{2k}}}{k} (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0 \sum_{i \in \mathcal{I}_{1,u}} \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}} |\langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{1,v} \rangle|. \end{aligned} \quad (\text{D.26})$$

Define $W_{3i} = \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}} |\langle \mathbf{X}_i, \bar{\boldsymbol{\Theta}}_{1,v} \rangle|$ and $V_{3i} = W_{3i} - \mathbb{E}W_{3i}$. Then, we derive the upper bounds for the moments and easy to obtain

$$\mathbb{E}W_{3i} \leq c'_1 (d_1 + d_2)^{1-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1}, \mathbb{E}W_{3i}^2 \leq c'_2 (d_1 + d_2)^{1-\frac{1}{k}} N_{1,u}^{\frac{1}{k}-1}, \mathbb{E}W_{3i}^{2k} \leq c'_3 2^{2k}.$$

And the moments for V_{4i} can be derived as

$$\mathbb{E}V_{3i}^2 \leq c'_4 (d_1 + d_2)^{1-\frac{1}{k}} N_{1,u}^{\frac{1}{k}-1}, \mathbb{E}V_{3i}^{2k} \leq c'_5 2^{2k}.$$

By Rosenthal' inequality, we have

$$\begin{aligned} \mathbb{E} \left| \sum_{i \in \mathcal{I}_{1,u}} V_{3i} \right|^{2k} &\leq 2^{2k} \left[\left(\sum_{i \in \mathcal{I}_{1,u}} \mathbb{E}V_{3i}^2 \right)^k + \sum_{i \in \mathcal{I}_{1,u}} \mathbb{E}V_{3i}^{2k} \right] \\ &\leq 2^{2k} [(d_1 + d_2)^{k-1} N_{1,u} + 2^{2k} N_{1,u}]. \end{aligned}$$

Then, by Markov's inequality, we have

$$\begin{aligned} \mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_{1,u}} V_{3i} \right| > c'_6 (d_1 + d_2)^{1-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}} \right\} &\leq \frac{\mathbb{E} \left| \sum_{i \in \mathcal{I}_{1,u}} V_{3i} \right|^{2k}}{c'^{2k}_6 (d_1 + d_2)^{2k-1} N_{1,u}} \\ &\leq \left(\frac{C_3}{d_1 + d_2} \right)^k. \end{aligned}$$

Recalling (D.26), its upper bound can be written as

$$|U_3| \leq c_3 (d_1 + d_2)^{1-\frac{1}{k}} N_{1,u}^{\frac{1}{k}}.$$

For the forth term $|U_4|$, we derive its upper bound as

$$\begin{aligned}
|U_4| &\leq 2 \left\| \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) X_i \right\|_{op} \|\bar{\Delta}_{1,v}\|_* \\
&\leq_{(1)} 2c_* R_{\max} (d_1 + d_2)^{\frac{1}{2} - \frac{q}{2}} N_{1,v}^{\frac{1}{2} - \frac{q}{2}} \left\| \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) X_i \right\|_{op} \\
&\leq_{(2)} c'_1 \|\Delta^*\|_F \left\| \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) X_i \right\|_{op}, \tag{D.27}
\end{aligned}$$

where the inequality (1) is derived by Theorem 4.1; the inequality (2) is derived by Assumption 3. The remain needed to be controlled is the operator norm. By Lemma Appendix B.1, the operator norm can be bounded by $\max_{\mathbf{u} \in \mathcal{N}^{d_1-1}} \max_{\mathbf{v} \in \mathcal{N}^{d_2-1}} \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) Z_i$. By Assumption 1 and the definition of sub-Gaussian random variable, Z_i is a sub-Gaussian random variable with $\|Z_i\|_{\psi_2} \lesssim K_0$. Because the truncated samples are bounded, we have

$$\left\| \tilde{Y}_i(v) - \tilde{Y}_i(u) \right\|_{\psi_2} \lesssim (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0.$$

Thus, $(\tilde{Y}_i(v) - \tilde{Y}_i(u))Z_i$ is a sub-Exponential random variable with $\|(\tilde{Y}_i(v) - \tilde{Y}_i(u))Z_i\|_{\psi_1} \lesssim (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0 K_0$.

Then, we derive the upper bound for its expectation as

$$\begin{aligned}
\mathbb{E} \left[\left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) Z_i \right] &\lesssim_{(1)} (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0 \mathbb{E}[|Z_i| \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}}] \\
&\lesssim_{(2)} (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0 \sqrt{\mathbb{E} Z_i^2} \sqrt{\mathbb{E} \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}}} \\
&\lesssim_{(3)} (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0^{\frac{1}{2}},
\end{aligned}$$

where the inequality (1) is derived by Lemma Appendix C.4; the inequality (2) is derived by Cauchy-Schwarz's inequality; the inequality (3) is derived by Markov's inequality. By Bernstein's inequality (see, eg. Theorem 2.8.1 in Vershynin (2018)), we have

$$\begin{aligned}
&\mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) Z_i \right| > c'_1 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,u}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \right\} \\
&\leq 2 \exp \left\{ -C_4 \frac{(d_1 + d_2)^{1 - \frac{1}{k}} N_{1,u}^{\frac{1}{k}} N_0}{N_{1,u} (d_1 + d_2)^{-\frac{1}{k}} N_{1,u}^{\frac{1}{k}-2} N_0^2} \wedge \frac{(d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,u}^{\frac{1}{2k}} N_0^{\frac{1}{2}}}{(d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0} \right\} \\
&\leq 2 \exp \{-C_4 (d_1 + d_2)\}.
\end{aligned}$$

Taking the union bound, we can derive that

$$\begin{aligned} & \mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_{1,u}} \left(\tilde{Y}_i(v) - \tilde{Y}_i(u) \right) Z_i \right| > c'_1 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,u}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \right\} \\ & \leq 2 \exp\{-(C_4 - \log 8)(d_1 + d_2)\}. \end{aligned}$$

Recalling (D.27), we can obtain

$$|U_4| \leq c_4 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,u}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \|\Delta^*\|_F.$$

The remaining two terms $|U_5|$ and $|U_6|$ are symmetric to $|U_3|$ and $|U_4|$. Thus, following the same proof as above, we obtain that

$$\mathbb{P} \left\{ |U_5| \leq c_5 (d_1 + d_2)^{1 - \frac{1}{k}} N_{2,v}^{\frac{1}{k}} \right\} \geq 1 - \left(\frac{C_5}{d_1 + d_2} \right)^k,$$

and

$$\mathbb{P} \left\{ |U_6| \leq c_6 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{2,v}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \|\Delta^*\|_F \right\} \geq 1 - 2 \exp\{-(C_6 - \log 8)(d_1 + d_2)\}.$$

Recalling (D.18) and all of the upper bounds we derive for each term, we have

$$\begin{aligned} II & \geq -c_1 (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}} - c_2 (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}} \\ & \quad - \frac{1}{4} c_1^* N_0 \|\Delta^*\|_F^2 - c_3 (d_1 + d_2)^{1 - \frac{1}{k}} N_{1,u}^{\frac{1}{k}} - c_4 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{1,u}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \|\Delta^*\|_F \\ & \quad - c_5 (d_1 + d_2)^{1 - \frac{1}{k}} N_{2,v}^{\frac{1}{k}} - c_6 (d_1 + d_2)^{\frac{1}{2} - \frac{1}{2k}} N_{2,v}^{\frac{1}{2k}} N_0^{\frac{1}{2}} \|\Delta^*\|_F \\ & \geq_{(1)} -\frac{1}{4} c_1^* N_0 \|\Delta^*\|_F^2 - (c_1 + c_2 + c_4 + c_5) (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}} \\ & \quad - \frac{3}{8} c_1^* N_0 \|\Delta^*\|_F^2 - \frac{2(c_4^2 + c_6^2)}{c_1^*} (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}} \\ & \geq_{(2)} -\frac{5}{8} c_1^* N_0 \|\Delta^*\|_F^2 - c_3^* (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}}, \end{aligned}$$

where the inequality (1) is derived by the mean value inequality $2ab \leq a^2 + b^2$ for any $a, b \in \mathbb{R}$; the constant c_3^* of inequality (2) absorbs the constants before. Combining (D.16) and (D.17), the lower bound for the loss

function can be written as

$$\begin{aligned}
f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u}) &\geq c_1^* N_0 \|\Delta^*\|_F^2 - c_2^* R_{\max}(d_1 + d_2)^{1 - \frac{q}{2} \vee \frac{1}{k}} N_{1,v}^{\frac{q}{2} \vee \frac{1}{k}} \\
&\quad - \frac{5}{8} c_1^* N_0 \|\Delta^*\|_F^2 - c_3^* (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}} \\
&\geq \frac{3}{8} c_1^* N_0 \|\Delta^*\|_F^2 - c_2^* R_{\max}(d_1 + d_2)^{1 - \frac{q}{2} \vee \frac{1}{k}} N_{1,v}^{\frac{q}{2} \vee \frac{1}{k}} \\
&\quad - c_3^* (d_1 + d_2)^{1 - \frac{1}{k}} [\max(N_{1,v}, N_{2,u})]^{\frac{1}{k}}.
\end{aligned}$$

Because of $N_0 > \tau^* N/2$ and Assumption 3, the lower bound of $f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u})$ is larger than 0 absolutely for a fixed pair of (u, v) . Then, we consider to take the union bound.

To apply a union bound, we analyze the cardinality of the set $\mathcal{D}_1$, following the argument in Kovács et al. (2024), and obtain $|\mathcal{D}_1| \leq 4 \log N$. Consequently, the above result can be expressed with high probability over the entire search domain $\mathcal{D}_1$ as

$$\begin{aligned}
&\mathbb{P}\{f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u}) \geq 0 \text{ for any } (u, v) \in \mathcal{D}_1\} \\
&\geq 1 - C_1 \log N \exp\{-C_2(d_1 + d_2)\} - C_3 \log N \left(\frac{C_4}{d_1 + d_2}\right)^k,
\end{aligned}$$

which implies that the left-hand interval of the probe point v will be retained, and the true change-point τ^* remains within the preserved subinterval.

Step 2. In this step, we analyze whether the true change-point τ^* remains within the interval as $k > k^*$. Note that the true change-point τ^* falls outside the interval if and only if the probe points lie on the same side of τ^* . To characterize this event, we define the probe points sets as

$$\begin{aligned}
\mathcal{D}_2 &= \{(u, v) : u, v \text{ are the probe points of } I_{k_{os}} \text{ for } k > k^* \text{ and } u, v \text{ are in the same side of } \tau^* \\
&\quad \text{with } |u - \tau^*| < |v - \tau^*|\}.
\end{aligned}$$

We also denote $\mathcal{I}_0 = \{i \mid \min(u, v) \leq t_i \leq \max(u, v)\}$, $N_0 = |\mathcal{I}_0|$ and let

$$\eta_0 = c R_{\max}(d_1 + d_2)^{1 - \frac{q}{2} \vee \frac{1}{k}} N_{1,v}^{\frac{q}{2} \vee \frac{1}{k}} \|\Delta^*\|_F^{-2}.$$

Without loss of generality, we assume that the probe points (u, v) lie to the right of the true change-point

τ^* . Then, following the proof in Step 1, we have

$$\begin{aligned} & \mathbb{P}\{f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u}) < 0 \text{ with } |u - v|N \geq \eta_0\} \\ & \leq C_1 \exp\{-C_2(d_1 + d_2)\} + C_3 \left(\frac{C_4}{d_1 + d_2}\right)^k. \end{aligned}$$

To obtain a union bound, we analyze the cardinality of the set $\mathcal{D}_2$ following the approach in [Kovács et al. \(2024\)](#) and establish that $|\mathcal{D}_2| \leq (4 \log N)^2$. Then, we have

$$\begin{aligned} & \mathbb{P}\{f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u}) \geq 0 \text{ for any } (u, v) \in \mathcal{D}_2 \text{ with } |u - v|N \geq \eta_0\} \\ & \leq 1 - C_1(\log N)^2 \exp\{-C_2(d_1 + d_2)\} - C_3(\log N)^2 \left(\frac{C_4}{d_1 + d_2}\right)^k, \end{aligned}$$

which implies that, with high probability, the true change-point τ^* remains within the retained interval provided that the distance between the two probe points is sufficiently large. However, the bias $|\tau - \tau^*|$ must also be considered when the distance becomes smaller than η_* . According to the OS algorithm, the distance between the two probe points $|u - v|$ decreases as the iteration number k increases. At the first time the distance falls below η_0 , the total length of the retained interval is less than $4\eta_0$, and with high probability, the true change-point still lies within this interval. Therefore, we conclude that $|\tau - \tau^*| < 4\eta_0$. Then, we combine the probabilities and completes the proof.

Appendix E. Proof of Lemmas for Section [Appendix D](#)

Appendix E.1. Proof of Lemma [Appendix C.1](#)

To prove Lemma [Appendix C.1](#), we establish separate upper and lower bounds for $\sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \Delta \rangle^2$. We begin with a fixed τ . Let $\mathcal{R}(t) = \{\Delta \in \mathbb{R}^{d_1 \times d_2} \mid \|\Delta\|_* \leq t\|\Delta\|_F\}$ and define $\overline{\Delta} = \frac{\Delta}{\|\Delta\|_F}$. We first find the upper bound of

$$\begin{aligned} & \sup_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \\ & \leq \sup_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \mathbb{E} \left[\sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] + \sup_{\Delta \in \mathcal{R}(t)} \left\{ \frac{1}{N_{s,\tau}} \sum \left[\langle \mathbf{X}_i, \overline{\Delta} \rangle^2 - \mathbb{E} \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] \right\}. \end{aligned} \tag{E.1}$$

For the first term, by Assumption [1](#), it is naturally to obtain

$$\sup_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \mathbb{E} \left[\sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] \leq \bar{\lambda}.$$

For the second term, we consider to utilize Theorem 5.5 of [Dirksen \(2015\)](#) and first derive that

$$\sigma = \sup_{\Delta \in \mathcal{R}(t)} \left[\frac{1}{N_{s,\tau}} \sum \|\langle \mathbf{X}_i, \overline{\Delta} \rangle\|_{\psi_2}^4 \right]^{\frac{1}{2}} \lesssim K_0^2, b = \sup_{\Delta \in \mathcal{R}(t)} \max_{i \in \mathcal{I}_{s,\tau}} \|\langle \mathbf{X}_i, \overline{\Delta} \rangle\|_{\psi_2}^2 \lesssim K_0^2,$$

where the inequalities are derived by Assumption 1 and properties of sub-Gaussian random variables. Then, for $q = 2, 3, \dots$, we can calculate that

$$\sup_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \sum \mathbb{E} \left| \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 - \mathbb{E} \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right|^q \leq \sup_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \sum \mathbb{E} \langle \mathbf{X}_i, \overline{\Delta} \rangle^{2q} \lesssim \frac{q!}{2} \sigma^2 b^{q-2}.$$

Thus, the condition is satisfied and we can obtain

$$\mathbb{P} \left\{ \sup_{\Delta \in \mathcal{R}(t)} \left| \frac{1}{N_{s,\tau}} \sum \left[\langle \mathbf{X}_i, \overline{\Delta} \rangle^2 - \mathbb{E} \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] \right| \geq c_1 \left[\frac{1}{N_{s,\tau}} \gamma_2^2(\mathcal{R}(t), d_{\psi_2}) + \frac{\overline{\Delta}_{\psi_2}(\mathcal{R}(t))}{\sqrt{N}} \gamma_2^2(\mathcal{R}(t), d_{\psi_2}) \right. \right. \\ \left. \left. + \frac{\sigma \sqrt{d_1 + d_2}}{N_{s,\tau}} + \frac{b(d_1 + d_2)}{N_{s,\tau}} \right] \right\} \leq \exp\{-C_1(d_1 + d_2)\}, \quad (\text{E.2})$$

where the constants $c_1, C_1 > 0$ and the associated radius of $\mathcal{R}(t)$ is defined as

$$\overline{\Delta}_{\psi_2}(\mathcal{R}(t)) = \sup_{\Delta \in \mathcal{R}(t)} \max_{i \in \mathcal{I}_{s,\tau}} \|\langle \mathbf{X}_i, \overline{\Delta} \rangle\|_{\psi_2} \lesssim K_0.$$

For the γ_2 -functional of $(\mathcal{R}(t), d_{\psi_2})$, we first derive the upper bound for metric as

$$d_{\psi_2}(s_1, s_2) = \max_{i \in \mathcal{I}_{s,\tau}} \|\langle \mathbf{X}_i, \overline{\Delta}_{s_1} \rangle - \langle \mathbf{X}_i, \overline{\Delta}_{s_2} \rangle\|_{\psi_2} \leq K_0 \max_{i \in \mathcal{I}_{s,\tau}} \|\langle \mathbf{X}'_i, \overline{\Delta}_{s_1} \rangle - \langle \mathbf{X}'_i, \overline{\Delta}_{s_2} \rangle\|_2 \leq K_0 d(s_1, s_2),$$

where $\mathbf{X}'_i \in \mathbb{R}^{d_1 \times d_2}$ satisfying that $\text{vec}(\mathbf{X}'_i) \sim N_{d_1 d_2}(\mathbf{0}, \mathbf{I}_{d_1 d_2})$. Thus, by Talagrand's majorizing measure theorem (eg. Theorem 8.6.1 of [Vershynin \(2018\)](#)), we can derive

$$\gamma_2(\mathcal{R}(t), d_{\psi_2}) \leq K_0 \gamma_2(\mathcal{R}(t), d) \lesssim \mathbb{E} \sup_{\Delta \in \mathcal{R}(t)} \langle \mathbf{X}'_i, \overline{\Delta} \rangle \leq 2 \sup_{\Delta \in \mathcal{R}(t)} \mathbb{E} |\langle \mathbf{X}'_i, \overline{\Delta} \rangle| \leq 2 \sup_{\Delta \in \mathcal{R}(t)} \|\overline{\Delta}\|_* \mathbb{E} \|\mathbf{X}'_i\|_{op}.$$

By Lemma H.1 of [Negahban and Wainwright \(2011\)](#), we have $\mathbb{E} \|\mathbf{X}'_i\|_{op} \lesssim \sqrt{d_1 + d_2}$. By the definition of $\mathcal{R}(t)$, we have $\sup_{\Delta \in \mathcal{R}(t)} \|\overline{\Delta}\|_* \leq t$. Then, the upper bound for the γ_2 -functional is derived as $\gamma_2(\mathcal{R}(t), d_{\psi_2}) \lesssim t \sqrt{d_1 + d_2}$.

Recalling (E.2), with probability at least $1 - \exp\{-C_1(d_1 + d_2)\}$, we have

$$\begin{aligned} \sup_{\Delta \in \mathcal{R}(t)} \left| \frac{1}{N_{s,\tau}} \sum \left[\langle \mathbf{X}_i, \overline{\Delta} \rangle^2 - \mathbb{E} \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] \right| &\leq c_1 \left[\sqrt{\frac{d_1 + d_2}{N_{s,\tau}}} t + \frac{(d_1 + d_2)t^2}{N_{s,\tau}} \right] \\ &\leq \frac{1}{2} \bar{\lambda} + \frac{c'(d_1 + d_2)}{N_{s,\tau}} t^2 \quad (\text{by } 2ab \leq a^2 + b^2, \forall a, b \in \mathbb{R}). \end{aligned}$$

Recalling (E.1), we can derive that with probability at least $1 - \exp\{-C_1(d_1 + d_2)t^2\}$, we have

$$\sup_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \leq \frac{3}{2} \bar{\lambda} + \frac{c'(d_1 + d_2)}{N_{s,\tau}} t^2.$$

To obtain a uniform upper bound over all $\Delta \in \mathbb{R}^{d_1 \times d_2}$, we employ a peeling argument. Define $\mu := \frac{c'(d_1 + d_2)}{N_{s,\tau}}$ and we have $\frac{c'(d_1 + d_2)}{N_{s,\tau}} t^2 \in [\mu, \infty)$. Define a series of set

$$\mathcal{R}_m = \left\{ \Delta \in \mathbb{R}^{d_1 \times d_2} \mid 2^{m-1} \mu \leq \frac{c'(d_1 + d_2) \|\Delta\|_*^2}{N_{s,\tau} \|\Delta\|_F^2} \leq 2^m \mu \right\}, m = 1, 2, \dots$$

Define an event as

$$\mathcal{E} = \left\{ \sup_{\Delta \in \mathbb{R}^{d_1 \times d_2}} \frac{1}{N_{s,\tau}} \sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 > \frac{3}{2} \bar{\lambda} + \frac{c'(d_1 + d_2) \|\Delta\|_*^2}{N_{s,\tau} \|\Delta\|_F^2} \right\}.$$

Thus, we have

$$\begin{aligned} \mathbb{P}(\mathcal{E}) &\leq \sum_{m=1}^{\infty} \mathbb{P} \left\{ \sup_{\Delta \in \mathcal{R}_m} \frac{1}{N_{s,\tau}} \sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 > \frac{3}{2} \bar{\lambda} + \frac{c'(d_1 + d_2) \|\Delta\|_*^2}{N_{s,\tau} \|\Delta\|_F^2} \right\} \\ &\leq \sum_{m=1}^{\infty} \mathbb{P} \left\{ \sup_{\|\Delta\|_*^2 \leq 2^m \|\Delta\|_F^2} \frac{1}{N_{s,\tau}} \sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 > \frac{3}{2} \bar{\lambda} + 2^{m-1} \mu \right\} \\ &\leq \sum_{m=1}^{\infty} \exp\{-C_1(d_1 + d_2)2^m\} \leq \frac{\exp\{-C_1(d_1 + d_2)\}}{1 - \exp\{-C_1(d_1 + d_2)\}}. \end{aligned}$$

According to Assumption 3, the dimensions $(d_1 + d_2)$ is large enough and we have $\mathbb{P}(\mathcal{E}) \leq \exp\{-C_1(d_1 + d_2)\}$.

For the lower bound, we have

$$\begin{aligned} &\inf_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \\ &\geq \inf_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \mathbb{E} \left[\sum \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] - \sup_{\Delta \in \mathcal{R}(t)} \left| \frac{1}{N_{s,\tau}} \sum \left[\langle \mathbf{X}_i, \overline{\Delta} \rangle^2 - \mathbb{E} \langle \mathbf{X}_i, \overline{\Delta} \rangle^2 \right] \right|. \end{aligned}$$

The lower bound for the first term in the right-hand side can be derived by Assumption 1 as

$$\inf_{\Delta \in \mathcal{R}(t)} \frac{1}{N_{s,\tau}} \mathbb{E} \left[\sum \langle \mathbf{X}_i, \Delta \rangle^2 \right] \geq \lambda.$$

And the second term can be controlled by the same way as before. Thus, regarding the union bound for any $\tau \in \mathcal{T}$, we have

$$\mathbb{P} \left\{ \frac{1}{2} N_{s,\tau} \lambda \|\Delta\|_F^2 - c'(d_1 + d_2) \|\Delta\|_*^2 \leq \sum_{i \in \mathcal{I}_{s,\tau}} \langle \mathbf{X}_i, \Delta \rangle^2 \leq \frac{3}{2} N_{s,\tau} \lambda \|\Delta\|_F^2 + c'(d_1 + d_2) \|\Delta\|_*^2, \right. \\ \left. \forall \tau \in \mathcal{T}, \forall \Delta \in \mathbb{R}^{d_1 \times d_2} \right\} \leq 1 - 2N \exp\{-C_1(d_1 + d_2)\}.$$

Appendix E.2. Proof of Lemma Appendix C.2

For simplicity, we write $\tilde{Y}_i(\tau)$ as $\tilde{Y}_i$. Without loss of generality, we consider $\tau > \tau^*$ and the gradient term can be decomposed as

$$\begin{aligned} \nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) &= -2 \sum_{i \in \mathcal{I}_{1,\tau}} \tilde{Y}_i \mathbf{X}_i + 2 \sum_{i \in \mathcal{I}_{1,\tau}} \mathbf{X}_i \langle \mathbf{X}_i, \Theta_1^* \rangle \\ &= -2 \left\{ \underbrace{\sum_{i \in \mathcal{I}_{1,\tau}} [\tilde{Y}_i \mathbf{X}_i - \mathbb{E}(\tilde{Y}_i \mathbf{X}_i)]}_{U_1} + \underbrace{\sum_{i \in \mathcal{I}_{1,\tau}} \mathbb{E}[(\tilde{Y}_i - Y_i) \mathbf{X}_i]}_{U_2} \right. \\ &\quad + \underbrace{\sum_{i \in \mathcal{I}_{1,\tau^*}} [\mathbb{E}(Y_i \mathbf{X}_i) - \langle \mathbf{X}_i, \Theta_1^* \rangle \mathbf{X}_i]}_{U_3} + \underbrace{\sum_{i \in \mathcal{I}_0} [\mathbb{E}(Y_i \mathbf{X}_i) - \langle \mathbf{X}_i, \Theta_2^* \rangle \mathbf{X}_i]}_{U_4} \\ &\quad \left. - \underbrace{\frac{\tau^*}{\tau} \sum_{i \in \mathcal{I}_0} \langle \mathbf{X}_i, \Delta^* \rangle \mathbf{X}_i + \frac{\tau - \tau^*}{\tau} \sum_{i \in \mathcal{I}_{1,\tau^*}} \langle \mathbf{X}_i, \Delta^* \rangle \mathbf{X}_i}_{U_5} \right\}. \end{aligned} \tag{E.3}$$

By triangle inequality, we can also decompose the operator norm of $\nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau})$ as

$$\left\| \nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) \right\|_{op} \leq 2\|U_1\|_{op} + 2\|U_2\|_{op} + 2\|U_3 + U_4\|_{op} + 2\|U_5\|_{op}.$$

Otherwise, the upper bound for operator norm can be transformed by covering net theory. By Lemma [Appendix B.1](#), we can obtain that

$$\begin{aligned} \left\| \nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) \right\|_{op} &\leq \frac{32}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \mathbf{u}^\top U_1 \mathbf{v} + \frac{32}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \mathbf{u}^\top U_2 \mathbf{v} \\ &\quad + \frac{32}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \mathbf{u}^\top (U_3 + U_4) \mathbf{v} + \frac{32}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \mathbf{u}^\top U_5 \mathbf{v}. \end{aligned} \quad (\text{E.4})$$

For fixed $\mathbf{u}$ and $\mathbf{v}$, we write $\mathbf{u}^\top \mathbf{X}_i \mathbf{v}$ as Z_i . And by Assumption [1](#) and $Z_i = \text{vec}(\mathbf{X}_i)^\top \text{vec}(\mathbf{u}\mathbf{v}^\top)$, Z_i is an sub-Gaussian random variable with $\|Z_i\|_{\psi_2} \lesssim K_0$. Then, we bound each term of the RHS above.

For the first term $\mathbf{u}^\top U_1 \mathbf{v}$, we consider to utilize the Bernstein's inequality (see, eg. Theorem 2.10 in [Boucheron et al. \(2013\)](#)) because of the bounded $\tilde{Y}_i$. We start with deriving the upper bound for the p -th moment of $\tilde{Y}_i Z_i$.

$$\mathbb{E}|\tilde{Y}_i Z_i|^p \leq_{(1)} \gamma_{1,\tau}^{p-2} \mathbb{E}(Y_i^2 |Z_i|^p) =_{(2)} \gamma_{1,\tau}^{p-2} \mathbb{E} \left[\langle \mathbf{X}_i, \Theta_{(i)}^* \rangle^2 |Z_i|^p + \varepsilon_i^2 |Z_i|^p \right], \quad (\text{E.5})$$

where the inequality (1) is derived by the definition of truncation; the equation (2) is derived by Assumption [1](#) that $\mathbb{E}(\mathbf{X}_i) = 0$ and $\mathbb{E}(\varepsilon_i | \mathbf{X}_i) = 0$. Then, we derive the upper bound for the two terms of RHS.

$$\begin{aligned} \mathbb{E} \left[\langle \mathbf{X}_i, \Theta_{(i)}^* \rangle^2 |Z_i|^p \right] &\leq_{(1)} \sqrt{\mathbb{E} \langle \mathbf{X}_i, \Theta_{(i)}^* \rangle^4} \sqrt{\mathbb{E} |Z_i|^{2p}} \lesssim_{(2)} K_0^2 \|\Theta_{(i)}^*\|_F^2 K_0^p (\sqrt{2p})^p \\ &\lesssim_{(3)} K_0^2 \frac{M^{\frac{1}{k}}}{\lambda} (K_0 \sqrt{2p})^p, \end{aligned}$$

where the inequality (1) is derived by Cauchy-Schwarz inequality; the inequality (2) is derived by Assumption [1](#) and Proposition 2.5.2 in [Vershynin \(2018\)](#); the inequality (3) is based on Assumption [1](#) and [2](#) and we can derive that $M^{\frac{1}{k}} \geq \mathbb{E} Y_i^2 \geq \mathbb{E} \langle \mathbf{X}_i, \Theta_{(i)}^* \rangle^2 \geq \lambda \|\Theta_{(i)}^*\|_F^2$.

$$\begin{aligned} \mathbb{E}[\varepsilon_i^2 |Z_i|^p] &=_{(1)} \mathbb{E} \left[\mathbb{E}(\varepsilon_i^2 | \mathbf{X}_i) |Z_i|^p \right] \leq_{(2)} \left\{ \mathbb{E} \left[\mathbb{E}(\varepsilon_i^2 | \mathbf{X}_i) \right]^k \right\}^{\frac{1}{k}} \left[\mathbb{E} |Z_i|^{\frac{pk}{k-1}} \right]^{\frac{k-1}{k}} \\ &\leq_{(3)} \left\{ \mathbb{E} \left[\mathbb{E}(Y_i^2 | \mathbf{X}_i) \right]^k \right\}^{\frac{1}{k}} \left[\mathbb{E} |Z_i|^{\frac{pk}{k-1}} \right]^{\frac{k-1}{k}} \lesssim_{(4)} M^{\frac{1}{k}} \left(K_0 \sqrt{\frac{pk}{k-1}} \right)^p, \end{aligned}$$

where the equation (1) is derived by law of iterated expectations; the inequality (2) is derived by Hölder's inequality; the inequality (3) is derived by the equation $\mathbb{E}(Y_i^2 | \mathbf{X}_i) = \langle \mathbf{X}_i, \Theta_{(i)}^* \rangle^2 + \mathbb{E}(\varepsilon_i^2 | \mathbf{X}_i)$; the inequality (4) is derived by Assumption [1](#), Proposition 2.5.2 in [Vershynin \(2018\)](#) and the inequality $\mathbb{E} \left[\mathbb{E}(Y_i^2 | \mathbf{X}_i) \right]^k \leq \mathbb{E} \left[\mathbb{E}(Y_i^{2k} | \mathbf{X}_i) \right] = \mathbb{E} Y_i^{2k} \leq M$.

Recalling (E.5), the upper bound for p -th moment can be written as

$$\begin{aligned}\mathbb{E} \left| \tilde{Y}_i Z_i \right|^p &\lesssim \gamma_{1,\tau}^{p-2} \left[K_0^2 \frac{M^{\frac{1}{k}}}{\lambda} (K_0 \sqrt{2p})^p + M^{\frac{1}{k}} \left(K_0 \sqrt{\frac{pk}{k-1}} \right)^p \right] \\ &\lesssim \frac{p! M^{\frac{1}{k}}}{2} \left(\gamma_{1,\tau} K_0 e \sqrt{\frac{2k}{k-1}} \right)^{p-2},\end{aligned}$$

where the last inequality is derived by Stirling's approximation. Then, by Bernstein's inequality (see, eg. Theorem 2.10 in [Boucheron et al. \(2013\)](#)), we obtain that

$$\mathbb{P} \left\{ \sum_{i \in \mathcal{I}_{1,\tau}} [\tilde{Y}_i Z_i - \mathbb{E}(\tilde{Y}_i Z_i)] \geq c_1 \left(\sqrt{2N_{1,\tau} M^{\frac{1}{k}} t} + \gamma_{1,\tau} t \right) \right\} \leq 2 \exp(-t), \quad \forall t > 0.$$

Taking $t = c_1(d_1 + d_2)$ and $\gamma_{1,\tau} = c_\gamma \left(\frac{MN_{1,\tau}}{d_1 + d_2} \right)^{\frac{1}{2k}}$, with probability at most $2 \exp\{-C_1(d_1 + d_2)\}$, we have

$$\begin{aligned}\sum_{i \in \mathcal{I}_{1,\tau}} [\tilde{Y}_i Z_i - \mathbb{E}(\tilde{Y}_i Z_i)] &\geq c_1 \sqrt{2N_{1,\tau} M^{\frac{1}{k}} (d_1 + d_2)} + c_2 \sqrt{M^{\frac{1}{k}} N_{1,\tau} (d_1 + d_2)} \left(\frac{d_1 + d_2}{N_{1,\tau}} \right)^{\frac{k-1}{2k}} \\ &\geq_{(1)} c_1 \sqrt{2N_{1,\tau} M^{\frac{1}{k}} (d_1 + d_2)} \left[1 + \frac{c_2(1 - c_\tau)^{\frac{k-1}{2k}}}{c_1} \left(\frac{d_1 + d_2}{N_{1,\tau^*}} \right)^{\frac{k-1}{2k}} \right] \\ &\gtrsim_{(2)} \sqrt{M^{\frac{1}{k}} N_{1,\tau} (d_1 + d_2)},\end{aligned}$$

where the inequality (1) is derived by search domain condition; the inequality (2) is derived by Assumption 3; and c_1, c_2 are some constants. Based on above inequality, a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$ yields that

$$\mathbb{P} \left\{ \|U_1\|_{op} \geq c_1 \sqrt{M^{\frac{1}{k}} N_{1,\tau} (d_1 + d_2)} \right\} \leq 2 \exp \{ -(C_1 - \log 8)(d_1 + d_2) \}.$$

For the second term $\mathbf{u}^\top U_2 \mathbf{v}$, we can derive

$$\mathbb{E} \left[(\tilde{Y}_i - Y_i) Z_i \right] \leq_{(1)} \mathbb{E} \left[|Y_i Z_i| \mathbf{1}_{\{|Y_i| > \gamma_{1,\tau}\}} \right] \leq_{(2)} \sqrt{\mathbb{E}(Y_i^2 Z_i^2) \mathbb{P}(|Y_i| > \gamma_{1,\tau})}, \quad (\text{E.6})$$

where the inequality (1) is derived by the definition of truncation $\tilde{Y}_i = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{1,\tau})$; the inequality is derived by Cauchy-Schwarz's inequality. Focusing on the RHS above, we can obtain $\mathbb{E}(Y_i^2 Z_i^2) \lesssim M^{\frac{1}{k}}$ by

the proof of the p -th moment for the first term. Moreover, by Markov's inequality, we have

$$\mathbb{P}\{|Y_i| > \gamma_{1,\tau}\} \leq \frac{\mathbb{E}|Y_i|^{2k}}{\gamma_{1,\tau}^{2k}} \lesssim \frac{d_1 + d_2}{N_{1,\tau}}.$$

Recalling (E.6), we can derive the upper bound as

$$\mathbb{E}\left[(\tilde{Y}_i - Y_i)Z_i\right] \lesssim \sqrt{\frac{M^{\frac{1}{k}}(d_1 + d_2)}{N_{1,\tau}}}. \quad (\text{E.7})$$

Thus, we have

$$\|U_2\|_{op} \leq c_2 \sqrt{M^{\frac{1}{k}}(d_1 + d_2)N_{1,\tau}},$$

where c_2 is some constant.

For the third and forth term $\mathbf{u}^\top(U_3 + U_4)\mathbf{v}$, we can obtain that $\langle \mathbf{X}_i, \boldsymbol{\Theta}_1^* \rangle Z_i$ and $\langle \mathbf{X}_i, \boldsymbol{\Theta}_2^* \rangle Z_i$ are both sub-Exponential random variable with $\|\langle \mathbf{X}_i, \boldsymbol{\Theta}_s^* \rangle Z_i\|_{\psi_1} \lesssim K_0^2 \|\boldsymbol{\Theta}_s^*\|_F$ by the relationship between sub-Gaussian and sub-Exponential (see, eg. Lemma 2.7.7 in Vershynin (2018)). Thus, by Bernstein's inequality (see, eg. Theorem 2.8.1 in Vershynin (2018)), we have

$$\mathbb{P}\left\{|\mathbf{u}^\top(U_3 + U_4)\mathbf{v}| \geq c_3 \sqrt{M^{\frac{1}{k}}(d_1 + d_2)N_{1,\tau}}\right\} \leq 2 \exp\{-C_2(d_1 + d_2)\}.$$

As same as the proof for the first term, a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$ yields that

$$\mathbb{P}\left\{\|U_3 + U_4\|_{op} \geq c_3 \sqrt{M^{\frac{1}{k}}(d_1 + d_2)N_{1,\tau}}\right\} \leq 2 \exp\{-(C_2 - \log 8)(d_1 + d_2)\}.$$

For the fifth term $\mathbf{u}^\top U_5 \mathbf{v}$, we define the coefficient before each one as a vector

$$\boldsymbol{\alpha} = \left[\frac{\tau - \tau^*}{\tau}, \dots, \frac{\tau - \tau^*}{\tau}, -\frac{\tau^*}{\tau}, \dots, -\frac{\tau^*}{\tau} \right]^\top,$$

and we have $\|\boldsymbol{\alpha}\|_\infty \leq 1$, $\|\boldsymbol{\alpha}\|_2^2 \leq N_{1,\tau}$. By the relationship between sub-Gaussian and sub-Exponential (see, eg. Lemma 2.7.7 in Vershynin (2018)), $\langle \mathbf{X}_i, \boldsymbol{\Delta}^* \rangle Z_i$ is a sub-Exponential random variable with $\|\langle \mathbf{X}_i, \boldsymbol{\Delta}^* \rangle Z_i\|_{\psi_1} \lesssim K_0^2 \|\boldsymbol{\Delta}^*\|_F$. Then, by Bernstein's inequality (see, eg. Theorem 2.5.2 in Vershynin (2018)), we have

$$\mathbb{P}\left\{\left|\sum_{i \in \mathcal{I}_{1,\tau}} \alpha_i \langle \mathbf{X}_i, \boldsymbol{\Delta}^* \rangle Z_i\right| \geq c_4 \sqrt{M^{\frac{1}{k}}(d_1 + d_2)N_{1,\tau}}\right\} \leq 2 \exp\{-C_3(d_1 + d_2)\},$$

where c_3 and C_4 are some constants. As same as the proof for the first term, a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$ yields that

$$\mathbb{P} \left\{ \|U_5\|_{op} \geq c_4 \sqrt{M^{\frac{1}{k}}(d_1 + d_2)N_{1,\tau}} \right\} \leq 2 \exp\{-(C_3 - \log 8)(d_1 + d_2)\}.$$

Then, recalling (E.4), with probability at least $1 - C_1 \exp\{-C_2(d_1 + d_2)\}$, we have

$$\left\| \nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) \right\|_{op} \leq \frac{1}{2} c_\lambda \sqrt{(d_1 + d_2)N_{1,\tau}} = \frac{1}{2} \lambda_{1,\tau},$$

where the probability absorbs all of the probabilities derived for U_1 - U_5 and the constant c_λ absorbs the constants c_1 - c_4 and $\sqrt{M^{\frac{1}{k}}}$.

Otherwise, the upper bound for $\left\| \nabla \tilde{L}_{2,\tau}(\bar{\Theta}_{2,\tau}) \right\|_{op}$ and the case $\tau \leq \tau^*$ can also be derived as same. For any $\tau \in \mathcal{T}$, we take the union bound as $|\mathcal{T}| = O(N)$ and complete the proof.

Appendix E.3. Proof of Lemma Appendix C.3

By introducing the intermediate term $\mathbb{E}(\tilde{Y}_i | \mathbf{X}_i)$, we obtain the following decomposition.

$$\tilde{Y}_i - \langle \mathbf{X}_i, \Theta^* \rangle = \underbrace{\tilde{Y}_i - \mathbb{E}(\tilde{Y}_i | \mathbf{X}_i)}_{U_{1i}} + \underbrace{\mathbb{E}(\tilde{Y}_i | \mathbf{X}_i) - \langle \mathbf{X}_i, \Theta^* \rangle}_{U_{2i}}.$$

Then, we have

$$\left| \sum_{i \in \mathcal{I}} (\tilde{Y}_i - \langle \mathbf{X}_i, \Theta^* \rangle) Z_i \right| \leq \left| \sum_{i \in \mathcal{I}} U_{1i} Z_i \right| + \left| \sum_{i \in \mathcal{I}} U_{2i} Z_i \right|. \quad (\text{E.8})$$

To bound the RHS, we control two terms separately. For the first term, we can derive the upper bound for $|U_{1i}|$ as

$$|U_{1i}| \leq_{(1)} |\tilde{Y}_i| + |\mathbb{E}(\tilde{Y}_i | \mathbf{X}_i)| \leq_{(2)} 2\gamma_{s,\tau},$$

where the inequality (1) is derived by the triangle inequality; the inequality (2) is derived by the definition of truncation. It is easily to obtain that U_{1i} is a sub-Gaussian random variable from the definition of sub-Gaussian variable (see, eg. Definition 2.5.6 in Vershynin (2018)). And its sub-Gaussian norm is controlled by $\gamma_{s,\tau}$ as $\|U_{1i}\|_{\psi_2} \lesssim \gamma_{s,\tau}$. Thus, by the relationship between sub-Gaussian and sub-Exponential random variable (see, eg. Lemma 2.7.7 in Vershynin (2018)), we can obtain that $U_{1i}Z_i$ is a sub-Exponential random variable with $K_0\gamma_{s,\tau}$. Then, we utilize the Bernstein's inequality to bound $|\sum_{i \in \mathcal{I}} U_{1i}Z_i|$ and compute its

expectation first.

$$\mathbb{E}(U_{1i}Z_i) = \mathbb{E}[\mathbb{E}(U_{1i}Z_i|\mathbf{X}_i)] = \mathbb{E}\left\{Z_i\mathbb{E}\left[\tilde{Y}_i - \mathbb{E}(\tilde{Y}_i|\mathbf{X}_i)\middle|\mathbf{X}_i\right]\right\} = 0.$$

Then, by Bernstein's inequality (see, eg. Theorem 2.8.1 in [Vershynin \(2018\)](#)), we can derive the upper bound as

$$\mathbb{P}\left\{\left|\sum_{i \in \mathcal{I}} U_{1i}Z_i\right| \geq c_1 K_0 \gamma_{s,\tau} (\sqrt{N_I t} + t)\right\} \leq 2 \exp\{-C_1 t\}, \forall t > 0,$$

where c_1 and C_1 are some suitable constants. Taking $t = d_1 + d_2$, the above inequality can be transformed into

$$\mathbb{P}\left\{\left|\sum_{i \in \mathcal{I}} U_{1i}Z_i\right| \geq c_1 K_0 \gamma_{s,\tau} \sqrt{N_I(d_1 + d_2)}\right\} \leq 2 \exp\{-C_1(d_1 + d_2)\}.$$

For the second term in the RHS of (E.8), we consider the similar proof to the first term. And we can derive that

$$\|U_{2i}\|_{\psi_2} \leq \left\|\mathbb{E}(\tilde{Y}_i|\mathbf{X}_i)\right\|_{\psi_2} + \|\langle \mathbf{X}_i, \boldsymbol{\Theta}^* \rangle\|_{\psi_2} \lesssim \gamma_{s,\tau} + K_0 \|\boldsymbol{\Theta}^*\|_F,$$

which implies that U_{2i} is a sub-Gaussian random variable. From the relationship between sub-Gaussian and sub-Exponential random variable (see, eg. Lemma 2.7.7 in [Vershynin \(2018\)](#)), $U_{2i}Z_i$ is sub-Exponential random variable with $\|U_{2i}Z_i\|_{\psi_1} \lesssim K_0(\gamma_{s,\tau} + K_0 \|\boldsymbol{\Theta}^*\|_F)$. Then, we derive the upper bound for its expectation.

$$\mathbb{E}(U_{2i}Z_i) = \mathbb{E}\left[\mathbb{E}(\tilde{Y}_i - Y_i|\mathbf{X}_i)Z_i\right] = \mathbb{E}\left[(\tilde{Y}_i - Y_i)Z_i\right] \lesssim \sqrt{\frac{d_1 + d_2}{N_{s,\tau}}},$$

where the last inequality is derived by (E.7). By Bernstein's inequality, we can derive that

$$\mathbb{P}\left\{\left|\sum_{i \in \mathcal{I}} [U_{2i}Z_i - \mathbb{E}(U_{2i}Z_i)]\right| \geq t\right\} \leq 2 \exp\left\{-\frac{t^2}{N_I K_0^2 (\gamma_{s,\tau} + K_0 \|\boldsymbol{\Theta}^*\|_F)^2} \wedge \frac{t}{K_0 (\gamma_{s,\tau} + K_0 \|\boldsymbol{\Theta}^*\|_F)}\right\}, \forall t > 0,$$

where c_2 and C_2 are some constants. Taking $t = c_2 \gamma_{s,\tau} \sqrt{N_I(d_1 + d_2)}$, the above inequality can be transformed into

$$\mathbb{P}\left\{\left|\sum_{i \in \mathcal{I}} U_{2i}Z_i\right| \geq c_2 \gamma_{s,\tau} \sqrt{N_I(d_1 + d_2)}\right\} \leq 2 \exp\{-C_2(d_1 + d_2)\}.$$

Recalling (E.8), we can obtain that

$$\left| \sum_{i \in \mathcal{I}} (\tilde{Y}_i - \langle \mathbf{X}_i, \boldsymbol{\Theta}^* \rangle) Z_i \right| \lesssim \gamma_{s,\tau} \sqrt{N_I(d_1 + d_2)}.$$

Appendix E.4. Proof of Lemma Appendix C.4

The proof consists of three parts corresponding to the samples belonging to different segments.

We first consider the case with $i \in \mathcal{I}_{1,u}$ and the truncated response variable can be written as

$$\tilde{Y}_i(v) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{1,v}), \quad \text{and} \quad \tilde{Y}_i(u) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{1,u}).$$

Then, we can derive the upper bound for the difference between truncated samples as

$$\begin{aligned} |\tilde{Y}_i(v) - \tilde{Y}_i(u)| &\leq_{(1)} (\gamma_{1,v} - \gamma_{1,u}) \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}} \\ &\leq_{(2)} c_\gamma \left(\frac{MN}{d_1 + d_2} \right)^{\frac{1}{2k}} \left(v^{\frac{1}{2k}} - u^{\frac{1}{2k}} \right) \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}} \\ &\leq_{(3)} \frac{c_\gamma}{2k} \left(\frac{MN}{d_1 + d_2} \right)^{\frac{1}{2k}} \zeta^{\frac{1}{2k}-1} (v - u) \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}} \\ &\leq_{(4)} \frac{c_\gamma}{2k} \left(\frac{MN}{d_1 + d_2} \right)^{\frac{1}{2k}} u^{\frac{1}{2k}-1} (v - u) \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}} \\ &\leq_{(5)} \frac{c_\gamma M^{\frac{1}{2k}}}{2k} (d_1 + d_2)^{-\frac{1}{2k}} N_{1,u}^{\frac{1}{2k}-1} N_0 \mathbf{1}_{\{|Y_i| > \gamma_{1,u}\}}, \end{aligned}$$

where the inequality (1) is derived by the definition of truncation; the inequality (2) is derived by the definition of truncation parameter $\gamma_{1,v}$ and $\gamma_{1,u}$; the inequality (3) is derived by the mean value theorem and $\zeta \in (u, v)$; the inequality (4) is derived by $u < \zeta$ and $1/2k < 1$; the inequality (5) is derived by multiplying and dividing N .

Next, we consider the case with $i \in \mathcal{I}_0$ and the truncated response variable can be written as

$$\tilde{Y}_i(v) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{1,v}), \quad \text{and} \quad \tilde{Y}_i(u) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{2,u}).$$

And define $\gamma_{\min} = \min(\gamma_{1,v}, \gamma_{2,u})$. Then, we can derive the upper bound for the difference between truncated

samples as

$$\begin{aligned} |\tilde{Y}_i(v) - \tilde{Y}_i(u)| &\leq_{(1)} |\gamma_{1,v} - \gamma_{2,u}| \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}} \\ &\leq_{(2)} c_\gamma \left(\frac{M}{d_1 + d_2} \right)^{\frac{1}{2k}} |N_{1,v} - N_{2,u}|^{\frac{1}{2k}} \mathbf{1}_{\{|Y_i| > \gamma_{\min}\}}, \end{aligned}$$

where the inequality (1) is derived by the definition of truncation; the inequality (2) is derived by the definition of truncation parameter $\gamma_{1,v}$ and $\gamma_{2,u}$ and Hölder's inequality.

Finally, we consider the case with $i \in \mathcal{I}_{2,v}$ and the truncated response variable can be written as

$$\tilde{Y}_i(v) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{2,v}), \quad \text{and} \quad \tilde{Y}_i(u) = \text{sgn}(Y_i) \min(|Y_i|, \gamma_{2,u}).$$

Then, we can derive the upper bound for the difference between truncated sample as

$$\begin{aligned} |\tilde{Y}_i(v) - \tilde{Y}_i(u)| &\leq_{(1)} (\gamma_{2,u} - \gamma_{2,v}) \mathbf{1}_{\{|Y_i| > \gamma_{2,v}\}} \\ &\leq_{(2)} c_\gamma \left(\frac{MN}{d_1 + d_2} \right)^{\frac{1}{2k}} \left[(1-u)^{\frac{1}{2k}} - (1-v)^{\frac{1}{2k}} \right] \mathbf{1}_{\{|Y_i| > \gamma_{2,v}\}} \\ &\leq_{(3)} \frac{c_\gamma M^{\frac{1}{2k}}}{2k} \left(\frac{M}{d_1 + d_2} \right)^{\frac{1}{2k}} (d_1 + d_2)^{-\frac{1}{2k}} N_{2,v}^{\frac{1}{2k}-1} N_0 \mathbf{1}_{\{|Y_i| > \gamma_{2,v}\}}, \end{aligned}$$

where the inequality (1) is derived by the definition of truncation; the inequality (2) is derived by the definition of truncation parameter $\gamma_{2,v}$ and $\gamma_{2,u}$; the inequality (3) is derived by the mean value theorem.

Appendix F. Lemmas for Section [Appendix G](#)

Lemma Appendix F.1. Suppose Assumption [4](#) and [6](#) hold. There exist constants C_1, C_2 and then with the probability at least $1 - C_1 \exp\{-C_2 d_1\}$, we have

$$\frac{1}{2} n_{s,\tau} \lambda \|\Delta\|_F^2 - c'(d_1 + d_2) \|\Delta\|_*^2 \leq \sum_{i \in \mathcal{I}_{s,\tau}} \|\Delta^\top \mathbf{x}_i\|_2^2 \leq \frac{3}{2} n_{s,\tau} \bar{\lambda} \|\Delta\|_F^2 + c'(d_1 + d_2) \|\Delta\|_*^2,$$

for all $\tau \in \mathcal{T}$ and $s = 1, 2$, where $\Delta \in \mathbb{R}^{d_1 \times d_2}$ is any matrix and c' is some positive constant.

Lemma Appendix F.2. Suppose that Assumption [4-6](#) hold. There exists some positive constants C_1-C_4 and with probability at least $1 - C_1 d^{-C_2} - C_3 \exp\{-C_4 d\}$, the inequality

$$\|\nabla \tilde{L}_{s,\tau}(\bar{\Theta}_{s,\tau})\|_{\text{op}} \leq \frac{1}{2} \lambda_{s,\tau},$$

holds uniformly for all $\tau \in \mathcal{T}$ and $s = 1, 2$, where $\lambda_{s,\tau} = c_\lambda \sqrt{(d_1 + d_2)n_{s,\tau}}$ and $\gamma_{s,\tau} = c_\gamma \sqrt{d_2} \left(\frac{n_{s,\tau}}{d \log d} \right)^{1/2k}$ with constants c_λ and c_γ .

Lemma Appendix F.3. Suppose that we have samples $(\mathbf{y}_i, \mathbf{x}_i), i \in \mathcal{I}$ and $\mathbf{y}_i = \boldsymbol{\Theta}^{*\top} \mathbf{x}_i + \boldsymbol{\varepsilon}_i$. Define the truncated response vector as $\tilde{\mathbf{y}}_i = \min(\gamma_{s,\tau}, \|\mathbf{y}_i\|_2) \mathbf{y}_i / \|\mathbf{y}_i\|_2$ and $\gamma_{s,\tau} = c_\gamma \sqrt{d_2} \left(\frac{n_{s,\tau}}{d \log d} \right)^{1/2k}$. Define the random vector $\mathbf{z}_i \in \mathbb{R}^{d_2}$ as $\mathbf{A}^\top \mathbf{x}_i$, where $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$ is a matrix with $\|\mathbf{A}\|_F = 1$. If $n_I = |\mathcal{I}| \gtrsim d \log d$, then there exists some positive constants C_1 - C_4 and with probability at least $1 - C_1 d^{-C_2} - C_3 \exp\{-C_4 d\}$, we have

$$\sum_{i \in \mathcal{I}} \mathbf{z}_i^\top (\tilde{\mathbf{y}}_i - \boldsymbol{\Theta}^{*\top} \mathbf{x}_i) \lesssim (d \log d)^{\frac{1}{2} - \frac{1}{2k}} n_{s,\tau}^{\frac{1}{2k}} n_I^{\frac{1}{2}} + \|\mathbf{A}\|_* (d \log d)^{\frac{1}{2}} n_I^{\frac{1}{2}}.$$

Appendix G. Proof of main results for MR case

Appendix G.1. Proof of Theorem 4.5

The proof of Theorem 4.5 follows the proof of Theorem 4.1 in Section Appendix D.1 exactly, except that $\lambda_{s,\tau}$ is chosen as $c_\lambda \sqrt{n_{s,\tau} d \log d}$ and the loss comparison is written as

$$\begin{aligned} -2 \sum_{i \in \mathcal{I}_{s,\tau}} \mathbf{x}_i^\top \hat{\boldsymbol{\Theta}}_{s,\tau} \tilde{\mathbf{y}}_i + \sum_{i \in \mathcal{I}_{s,\tau}} \|\hat{\boldsymbol{\Theta}}_{s,\tau}^\top \mathbf{x}_i\|_2^2 &\leq -2 \sum_{i \in \mathcal{I}_{s,\tau}} \mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{s,\tau} \tilde{\mathbf{y}}_i + \sum_{i \in \mathcal{I}_{s,\tau}} \|\bar{\boldsymbol{\Theta}}_{s,\tau}^\top \mathbf{x}_i\|_2^2 \\ \langle \nabla \tilde{L}_{s,\tau}(\bar{\boldsymbol{\Theta}}_{s,\tau}), \bar{\boldsymbol{\Delta}}_{s,\tau} \rangle + \sum_{i \in \mathcal{I}_{s,\tau}} \|\bar{\boldsymbol{\Delta}}_{s,\tau}^\top \mathbf{x}_i\|_2^2 &\leq \lambda_{s,\tau} \left[\|\bar{\boldsymbol{\Theta}}_{s,\tau}\|_* - \|\hat{\boldsymbol{\Theta}}_{s,\tau}\|_* \right]. \end{aligned}$$

Appendix G.2. Proof of Theorem 4.6

The proof of Theorem 4.6 follows the proof sketch of Theorem 4.2 in Section Appendix D.2. But the details are different and we introduce it as follow. The symbols are same as Section Appendix D.2. Denote the truncated samples $\tilde{\mathbf{y}}_i(\tau)$ as $\tilde{\mathbf{y}}_i$ and denote $|\mathcal{I}_0|$ as n_0 . Without loss of generality, suppose $\tau > \tau^*$ and assume that

$$|\tau - \tau^*|n > cR_{\max} (d \log d)^{1 - \frac{q}{2} \vee \frac{1}{k}} n_{1,\tau}^{\frac{q}{2} \vee \frac{1}{k}} \|\boldsymbol{\Delta}^*\|_F^{-2}.$$

Then, we decompose the difference of loss function as

$$\begin{aligned}
0 &\geq \tilde{L}_{1,\tau}(\Theta_1) + \tilde{L}_{2,\tau}(\Theta_2) - \tilde{L}_{1,\tau^*}(\Theta_1) - \tilde{L}_{2,\tau^*}(\Theta_2) \\
&= 2 \underbrace{\sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \bar{\Delta}_2 (\tilde{\mathbf{y}}_i - \Theta_2^{*\top} \mathbf{x}_i)}_{U_2} - 2 \underbrace{\sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \bar{\Delta}_1 (\tilde{\mathbf{y}}_i - \Theta_2^{*\top} \mathbf{x}_i)}_{U_3} \\
&\quad + 2 \underbrace{\frac{\tau^*}{\tau} \sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \Delta^* (\tilde{\mathbf{y}}_i - \Theta_2^{*\top} \mathbf{x}_i)}_{U_4} + 2 \underbrace{\frac{\tau^*}{\tau} \sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \Delta^* \bar{\Delta}_1^\top \mathbf{x}_i}_{U_5} \\
&\quad - \underbrace{\sum_{i \in \mathcal{I}_0} \|\bar{\Delta}_2^\top \mathbf{x}_i\|_2^2}_{U_1} + \underbrace{\left(\frac{\tau^*}{\tau}\right)^2 \sum_{i \in \mathcal{I}_0} \|\Delta^{*\top} \mathbf{x}_i\|_2^2}_{U_6}.
\end{aligned}$$

Then, the above inequality can be simplified as

$$U_6 \leq U_1 + |U_2| + |U_3| + |U_4| + |U_5|. \quad (\text{G.1})$$

For term U_6 , we first derive its expectation as

$$\mathbb{E} \|\Delta^{*\top} \mathbf{x}_i\|_2^2 \geq \rho_{\min} [\mathbb{E}(\mathbf{x}_i \mathbf{x}_i^\top)] \|\Delta^*\|_F^2 \geq \lambda \|\Delta^*\|_F^2,$$

where the last inequality is derived by Assumption 4. It also follows Assumption 4 that $\|\Delta^{*\top} \mathbf{x}_i\|_2^2$ is sub-Exponential random variable with $\|\mathbf{x}_i^\top \Delta^* \Delta^{*\top} \mathbf{x}_i\|_{\psi_1} \lesssim K_0^2 \|\Delta^*\|_{op}^2$. By Bernstein's inequality, we have

$$\begin{aligned}
\mathbb{P}\{U_6 < c_6 n_0 \|\Delta^*\|_F^2\} &\leq \exp \left\{ -C'_1 \min \left[\frac{n_0^2 \|\Delta^*\|_F^4}{n_0 K_0^4 \|\Delta^*\|_{op}^4}, \frac{n_0 \|\Delta^*\|_F^2}{K_0^2 \|\Delta^*\|_{op}^2} \right] \right\} \\
&\leq \exp\{-C_1(d_1 + d_2)\}.
\end{aligned}$$

For the first term U_1 in RHS, it follows the proof of U_1 in Section Appendix D.2 and its upper bound is derived as

$$\begin{aligned}
\sum_{i \in \mathcal{I}_0} \|\bar{\Delta}_2^\top \mathbf{x}_i\|_2^2 &\leq \sum_{i \in \mathcal{I}_{2,\tau^*}} \|\bar{\Delta}_2^\top \mathbf{x}_i\|_2^2 \leq \frac{3}{2} \bar{\lambda} n_{2,\tau^*} \|\bar{\Delta}_2\|_F^2 + 2c'(d_1 + d_2) \|\bar{\Delta}_2\|_*^2 \\
&\leq \frac{1}{4} c_6 n_0 \|\Delta^*\|_F^2.
\end{aligned}$$

For the second term U_2 , we can derive the upper bound as

$$|U_2| = 2 \sum_{i \in \mathcal{I}_0} \langle \mathbf{x}_i (\tilde{\mathbf{y}}_i - \boldsymbol{\Theta}_2^{*\top} \mathbf{x}_i)^\top, \bar{\Delta}_2 \rangle \leq 2 \left\| \sum_{i \in \mathcal{I}_0} \mathbf{x}_i (\tilde{\mathbf{y}}_i - \boldsymbol{\Theta}_2^{*\top} \mathbf{x}_i)^\top \right\|_{op} \|\bar{\Delta}_2\|_*, \quad (\text{G.2})$$

For the first term in RHS, we decompose it as

$$\begin{aligned} \left\| \sum_{i \in \mathcal{I}_0} \mathbf{x}_i (\tilde{\mathbf{y}}_i - \boldsymbol{\Theta}_2^{*\top} \mathbf{x}_i)^\top \right\|_{op} &\leq \left\| \sum_{i \in \mathcal{I}_0} [\mathbf{x}_i \tilde{\mathbf{y}}_i^\top - \mathbb{E}(\mathbf{x}_i \tilde{\mathbf{y}}_i^\top)] \right\|_{op} + \left\| \sum_{i \in \mathcal{I}_0} \mathbb{E}[\mathbf{x}_i (\tilde{\mathbf{y}}_i - \mathbf{y}_i)^\top] \right\|_{op} \\ &\quad + \left\| \sum_{i \in \mathcal{I}_0} [\mathbb{E}(\mathbf{x}_i \mathbf{y}_i^\top) - \mathbf{x}_i \mathbf{x}_i^\top \boldsymbol{\Theta}_2^*] \right\|_{op} \\ &\leq c'_2 (d \log d)^{\frac{1}{2} - \frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} n_0^{\frac{1}{2}}, \end{aligned}$$

where the last inequality is derived by the proof in Section [Appendix H.2](#). Recalling [\(G.2\)](#), we have

$$\begin{aligned} |U_2| &\leq_{(1)} 2c'_2 (d \log d)^{\frac{1}{2} - \frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} n_0^{\frac{1}{2}} c_* R_{\max} (d \log d)^{\frac{1}{2} - \frac{q}{2}} n_{2,\tau}^{\frac{q}{2} - \frac{1}{2}} \\ &\leq_{(2)} c_2 (d \log d)^{1 - \frac{q}{2} - \frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} n_{2,\tau}^{\frac{q}{2} - \frac{1}{2}} n_0^{\frac{1}{2}}, \end{aligned}$$

where the inequality (1) is derived by Theorem [4.5](#) and the constant c_2 absorbs the constants before. For the third term U_3 , we can derive its upper bound as same as U_2 and obtain

$$|U_3| \leq c_3 (d \log d)^{1 - \frac{q}{2} - \frac{1}{2k}} n_{1,\tau}^{\frac{q}{2} + \frac{1}{2k} - \frac{1}{2}} n_0^{\frac{1}{2}}.$$

For the forth term U_4 , it follows Lemma [Appendix F.3](#) that

$$\begin{aligned} |U_4| &\leq 2 \frac{\tau^*}{\tau} \left| \sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \frac{\Delta^*}{\|\Delta^*\|_F} (\tilde{\mathbf{y}}_i - \boldsymbol{\Theta}_2^{*\top} \mathbf{x}_i) \right| \|\Delta^*\|_F \\ &\lesssim \|\Delta^*\|_F \left[(d \log d)^{\frac{1}{2} - \frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} n_0^{\frac{1}{2}} + \frac{\|\Delta^*\|_*}{\|\Delta^*\|_F} (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \right]. \end{aligned}$$

To bound RHS, we have

$$\begin{aligned} \frac{\|\Delta^*\|_*}{\|\Delta^*\|_F} &\leq_{(1)} \|\Delta^*\|_{S_q}^{\frac{q}{2-q}} \|\Delta^*\|_F^{-\frac{q}{2-q}} \leq [\|\boldsymbol{\Theta}_1^*\|_{S_q}^q + \|\boldsymbol{\Theta}_2^*\|_{S_q}^q]^{\frac{1}{2-q}} \|\Delta^*\|_F^{-\frac{q}{2-q}} \\ &\lesssim_{(2)} R_{\max}^{\frac{1}{2-q}} R_{\max}^{-\frac{q}{2(2-q)}} \left(\frac{d \log d}{\xi^* n} \right)^{-\frac{q}{2(2-q)} (1 - \frac{1}{k} - \frac{q}{2})} \\ &\lesssim_{(3)} R_{\max}^{\frac{1}{2}} (d \log d)^{-\frac{q}{4}} n_{1,\tau}^{\frac{q}{4}}, \end{aligned}$$

where the inequality (1) is derived by Lemma [Appendix B.2](#); the inequality (2) is derived by near low-rank condition and Assumption [6](#); the inequality (3) is derived by search domain condition. Thus, the upper bound for $|U_4|$ can be derived as

$$|U_4| \leq c_4 \left[(d \log d)^{\frac{1}{2} - \frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} + R_{\max}^{\frac{1}{2}} (d \log d)^{\frac{1}{2} - \frac{q}{4}} n_{1,\tau}^{\frac{q}{4}} \right] n_0^{\frac{1}{2}} \|\Delta^*\|_F.$$

For the fifth term U_5 , we can write its upper bound as

$$\begin{aligned} |U_5| &\leq \frac{2\tau^*}{\tau} \left\| \sum_{i \in \mathcal{I}_0} \mathbf{x}_i \mathbf{x}_i^\top \Delta^* \right\|_{op} \|\bar{\Delta}_1\|_* \\ &\leq \frac{32\tau^*}{7\tau} \|\bar{\Delta}_1\|_* \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \sum_{i \in \mathcal{I}_0} \mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \Delta^* \mathbf{v}, \end{aligned}$$

where the last inequality is derived by Lemma [Appendix B.1](#). For fixed $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$, it follows Bernstein's inequality that

$$\begin{aligned} &\mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} \mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \Delta^* \mathbf{v} \right| > c'_5 (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \|\Delta^*\|_F \right\} \\ &\leq 2 \exp \left\{ -C_5 \min \left[\frac{c_5'^2 d \log d n_0^2 \|\Delta^*\|_F^2}{n_0 K_0^4 \|\Delta^*\|_{op}^2}, \frac{c_5' (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \|\Delta^*\|_F}{K_0^2 \|\Delta^*\|_{op}} \right] \right\} \\ &\leq 2 \exp \{-C_5(d_1 + d_2)\}. \end{aligned}$$

Taking a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1} \times \mathcal{N}^{d_2}$, the above inequality can be written as

$$\mathbb{P} \left\{ \left\| \sum_{i \in \mathcal{I}_0} \mathbf{x}_i \mathbf{x}_i^\top \Delta^* \right\|_{op} > c'_5 (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \|\Delta^*\|_F \right\} \leq 2 \exp \{-(C_5 - \log 8)(d_1 + d_2)\}.$$

Thus, the upper bound for $|U_5|$ can be derived as

$$\begin{aligned} |U_5| &\leq c'_5 \frac{32\tau^*}{7\tau} \|\bar{\Delta}_1\|_* (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \|\Delta^*\|_F \\ &\leq c_5 R_{\max} (d \log d)^{1 - \frac{q}{2}} n_{1,\tau}^{\frac{q}{2} - \frac{1}{2}} n_0^{\frac{1}{2}} \|\Delta^*\|_F, \end{aligned}$$

where the last inequality is derived by Theorem [4.5](#).

Recalling (G.1), we have

$$\begin{aligned}
c_6 n_0 \|\Delta^*\|_F^2 &\leq \frac{1}{4} c_6 n_0 \|\Delta^*\|_F^2 + c_2 R_{\max} (d \log d)^{1-\frac{q}{2}-\frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} n_{2,\tau}^{\frac{q}{2}-\frac{1}{2}} n_0^{\frac{1}{2}} \\
&\quad + c_3 (d \log d)^{1-\frac{q}{2}-\frac{1}{2k}} n_{1,\tau}^{\frac{q}{2}+\frac{1}{2k}-\frac{1}{2}} n_0^{\frac{1}{2}} \\
&\quad + c_4 \left[(d \log d)^{\frac{1}{2}-\frac{1}{2k}} n_{1,\tau}^{\frac{1}{2k}} + R_{\max}^{\frac{1}{2}} (d \log d)^{\frac{1}{2}-\frac{q}{4}} n_{1,\tau}^{\frac{q}{4}} \right] n_0^{\frac{1}{2}} \|\Delta^*\|_F \\
&\quad + c_5 R_{\max} (d \log d)^{1-\frac{q}{2}} n_{1,\tau}^{\frac{q}{2}-\frac{1}{2}} n_0^{\frac{1}{2}} \|\Delta^*\|_F.
\end{aligned}$$

Following the same process of Section [Appendix D.2](#), the above inequality can be written as

$$\frac{1}{2} c_6 n_0 \|\Delta^*\|_F^2 \leq c'_0 R_{\max} (d \log d)^{1-\frac{q}{2}\vee\frac{1}{k}} n_{1,\tau}^{\frac{q}{2}\vee\frac{1}{k}}.$$

Thus, the assumption for $|\tau - \tau^*|n$ is wrong and we can obtain

$$\begin{aligned}
|\tau - \tau^*| &\leq c R_{\max} (d \log d)^{1-\frac{q}{2}\vee\frac{1}{k}} n_{1,\tau}^{\frac{q}{2}\vee\frac{1}{k}} n^{-1} \|\Delta^*\|_F^{-2} \\
&\leq c R_{\max} (d \log d)^{1-\frac{q}{2}\vee\frac{1}{k}} n^{\frac{q}{2}\vee\frac{1}{k}-1} \|\Delta^*\|_F^{-2}.
\end{aligned}$$

Based on above result, the proof process for coefficient estimation bias is almost as same as Section [Appendix D.2](#).

Appendix G.3. Proof of Theorem 4.7

We follow the same proof framework as in Section [Appendix D.3](#). The only difference arises from the MR setting, where the loss function requires a modified comparison. We therefore present the key arguments

below.

$$\begin{aligned}
& f_v(\boldsymbol{\Theta}_{1,v}, \boldsymbol{\Theta}_{2,v}) - f_u(\boldsymbol{\Theta}_{1,u}, \boldsymbol{\Theta}_{2,u}) \\
& \geq \underbrace{\sum_{s=1}^2 \left[\sum_{i \in \mathcal{I}_{s,v}} \|\tilde{\mathbf{y}}_i(v) - \boldsymbol{\Theta}_{s,v}^\top \mathbf{x}_i\|_2^2 - \sum_{i \in \mathcal{I}_{s,u}} \|\tilde{\mathbf{y}}_i(v) - \boldsymbol{\Theta}_{s,v}^\top \mathbf{x}_i\|_2^2 \right]}_I \\
& \quad - 2 \underbrace{\sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} [\tilde{\mathbf{y}}_i(v) - \tilde{\mathbf{y}}_i(u)]}_{U_1} - 2 \underbrace{\sum_{i \in \mathcal{I}_0} \mathbf{x}_i^\top \bar{\boldsymbol{\Delta}}_{2,v} [\tilde{\mathbf{y}}_i(v) - \tilde{\mathbf{y}}_i(u)]}_{U_2} \\
& \quad - 2 \underbrace{\sum_{i \in \mathcal{I}_{1,u}} \mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{1,v} [\tilde{\mathbf{y}}_i(v) - \tilde{\mathbf{y}}_i(u)]}_{U_3} - 2 \underbrace{\sum_{i \in \mathcal{I}_{1,u}} \mathbf{x}_i^\top \bar{\boldsymbol{\Delta}}_{1,v} [\tilde{\mathbf{y}}_i(v) - \tilde{\mathbf{y}}_i(u)]}_{U_4} \\
& \quad - 2 \underbrace{\sum_{i \in \mathcal{I}_{2,v}} \mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} [\tilde{\mathbf{y}}_i(v) - \tilde{\mathbf{y}}_i(u)]}_{U_5} - 2 \underbrace{\sum_{i \in \mathcal{I}_{2,v}} \mathbf{x}_i^\top \bar{\boldsymbol{\Delta}}_{2,v} [\tilde{\mathbf{y}}_i(v) - \tilde{\mathbf{y}}_i(u)]}_{U_6}.
\end{aligned}$$

To derive the lower bound for term I , we can follow the process of the proof of Theorem 4.7 in Section Appendix G.2. Then, we have

$$I \geq c_1^* n_0 \|\boldsymbol{\Delta}^*\|_F^2 - c_2^* R_{\max} (d \log d)^{1 - \frac{q}{2} \vee \frac{1}{k}} n_{1,v}^{\frac{q}{2} \vee \frac{1}{k}}.$$

For the term U_1 , its upper bound can be written as

$$\begin{aligned}
|U_1| & \leq 2|\gamma_{1,v} - \gamma_{2,u}| \left| \sum_{i \in \mathcal{I}_0} \frac{1}{\|\mathbf{y}_i\|_2} \mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \mathbf{y}_i \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right| \\
& \leq 2d_2^{\frac{1}{2}} (d \log d)^{-\frac{1}{2k}} n_{\min}^{\frac{1}{2k}} \underbrace{\sum_{i \in \mathcal{I}_0} \frac{1}{\|\mathbf{y}_i\|_2} |\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \mathbf{y}_i| \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}}}_{W_{1i}}, \tag{G.3}
\end{aligned}$$

where the last inequality is derived by the similar proof process of Lemma Appendix C.4. Define $V_{1i} = W_{1i} - \mathbb{E}W_{1i}$. Recalling Markov's inequality, for any $t > 0$, we have

$$\mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} [W_{1i} - \mathbb{E}W_{1i}] \right| > t \right\} \leq \frac{\mathbb{E}(\sum_{i \in \mathcal{I}_0} V_{1i})^{2k}}{t^{2k}}. \tag{G.4}$$

Then, we derive the upper bound for the moments. It follows Hölder's inequality that

$$\mathbb{E}W_{1i} \leq [\mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \mathbf{y}_i|^{2k}]^{\frac{1}{2k}} \left\{ \mathbb{E} \left[\|\mathbf{y}_i\|_2^{-\frac{2k}{2k-1}} \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right] \right\}^{\frac{2k-1}{2k}}. \tag{G.5}$$

For the first term, we decompose it as

$$\mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \mathbf{y}_i|^{2k} \leq 2^{2k-1} \left[\mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i|^{2k} + \mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \boldsymbol{\varepsilon}_i|^{2k} \right].$$

The upper bound for term $\mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i|^{2k}$ can be derived by the moment property of sub-Gaussian random variable. Denote the j -th element of $\bar{\boldsymbol{\Theta}}_{2,v}^\top \mathbf{x}_i$ as u_{ij} . We provide the details for the other term that

$$\begin{aligned} \mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \boldsymbol{\varepsilon}_i|^{2k} &= \mathbb{E} \left[\mathbb{E} (|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \boldsymbol{\varepsilon}_i|^{2k} | \mathbf{x}_i) \right] \\ &\leq_{(1)} 2^{2k} \mathbb{E} \left\{ \left[\sum_{j=1}^{d_2} u_{ij}^2 \mathbb{E}(\varepsilon_{ij}^2 | \mathbf{x}_i) \right]^k + \sum_{j=1}^{d_2} |u_{ij}|^{2k} \mathbb{E}[|\varepsilon_{ij}|^{2k} | \mathbf{x}_i] \right\} \\ &\lesssim_{(2)} 2^{2k} M K_0^{2k}, \end{aligned}$$

where the inequality (1) is derived by Rosenthal's inequality; the inequality (2) is derived by Assumption 4 and 5. And we also have

$$\mathbb{E} \left[\|\mathbf{y}_i\|_2^{-\frac{2k}{2k-1}} \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right] \leq \frac{\mathbb{E} \|\mathbf{y}_i\|_2^{2k - \frac{2k}{2k-1}}}{\gamma_{\min}^{2k}} \lesssim d_2^{-\frac{k}{k-1}} (d \log d) n_{\min}^{-1}.$$

Recalling (G.5), the upper bound for $\mathbb{E}W_{1i}$ can be written as

$$\mathbb{E}W_{1i} \leq c'_1 d_2^{-\frac{1}{2}} (d \log d)^{1 - \frac{1}{2k}} n_{\min}^{\frac{1}{2k} - 1}.$$

Following the above process, we can derive the upper bound for $\mathbb{E}W_{1i}^2$ as

$$\begin{aligned} \mathbb{E}W_{1i}^2 &= \mathbb{E} \left[\frac{1}{\|\mathbf{y}_i\|_2^2} |\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \mathbf{y}_i|^2 \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right] \\ &\leq [\mathbb{E}|\mathbf{x}_i^\top \bar{\boldsymbol{\Theta}}_{2,v} \mathbf{y}_i|^{2k}]^{\frac{1}{k}} \left\{ \mathbb{E} \left[\|\mathbf{y}_i\|_2^{-\frac{2k}{2k-1}} \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right] \right\} \\ &\leq c'_2 d_2^{-1} (d \log d)^{1 - \frac{1}{k}} n_{\min}^{\frac{1}{k} - 1}. \end{aligned}$$

We also derive the upper bound for $\mathbb{E}W_{1i}^{2k}$ as

$$\begin{aligned} \mathbb{E}W_{1i}^{2k} &\leq \mathbb{E} \left[\|\mathbf{x}_i \bar{\boldsymbol{\Theta}}_{2,v}\|_2^{2k} \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right] \\ &\leq \sqrt{\mathbb{E} \|\mathbf{x}_i \bar{\boldsymbol{\Theta}}_{2,v}\|_2^{4k}} \sqrt{\mathbb{P}\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \\ &\leq c'_3 (d \log d)^{\frac{1}{2}} n_{\min}^{-\frac{1}{2}}. \end{aligned}$$

Thus, we can obtain the moments of V_{1i} .

$$\mathbb{E}V_{1i}^2 \leq 2\mathbb{E}W_{1i}^2 \leq c'_4 d_2^{-1} (d \log d)^{1-\frac{1}{k}} n_{\min}^{\frac{1}{k}-1},$$

and

$$\mathbb{E}V_{1i}^{2k} \leq 2^{2k} \mathbb{E}W_{1i}^{2k} \leq c'_5 2^{2k} (d \log d)^{\frac{1}{2}} n_{\min}^{-\frac{1}{2}}.$$

It follows Rosenthal's inequality that

$$\begin{aligned} \mathbb{E} \left| \sum_{i \in \mathcal{I}_0} V_{1i} \right|^{2k} &\leq 2^{2k} \left[\left(\sum_{i \in \mathcal{I}_0} \mathbb{E}V_{1i}^2 \right)^k + \sum_{i \in \mathcal{I}_0} \mathbb{E}V_{1i}^{2k} \right] \\ &\leq 2^{2k} \left\{ \left[c'_4 d_2^{-1} (d \log d)^{1-\frac{1}{k}} n_{\min}^{\frac{1}{k}} \right]^k + c'_5 2^{2k} (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \right\}. \end{aligned}$$

Recalling (G.4), we have

$$\begin{aligned} \mathbb{P} \left\{ \left| \sum_{i \in \mathcal{I}_0} W_{1i} \right| > c'_6 d_2^{-\frac{1}{2}} (d \log d)^{1-\frac{1}{2k}} n_{\min}^{\frac{1}{2k}} \right\} \\ \leq \frac{\mathbb{E} \left| \sum_{i \in \mathcal{I}_0} V_{1i} \right|^{2k}}{c_6^{2k} d_2^{-k} (d \log d)^{2k-1} n_{\min}} \leq C_1 (d \log d)^{1-k}. \end{aligned}$$

Recalling (G.3), the upper bound for U_1 can be written as

$$|U_1| \leq c_1 (d \log d)^{1-\frac{1}{k}} n_{\min}^{\frac{1}{k}}.$$

For the term U_2 , its upper bound can be derived as

$$\begin{aligned} |U_2| &\leq 2|\gamma_{1,v} - \gamma_{2,u}| \left\| \sum_{i \in \mathcal{I}_0} \frac{1}{\|\mathbf{y}_2\|_2} \mathbf{x}_i \mathbf{y}_i^\top \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right\|_{op} \|\bar{\Delta}_{2,v}\|_* \\ &\leq c'_1 R_{\max} d_2^{\frac{1}{2}} \left(\frac{d \log d}{n_{\min}} \right)^{\frac{1}{2} - \frac{q}{2} - \frac{1}{2k}} \left\| \sum_{i \in \mathcal{I}_0} \underbrace{\frac{1}{\|\mathbf{y}_2\|_2} \mathbf{x}_i \mathbf{y}_i^\top \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}}}_{W_{2i}} \right\|_{op}, \end{aligned} \quad (\text{G.6})$$

where the last inequality is derived by Theorem 4.5. The remain thing is bounding the operator norm above.

$$\left\| \sum_{i \in \mathcal{I}_0} W_{2i} \right\|_{op} \leq \left\| \sum_{i \in \mathcal{I}_0} [W_{2i} - \mathbb{E}W_{2i}] \right\|_{op} + n_0 \|\mathbb{E}W_{2i}\|_{op}. \quad (\text{G.7})$$

It follows the proof process of Lemma 5 in [Fan et al. \(2021\)](#) that for any $t > 0$,

$$\begin{aligned} & \mathbb{P} \left\{ \left\| \sum_{i \in \mathcal{I}_0} [W_{2i} - \mathbb{E}W_{2i}] \right\|_{op} > t \right\} \\ & \leq 2(d_1 + d_2) \exp \left\{ -C'_2 \min \left[\frac{t^2}{n_0 d_1 / d_2}, \frac{t}{\sqrt{d_1}} \right] \right\}. \end{aligned}$$

Taking $t = c'_2 d_2^{-\frac{1}{2}} (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}}$, the above inequality can be written as

$$\mathbb{P} \left\{ \left\| \sum_{i \in \mathcal{I}_0} [W_{2i} - \mathbb{E}W_{2i}] \right\|_{op} > c'_2 d_2^{-\frac{1}{2}} (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \right\} \leq 2d^{-C_2}.$$

As for the upper bound for $\|\mathbb{E}W_{2i}\|_{op}$, it is easier to derive.

$$\begin{aligned} \|\mathbb{E}W_{2i}\|_{op} & \leq_{(1)} \frac{16}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \mathbb{E} \left[\frac{1}{\|\mathbf{y}_i\|_2} \mathbf{u}^\top \mathbf{x}_i \mathbf{y}_i^\top \mathbf{v} \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{\min}\}} \right] \\ & \leq_{(2)} c'_3 d_2^{-\frac{1}{2}} (d \log d)^{1 - \frac{1}{2k}} n_{\min}^{\frac{1}{2k} - 1}, \end{aligned}$$

where the inequality (1) is derived by Lemma [Appendix B.1](#); the inequality (2) follows the proof of the upper bound for $\mathbb{E}W_{1i}$. Recalling [\(G.6\)](#) and [\(G.7\)](#), we can obtain

$$\begin{aligned} |U_2| & \leq c'_1 R_{\max} d_2^{\frac{1}{2}} \left(\frac{d \log d}{n_{\min}} \right)^{\frac{1}{2} - \frac{q}{2} - \frac{1}{2k}} \left[c'_2 d_2^{-\frac{1}{2}} (d \log d)^{\frac{1}{2}} n_0^{\frac{1}{2}} \right. \\ & \quad \left. + c'_3 d_2^{-\frac{1}{2}} (d \log d)^{1 - \frac{1}{2k}} n_{\min}^{\frac{1}{2k} - 1} n_0 \right] \\ & \leq c_2 R_{\max} (d \log d)^{1 - \frac{q}{2} - \frac{1}{2k}} n_{\min}^{\frac{q}{2} + \frac{1}{2k} - \frac{1}{2}} n_0^{\frac{1}{2}}. \end{aligned}$$

The remaining terms U_3 – U_6 have structures analogous to those of U_1 and U_2 , and can be bounded using

the same arguments. We therefore omit the repetitive details and state their upper bounds directly. And

$$\begin{aligned}
& f_v(\Theta_{1,v}, \Theta_{2,v}) - f_u(\Theta_{1,u}, \Theta_{2,u}) \\
& \geq c_1^* n_0 \|\Delta^*\|_F^2 - c_2^* R_{\max}(d \log d)^{1-\frac{q}{2} \vee \frac{1}{k}} n_{1,v}^{\frac{q}{2} \vee \frac{1}{k}} \\
& \quad - c_1 (d \log d)^{1-\frac{1}{k}} n_{\min}^{\frac{1}{k}} - c_2 R_{\max}(d \log d)^{1-\frac{q}{2}-\frac{1}{2k}} n_{\min}^{\frac{q}{2}+\frac{1}{2k}-\frac{1}{2}} n_0^{\frac{1}{2}} \\
& \quad - c_3 (d \log d)^{1-\frac{1}{k}} n_{1,u}^{\frac{1}{k}} - c_4 R_{\max}(d \log d)^{1-\frac{q}{2}-\frac{1}{2k}} n_{1,u}^{\frac{q}{2}+\frac{1}{2k}-\frac{1}{2}} n_0^{\frac{1}{2}} \\
& \quad - c_5 (d \log d)^{1-\frac{1}{k}} n_{2,v}^{\frac{1}{k}} - c_2 R_{\max}(d \log d)^{1-\frac{q}{2}-\frac{1}{2k}} n_{2,v}^{\frac{q}{2}+\frac{1}{2k}-\frac{1}{2}} n_0^{\frac{1}{2}} \\
& \geq \frac{1}{4} c_1^* n_0 \|\Delta^*\|_F^2 - c_2^* R_{\max}(d \log d)^{1-\frac{q}{2} \vee \frac{1}{k}} n_{1,v}^{\frac{q}{2} \vee \frac{1}{k}} \\
& \quad - c_3^* (d \log d)^{1-\frac{1}{k}} [\max(n_{1,v}, n_{2,u})]^{\frac{1}{k}}.
\end{aligned}$$

Appendix H. Proof of Lemmas for Section [Appendix G](#)

Appendix H.1. Proof of Lemma [Appendix F.1](#)

It is easier to prove the URSC and URSM condition for MR case. First, we fix the candidate change-point τ and it is naturally to derive that

$$\rho_{\min} \left(\frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} \mathbf{x}_i \mathbf{x}_i^\top \right) \|\Delta\|_F^2 \leq \frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} \|\Delta^\top \mathbf{x}_i\|_2^2 \leq \rho_{\max} \left(\frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} \mathbf{x}_i \mathbf{x}_i^\top \right) \|\Delta\|_F^2. \quad (\text{H.1})$$

And we first analyze the lower bound of $\sum_{i \in \mathcal{I}_{s,\tau}} \|\Delta^\top \mathbf{x}_i\|_2^2 / n_{s,\tau}$ to obtain

$$\begin{aligned}
\rho_{\min} \left(\frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} \mathbf{x}_i \mathbf{x}_i^\top \right) &= \rho_{\min} \left[\frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} (\mathbb{E} \mathbf{x}_i \mathbf{x}_i^\top + \mathbf{x}_i \mathbf{x}_i^\top - \mathbb{E} \mathbf{x}_i \mathbf{x}_i^\top) \right] \\
&\geq_{(1)} \lambda - \left\| \frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} (\mathbf{x}_i \mathbf{x}_i^\top - \mathbb{E} \mathbf{x}_i \mathbf{x}_i^\top) \right\|_{op}, \quad (\text{H.2})
\end{aligned}$$

where the inequality (1) is derived by Assumption 4. The operator norm term can be controlled by Theorem 6.5 of [Wainwright \(2019\)](#) and we have

$$\mathbb{P} \left\{ \left\| \frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} (\mathbf{x}_i \mathbf{x}_i^\top - \mathbb{E} \mathbf{x}_i \mathbf{x}_i^\top) \right\|_{op} \geq c_1 K_0^2 \left(\frac{d_1}{n_{s,\tau}} + \sqrt{\frac{d_1}{n_{s,\tau}}} \right) \right\} \leq 2 \exp\{-C_1 d_1\},$$

where c_1 and C_1 are some constants. Recalling (H.1) and (H.2), with probability $1 - 2\exp(-C_1 d_1)$ we have

$$\begin{aligned}
\frac{1}{n_{s,\tau}} \sum_{i \in \mathcal{I}_{s,\tau}} \|\Delta^\top \mathbf{x}_i\|_2^2 &\geq \lambda \|\Delta\|_F^2 - c_1 K_0^2 \left(\frac{d_1}{n_{s,\tau}} + \sqrt{\frac{d_1}{n_{s,\tau}}} \right) \|\Delta\|_F^2 \\
&\geq_{(1)} \lambda \|\Delta\|_F^2 - \left[\frac{1}{2} \lambda + \frac{c_1^2 K_0^4 d_1}{2 \lambda n_{s,\tau}} + \frac{c_1 K_0^2 d_1}{n_{s,\tau}} \right] \|\Delta\|_F^2 \\
&\geq_{(2)} \frac{1}{2} \lambda \|\Delta\|_F^2 - c' \frac{d_1}{n_{s,\tau}} \|\Delta\|_*^2,
\end{aligned}$$

where the inequality (1) is derived by the mean value inequality that $2ab \leq a^2 + b^2$ for any $a, b \in \mathbb{R}$; the constant c' of inequality (2) absorbs the corresponding constants above.

The upper bound of $\sum_{i \in \mathcal{I}_{s,\tau}} \|\Delta^\top \mathbf{x}_i\|_2^2 / n_{s,\tau}$ can be derived as the similar process. For any $\tau \in \mathcal{T}$, we apply the union bound to complete the proof.

Appendix H.2. Proof of Lemma [Appendix F.2](#)

For simplicity, we write $\tilde{\mathbf{y}}_i(\tau)$ as $\tilde{\mathbf{y}}_i$. Without loss of generality, we consider $\tau > \tau^*$ and the gradient term can be decomposed as

$$\begin{aligned}
\nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) &= -2 \sum_{i \in \mathcal{I}_{1,\tau}} \mathbf{x}_i \tilde{\mathbf{y}}_i^\top + 2 \sum_{i \in \mathcal{I}_{1,\tau}} \mathbf{x}_i \mathbf{x}_i^\top \bar{\Theta}_{1,\tau} \\
&= -2 \left\{ \underbrace{\sum_{i \in \mathcal{I}_{1,\tau}} [\mathbf{x}_i \tilde{\mathbf{y}}_i^\top - \mathbb{E}(\mathbf{x}_i \tilde{\mathbf{y}}_i^\top)]}_{U_1} + \underbrace{\sum_{i \in \mathcal{I}_{1,\tau}} \mathbb{E}[\mathbf{x}_i (\tilde{\mathbf{y}}_i - \mathbf{y}_i)^\top]}_{U_2} \right. \\
&\quad \left. + \underbrace{\sum_{i \in \mathcal{I}_{1,\tau}} [\mathbb{E}(\mathbf{x}_i \mathbf{y}_i^\top) - \mathbf{x}_i \mathbf{x}_i^\top \Theta_{(i)}^*]}_{U_3} - \underbrace{\frac{\tau^*}{\tau} \sum_{i \in \mathcal{I}_0} \mathbf{x}_i \mathbf{x}_i^\top \Delta^* + \frac{\tau - \tau^*}{\tau} \sum_{i \in \mathcal{I}_{1,\tau^*}} \mathbf{x}_i \mathbf{x}_i^\top \Delta^*}_{U_4} \right\}.
\end{aligned}$$

By triangle inequality, we can derive the upper bound for the operator norm of $\nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau})$ as

$$\left\| \nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) \right\|_{op} \leq 2\|U_1\|_{op} + 2\|U_2\|_{op} + 2\|U_3\|_{op} + 2\|U_4\|_{op}. \quad (\text{H.3})$$

For the first term U_1 , we denote $\mathbf{Z}_i \in \mathbb{R}^{d_1 \times d_2}$ as $\mathbf{x}_i \tilde{\mathbf{y}}_i^\top - \mathbb{E}(\mathbf{x}_i \tilde{\mathbf{y}}_i^\top)$ and let

$$\mathbf{S}_i = \begin{bmatrix} \mathbf{0} & \mathbf{Z}_i^\top \\ \mathbf{Z}_i & \mathbf{0} \end{bmatrix}.$$

Hence, it follows the proof of Lemma 5 in [Fan et al. \(2021\)](#) that for any $t > 0$,

$$\begin{aligned} & \mathbb{P} \left\{ \|U_1\|_{op} > t \right\} \\ & \leq 2(d_1 + d_2) \exp \left\{ - \min \left[\frac{t^2}{4C_1'' n_{1,\tau}(v_1 \vee v_2)}, \frac{t}{4C_2''(\gamma_{1,\tau}\sqrt{d_1} + \sqrt{Rd_1} + M^{\frac{1}{k}}d)} \right] \right\}, \end{aligned}$$

where $v_1 = C_1' d_1 (M^{\frac{1}{k}} + 4R/\underline{\lambda})$, $v_2 = C_2'(M^{\frac{1}{k}} d_2 + Rd_1/\underline{\lambda})$ and C_1' , C_2' , C_1'' , C_2'' are some constants. Taking $t = c_1 \sqrt{n_{1,\tau} d \log d}$, we drive the terms of the probability as

$$\begin{aligned} \frac{t^2}{4C_1'' n_{1,\tau}(v_1 \vee v_2)} &= \frac{c_1^2 d \log d}{4C_1''(v_1 \vee v_2)} = C_1 \log d, \\ \frac{t}{4C_2''(\gamma_{1,\tau}\sqrt{d_1} + \sqrt{Rd_1} + M^{\frac{1}{k}}d)} &= C_2 n_{1,\tau}^{\frac{1}{2} - \frac{1}{2k}} (d \log d)^{\frac{1}{2k} - \frac{1}{2}} \log d \leq C_2 \log d. \end{aligned}$$

With enough large constants $\min(C_1, C_2) > 4$, we have

$$\mathbb{P} \left\{ \|U_1\|_{op} > c_1 \sqrt{n_{1,\tau} d \log d} \right\} \leq 2d^{-C_1}.$$

For the second term U_2 , it follows Lemma [Appendix B.1](#) that

$$\|U_2\|_{op} \leq \frac{32}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \sum_{i \in \mathcal{I}_{1,\tau}} \mathbb{E}[\mathbf{u}^\top \mathbf{x}_i (\tilde{\mathbf{y}}_i - \mathbf{y}_i)^\top \mathbf{v}].$$

Then, we fix $\mathbf{u}$ and $\mathbf{v}$ to derive the upper bound as

$$\begin{aligned} \mathbb{E}[\mathbf{u}^\top \mathbf{x}_i (\tilde{\mathbf{y}}_i - \mathbf{y}_i)^\top \mathbf{v}] &\leq_{(1)} \mathbb{E} \left[|\mathbf{u}^\top \mathbf{x}_i \mathbf{y}_i^\top \mathbf{v}| \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{1,\tau}\}} \right] \\ &\leq_{(2)} \sqrt{\mathbb{E} |\mathbf{u}^\top \mathbf{x}_i \mathbf{y}_i^\top \mathbf{v}|^2 \mathbb{P}(\|\mathbf{y}_i\|_2 > \gamma_{1,\tau})}, \end{aligned} \tag{H.4}$$

where the inequality (1) is derived by the definition of truncation; the inequality (2) is derived by Cauchy-Schwarz inequality. Focusing on RHS, we derive its upper bound respectively.

$$\begin{aligned} \mathbb{E} |\mathbf{u}^\top \mathbf{x}_i \mathbf{y}_i^\top \mathbf{v}|^2 &= \mathbb{E} \left\{ (\mathbf{u}^\top \mathbf{x}_i)^2 \mathbb{E}[(\mathbf{v}^\top \mathbf{y}_i)^2 | \mathbf{x}_i] \right\} \\ &= \mathbb{E} \left\{ \mathbb{E} \left[(\mathbf{v}^\top \boldsymbol{\Theta}_{(i)}^* \mathbf{x}_i)^2 (\mathbf{u}^\top \mathbf{x}_i)^2 \right] + (\mathbf{u}^\top \mathbf{x}_i)^2 \mathbb{E}[(\mathbf{v}^\top \boldsymbol{\varepsilon}_i)^2 | \mathbf{x}_i] \right\} \\ &\leq 4K_0^2 \left(2K_0 \sqrt{\frac{R}{\underline{\lambda}}} \right)^2 + \frac{2kK_0^2 M^{\frac{1}{k}}}{k-1} = v_3, \end{aligned}$$

where the inequality above is derived by Assumption [4](#) following Lemma 5 in [Fan et al. \(2021\)](#).

By Markov's inequality, we have

$$\mathbb{P}(\|\mathbf{y}_i\|_2 > \gamma_{1,\tau}) \leq \frac{\mathbb{E}\|\mathbf{y}_i\|_2^{2k}}{\gamma_{1,\tau}^{2k}}. \quad (\text{H.5})$$

To derive the upper bound for $\mathbb{E}\|\mathbf{y}_i\|_2^{2k}$, we decompose it as

$$\mathbb{E}\|\mathbf{y}_i\|_2^{2k} = \mathbb{E}\|\boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i + \boldsymbol{\varepsilon}_i\|_2^{2k} \leq 2^{2k-1} \left[\mathbb{E}\|\boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i\|_2^{2k} + \mathbb{E}\|\boldsymbol{\varepsilon}_i\|_2^{2k} \right]. \quad (\text{H.6})$$

We first derive the upper bound for the first term in RHS above. It follows Assumption 4 that

$$\|\boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i\|_{\psi_2} \leq K_0 \|\boldsymbol{\Theta}_{(i)}^*\|_{op} \leq K_0 \sqrt{\frac{R}{\underline{\lambda}}},$$

where the last inequality is derived by $R \geq \mathbb{E}\|\mathbf{y}_i^\top \mathbf{y}_i\|_{op} \geq \sup_{\mathbf{v} \in \mathcal{S}^{d_2-1}} \mathbb{E}[(\boldsymbol{\Theta}_{(i)}^* \mathbf{v})^\top \mathbf{x}_i \mathbf{x}_i^\top (\boldsymbol{\Theta}_{(i)}^* \mathbf{v})] \geq \underline{\lambda} \|\boldsymbol{\Theta}_{(i)}^*\|_{op}$.

Then, by moment behavior of sub-Gaussian random vector, we have

$$\mathbb{E}\|\boldsymbol{\Theta}_{(i)}^*{}^\top \mathbf{x}_i\|_2^{2k} \leq \left(K_0 \sqrt{\frac{R}{\underline{\lambda}}} \right)^{2k} (d_2 + k)^k = v_4.$$

Then, we drive that

$$\begin{aligned} \mathbb{E}\|\boldsymbol{\varepsilon}_i\|_2^{2k} &= \mathbb{E} \left(\sum_{j=1}^{d_2} \varepsilon_{ij}^2 \right)^k \leq d_2^{k-1} \mathbb{E} \left(\sum_{j=1}^{d_2} |\varepsilon_{ij}|^{2k} \right) \\ &\leq d_2^{k-1} \sum_{j=1}^{d_2} \mathbb{E}[\mathbb{E}(|\varepsilon_{ij}|^{2k} | \mathbf{x}_i)] \leq d_2^k M, \end{aligned}$$

where the last inequality is derived by Assumption 5. Recalling (H.5) and (H.6), we have

$$\mathbb{P}(\|\mathbf{y}_i\|_2 > \gamma_{1,\tau}) \leq \frac{2^{2k-1}(v_4 + d_2^k M)}{\gamma_{1,\tau}^{2k}} \leq \frac{c'_3 d \log d}{n_{1,\tau}}. \quad (\text{H.7})$$

Recalling (H.4), we obtain the upper bound as

$$\mathbb{E}[\mathbf{u}^\top \mathbf{x}_i (\tilde{\mathbf{y}}_i - \mathbf{y}_i)^\top \mathbf{v}] \leq \sqrt{v_3 c'_3 d \log d / n_{1,\tau}} = c'_4 (d \log d)^{\frac{1}{2}} n_{1,\tau}^{-\frac{1}{2}}.$$

Thus, the upper bound for $\|U_2\|_{op}$ can be derived as

$$\|U_2\|_{op} \leq \frac{32}{7} n_{1,\tau} c'_4 (d \log d)^{\frac{1}{2}} n_{1,\tau}^{-\frac{1}{2}} \leq c_2 (d \log d)^{\frac{1}{2}} n_{1,\tau}^{\frac{1}{2}}.$$

For the third term U_3 , we first derive the expectation term as

$$\mathbb{E}(\mathbf{x}_i \mathbf{y}_i^\top) = \mathbb{E} \left[\mathbf{x}_i \mathbb{E}(\mathbf{x}_i^\top \boldsymbol{\Theta}_{(i)}^* + \boldsymbol{\varepsilon}_i^\top | \mathbf{x}_i) \right] = \mathbb{E}(\mathbf{x}_i \mathbf{x}_i^\top \boldsymbol{\Theta}_{(i)}^*).$$

It follows Lemma [Appendix B.1](#) that

$$\|U_3\|_{op} \leq \frac{16}{7} \sup_{\mathbf{u} \in \mathcal{N}^{d_1-1}, \mathbf{v} \in \mathcal{N}^{d_2-1}} \sum_{i \in \mathcal{I}_{1,\tau}} \left[\mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \boldsymbol{\Theta}_{(i)}^* \mathbf{v} - \mathbb{E} \left(\mathbf{u}^\top \mathbf{x}_i \mathbf{x}_i^\top \boldsymbol{\Theta}_{(i)}^* \mathbf{v} \right) \right].$$

By Assumption 4, we can obtain that $\mathbf{u}^\top \mathbf{x}_i$ is sub-Gaussian random variable with $\|\mathbf{u}^\top \mathbf{x}_i\|_{\psi_2} \lesssim K_0$ and $\mathbf{x}_i^\top \boldsymbol{\Theta}_{(i)}^* \mathbf{v}$ is also sub-Gaussian random variable with $\|\mathbf{x}_i^\top \boldsymbol{\Theta}_{(i)}^* \mathbf{v}\|_{\psi_2} \lesssim K_0 \sqrt{R}/\sqrt{\lambda}$. Thus, it follows Bernstein's inequality (see, eg. Theorem 2.8.1 in [Vershynin \(2018\)](#)) that

$$\mathbb{P} \left\{ |\mathbf{u}^\top U_3 \mathbf{v}| > c'_3 \sqrt{(d_1 + d_2) n_{1,\tau}} \right\} \leq 2 \exp\{-C_3(d_1 + d_2)\}.$$

Taking a union bound over all $(\mathbf{u}, \mathbf{v}) \in \mathcal{N}^{d_1-1} \times \mathcal{N}^{d_2-1}$, we have

$$\mathbb{P} \left\{ \|U_3\|_{op} > c_3 \sqrt{(d_1 + d_2) n_{1,\tau}} \right\} \leq 2 \exp\{-(C_3 - \log 8)(d_1 + d_2)\}.$$

For the forth term U_4 , we define the coefficients as a vector

$$\boldsymbol{\alpha} = \left[\frac{\tau - \tau^*}{\tau}, \dots, \frac{\tau - \tau^*}{\tau}, -\frac{\tau^*}{\tau}, \dots, -\frac{\tau^*}{\tau} \right],$$

and follow the proof process of fifth term U_5 in Lemma [Appendix C.2](#) to derive

$$\mathbb{P} \left\{ \|U_4\|_{op} > c_4 \sqrt{(d_1 + d_2) n_{1,\tau}} \right\} \leq 2 \exp\{-(C_4 - \log 8)(d_1 + d_2)\}.$$

Recalling (H.3), the upper bound for the operator norm of the gradient is derived as

$$\begin{aligned} \left\| \nabla \tilde{L}_{1,\tau}(\bar{\Theta}_{1,\tau}) \right\|_{op} &\leq 2c_1 \sqrt{n_{1,\tau} d \log d} + 2c_2 \sqrt{n_{1,\tau} d \log d} \\ &\quad + 2c_3 \sqrt{n_{1,\tau}(d_1 + d_2)} + 2c_4 \sqrt{n_{1,\tau}(d_1 + d_2)} \\ &\leq \frac{1}{2} \lambda_{1,\tau}, \end{aligned}$$

where the constant c_λ satisfies $c_\lambda > 4(c_1 + c_2 + c_3 + c_4)$.

Otherwise, the upper bound for $\left\| \nabla \tilde{L}_{2,\tau}(\bar{\Theta}_{2,\tau}) \right\|_{op}$ and the case $\tau \leq \tau^*$ can also be derived as same. For any $\tau \in \mathcal{T}$, we take the union bound as $|\mathcal{T}| = O(n)$ and complete the proof.

Appendix H.3. Proof of Lemma Appendix F.3

We first decompose the target term as

$$\begin{aligned} \sum_{i \in \mathcal{I}} \mathbf{z}_i^\top (\tilde{\mathbf{y}}_i - \Theta^{*\top} \mathbf{x}_i) &= \underbrace{\sum_{i \in \mathcal{I}} [\mathbf{z}_i^\top \tilde{\mathbf{y}}_i - \mathbb{E}(\mathbf{z}_i^\top \tilde{\mathbf{y}}_i)]}_{U_1} + \underbrace{\sum_{i \in \mathcal{I}} \mathbb{E}[\mathbf{z}_i^\top (\tilde{\mathbf{y}}_i - \mathbf{y}_i)]}_{U_2} \\ &\quad + \underbrace{\sum_{i \in \mathcal{I}} [\mathbb{E}(\mathbf{x}_i^\top \Theta^{*\top} \mathbf{x}_i) - \mathbf{x}_i^\top \Theta^{*\top} \mathbf{x}_i]}_{U_3}. \end{aligned}$$

For the first term U_1 , we denote $\bar{\mathbf{z}}_i = \mathbf{z}_i / \|\mathbf{z}_i\|_2$ and the upper bound for p -th moment can be derived as

$$\begin{aligned} \mathbb{E} |\mathbf{z}_i^\top \tilde{\mathbf{y}}_i|^p &= {}_{(1)} \mathbb{E} \{ \mathbb{E} [|\mathbf{z}_i^\top \tilde{\mathbf{y}}_i|^p | \mathbf{x}_i] \} \\ &\leq {}_{(2)} \gamma_{s,\tau}^{p-2} \mathbb{E} \{ \|\mathbf{z}_i\|_2^p \mathbb{E} [(\bar{\mathbf{z}}_i^\top \tilde{\mathbf{y}}_i)^2 | \mathbf{x}_i] \} \\ &\leq {}_{(3)} \gamma_{s,\tau}^{p-2} \mathbb{E} \left\{ \|\mathbf{x}_i\|_2^p (\bar{\mathbf{z}}_i^\top \Theta^{*\top} \mathbf{x}_i)^2 + \|\mathbf{x}_i\|_2^p \mathbb{E} [(\bar{\mathbf{z}}_i^\top \boldsymbol{\varepsilon}_i)^2 | \mathbf{x}_i] \right\}, \end{aligned} \tag{H.8}$$

where the inequality (1) is derived by law of iterated expectations; the inequality (2) is derived by the definition of truncation; the inequality (3) is derived by the inequality $\|\mathbf{z}_i\|_2 \leq \|\mathbf{A}\|_F \|\mathbf{x}_i\|_2 \leq \|\mathbf{x}_i\|_2$. Then, we derive the upper bound for the two terms of RHS.

$$\mathbb{E} \left[\|\mathbf{x}_i\|_2^p (\bar{\mathbf{z}}_i^\top \Theta^{*\top} \mathbf{x}_i)^2 \right] \leq {}_{(1)} \sqrt{\mathbb{E}(\bar{\mathbf{z}}_i^\top \Theta^{*\top} \mathbf{x}_i)^4} \sqrt{\mathbb{E} \|\mathbf{x}_i\|_2^{2p}} \lesssim {}_{(2)} \frac{K_0^4 R}{\underline{\lambda}} p! d_1 (\sqrt{d_1})^{p-2},$$

where the inequality (1) is derived by Cauchy-Schwarz inequality; the inequality (2) is derived by Assump-

tion 4 and the moment property of sub-Gaussian random vector.

$$\begin{aligned} \mathbb{E} \left\{ \|\mathbf{x}_i\|_2^p \mathbb{E} \left[(\bar{\mathbf{z}}_i^\top \boldsymbol{\varepsilon}_i)^2 | \mathbf{x}_i \right] \right\} &\leq_{(1)} \left[\mathbb{E} \|\mathbf{x}_i\|_2^{\frac{pk}{k-1}} \right]^{\frac{k-1}{k}} \left\{ \mathbb{E} \left[\mathbb{E} \left((\bar{\mathbf{z}}_i^\top \boldsymbol{\varepsilon}_i)^2 | \mathbf{x}_i \right) \right]^k \right\}^{\frac{1}{k}} \\ &\lesssim_{(2)} p! d_1^p M^{\frac{1}{k}} = p! d_1 (\sqrt{d_1})^{p-2}, \end{aligned}$$

where the inequality (1) is derived by Hölder's inequality; the inequality (2) is derived by Assumption 4 and 5. Recalling (H.8), we have $\mathbb{E} |\mathbf{z}_i^\top \tilde{\mathbf{y}}_i|^p \lesssim p! d_1 (\gamma_{s,\tau} \sqrt{d_1})^{p-2}$. It follows Bernstein's inequality (see, eg. Theorem 2.10 in Boucheron et al. (2013)) that for any $t > 0$,

$$\mathbb{P} \left\{ |U_1| > c'_1 \left(\sqrt{n_I d_1 t} + \gamma_{s,\tau} \sqrt{d_1 t} \right) \right\} \leq 2 \exp\{-t\}.$$

Taking $t = C_1 \log d$, the upper bound can be transformed into

$$\begin{aligned} c'_1 \left(\sqrt{n_I d_1 t} + \gamma_{s,\tau} \sqrt{d_1 t} \right) &= c'_1 \left(\sqrt{C_1 n_I d_1 \log d} + C_1 \gamma_{s,\tau} \sqrt{d_1 \log d} \right) \\ &\leq c'_1 \sqrt{C_1 n_I}^{\frac{1}{2}} (d \log d)^{\frac{1}{2}} + c'_1 C_1 (d \log d)^{1 - \frac{1}{2k}} n_{s,\tau}^{\frac{1}{2k}} \\ &\leq c_1 (d \log d)^{\frac{1}{2} - \frac{1}{2k}} n_I^{\frac{1}{2}} n_{s,\tau}^{\frac{1}{2k}}, \end{aligned}$$

where the last inequality is derived by $n_I \gtrsim d \log d$ and the search domain condition.

For the second term U_2 , we can derive that

$$\mathbb{E} [\mathbf{z}_i^\top (\tilde{\mathbf{y}}_i - \mathbf{y}_i)] \leq \mathbb{E} [\|\mathbf{z}_i^\top \mathbf{y}_i\| \mathbf{1}_{\{\|\mathbf{y}_i\|_2 > \gamma_{s,\tau}\}}] \leq \sqrt{\mathbb{E} |\mathbf{z}_i^\top \mathbf{y}_i|^2 \mathbb{P}\{\|\mathbf{y}_i\|_2 > \gamma_{s,\tau}\}},$$

where the last inequality is derived by Cauchy-Schwarz inequality. Then, we derive the upper bound for each term.

$$\mathbb{E} |\mathbf{z}_i^\top \mathbf{y}_i|^2 \leq \|\mathbf{A}\|_*^2 \mathbb{E} \|\mathbf{x}_i \mathbf{y}_i^\top\|_{op}^2 \leq \|\mathbf{A}\|_*^2 v_1,$$

where the last inequality is derived by the proof of Lemma 5 in Fan et al. (2021) and $v_1 = 4K_0^2(2K_0\sqrt{R/\underline{\lambda}}) + 2kK_0^2M^{\frac{1}{k}}/(k-1)$.

Recalling (H.7), the upper bound for $\mathbb{E} [\mathbf{z}_i^\top (\tilde{\mathbf{y}}_i - \mathbf{y}_i)]$ can be written as

$$\mathbb{E} [\mathbf{z}_i^\top (\tilde{\mathbf{y}}_i - \mathbf{y}_i)] \lesssim \|\mathbf{A}\|_* n_{s,\tau}^{-\frac{1}{2}} (d \log d)^{\frac{1}{2}}.$$

Thus, we have $U_2 \leq c_2 \|\mathbf{A}\|_* (d \log d)^{\frac{1}{2}} n_I^{\frac{1}{2}}$.

For the third term U_3 , we first prove that $\mathbf{x}_i^\top \boldsymbol{\Theta}^* \mathbf{x}_i$ is sub-Exponential random variable. The SVD of $\boldsymbol{\Theta}^*$ is given by $\mathbf{U} \boldsymbol{\Sigma} \mathbf{V}^\top$, where $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_{d_1 \wedge d_2})$, $\mathbf{U} = (\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{d_1 \wedge d_2}) \in \mathbb{R}^{d_1 \times (d_1 \wedge d_2)}$, and $\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{d_1 \wedge d_2}) \in \mathbb{R}^{d_2 \times (d_1 \wedge d_2)}$. Thus, term $\mathbf{x}_i^\top \boldsymbol{\Theta}^* \mathbf{x}_i$ can be decomposed as

$$\mathbf{x}_i^\top \boldsymbol{\Theta}^* \mathbf{x}_i = \sum_{j=1}^{d_1 \wedge d_2} \sigma_j (\mathbf{u}_j^\top \mathbf{x}_i) (\mathbf{v}_j^\top \mathbf{x}_i).$$

And we can derive that

$$\begin{aligned} \|\mathbf{x}_i^\top \boldsymbol{\Theta}^* \mathbf{x}_i\|_{\psi_1} &\leq \sum_{j=1}^{d_1 \wedge d_2} \sigma_j \|(\mathbf{u}_j^\top \mathbf{x}_i) (\mathbf{v}_j^\top \mathbf{x}_i)\|_{\psi_1} \lesssim \sum_{j=1}^{d_1 \wedge d_2} \sigma_j K_0^2 \\ &\lesssim \|\boldsymbol{\Theta}^*\|_* K_0^2 \lesssim R_{\max} K_0^2 \left(\frac{R}{\underline{\lambda}}\right)^{1-q}, \end{aligned}$$

where the last inequality is derived by the inequality $\|\boldsymbol{\Theta}^*\|_* \leq \|\boldsymbol{\Theta}^*\|_{op}^{1-q} \sum_{j=1}^{d_1 \wedge d_2} \sigma_j (\boldsymbol{\Theta}^*)^q \leq R_{\max} \|\boldsymbol{\Theta}^*\|_{op}^{1-q} \leq R_{\max} R^{1-q} / \underline{\lambda}^{1-q}$. Then, it follows Bernstein's inequality that

$$\begin{aligned} \mathbb{P} \left\{ |U_3| > c_3 n_I^{\frac{1}{2}} (d \log d)^{\frac{1}{2}} \right\} &\leq 2 \exp \left\{ -C'_3 \min \left[\frac{c_3^2 n_I d \log d}{n_I R_{\max}}, \frac{c_3 n_I^{\frac{1}{2}} (d \log d)^{\frac{1}{2}}}{R_{\max}} \right] \right\} \\ &\leq 2 \exp \{-C_3 d\}. \end{aligned}$$